Raising Revenues:

Evidence from Property Tax Assessments in India

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Abstract

How do property-owners respond to increases in property-tax liabilities? Particularly, so these responses differ across the initial-liability distribution, and how does accounting for this heterogeneity affect the revenue-maximizing tax rate? I construct a novel dataset of geolocated properties and digitised parcel-level maps from an Indian municipality to answer this question. I exploit variation in liabilities that arise from area-based guidance values that are fixed as of 2011, known as 'circle rates'. First, I find that a 1% increase in the circle rate reduces compliance by 0.49% but increases total revenues by 0.83%. This implies that revenues are maximized by scaling every circle rate by a common factor of 3.8. This preserves the existing relative assessment schedule while increasing liabilities proportionally across properties, and is equivalent to increasing the statutory tax rate from 0.25% to 0.95%. Second, I show that this estimate masks substantial heterogeneity in behavioural responses. Properties at higher circle rates are more elastic than those at lower circle rates. Accounting for this, reduces the revenue-maximizing common scaling factor to 2.25, equivalent to a statutory tax rate of 0.56%. Third, I find that updating liabilities to reflect the revenue maximising schedule leads to the largest gains in revenues but that simply updating outdated circle rates to more recent values recovers a substantial share of these gains.

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Property tax revenue as a percentage of GDP is around 0.6% in LMICs, four times lower than the OECD average of 2.5% (Bahl and Martinez-Vazquez, 2007). Among other reasons, low revenues may reflect low tax liabilities (Awasthi et al., 2021). Raising liabilities in low-enforcement environments however, may come with increased non-compliance[1] (Slemrod, 2019). More importantly, this response need not be the same throughout the tax base. If taxpayers facing different initial liabilities respond differently to an increase in taxes, a single average elasticity may mask substantial variation in the revenue consequences of raising liabilities.

This heterogeneity matters for policy. Existing work on income taxation shows both that behavioral elasticities can differ substantially across taxpayers (Gruber and Saez, 2002) and that pooled elasticity estimates combine underlying heterogeneous responses (Kumar and Liang, 2020). How much heterogeneity then exists in property-tax responses, and how does accounting for it change the amount by which a municipality can raise liabilities before additional increases begin to reduce revenue? I answer these questions using variation in *property assessments*. In the municipality I study, the statutory tax rate is common, while assessed values depend on administratively assigned, area-based guidance values that vary spatially and have remained fixed since 2011. This assessment system generates spatial variation in liabilities at many points of the existing tax base. I use this variation to estimate how compliance and the amount paid, conditional on payment[2], responds to higher assessed values, and how these responses differ across the guidance-value distribution.

I then use these estimates to evaluate alternative ways of updating the assessment schedule. First, I estimate the common proportional increase in guidance values that maximizes revenue using the pooled behavioral response. Second, I allow behavioral responses to vary across the guidance-value distribution and re-estimate the common scaling factor. Finally, I compare these revenue-maximizing updates with a more conventional policy of updating outdated guidance values to their values in a more recent year. I study these questions in Kolhapur (popn 800,000), India, where assessed values for residential, owner-occupied properties have remained unchanged since 2011.

I combine three new sources of data for this analysis. First, I collect administrative property-level data on tax liabilities, payments and demographic information across 12 years. I translate all names

[1]Non-compliance in this case refers to complete non-payment of the tax. There could be other behavioral responses to consider, typically characterised as changes in the tax base or avoidance (Saez et al., 2012). This includes actions such as filing for property re-assessments, changes in the decison to own property, moving to other low-liability areas or constructing properties in low-liability areas. I don't find evidence of these behaviors, making the decision to just not pay the tax the most important response.

[2]This estimate is important as households are not required to pay the full liability and often have penalties imposed due to non-compliance in a previous year, often leading to payments much larger than their liability. Penalties are 2% of the liability, so larger liabilities lead to larger penalties

and addresses from Marathi to English and mark properties as being residential and owner-occupied. I then geolocate a sample of 6,345 properties using data from an official Kolhapur Municipal Corporation (KMC) geotagging exercise. Third, I georeference and digitize scanned parcel and "circle"-level maps of the city and link them to "circle rate" data. Together, this data allows me to observe each property's tax liability, payment behavior and granular location.

To causally measure the elasticities of compliance and the amount of taxes paid to liabilities, I make use of a variation that arises from an area-based property valuation formula. Each property is assessed based on a function with a fixed set of weights for each property characteristic and an area-based guidance value known as the "circle rate". The tax liability is then 0.25% of the property valuation. The circle rate is a $ amount per square metre assigned to each property in a given "circle" [3] and is a floor price meant to guide the reporting of property sale prices (Anagol et al., 2024). The KMC pegs it at its 2011 value in the property-valuation formula[4]. Effectively, this results in *no* changes to tax liabilities since 2011. Since circle boundaries have stayed constant over the years and do not follow any local administrative divisions, I define the 2011 circle rate as the baseline circle rate in my main empirical specification.

Specifically, two *similar* properties that are located sufficiently close to each other, could lie on opposite sides of the circle boundary, get assigned varying circle rates and have different liabilities[5]. I first match properties using a nearest-neighbour matching algorithm; I retain property-pairs that are near-neighbors with a distance of 20 metres between them; and *show* that these property-pairs are indeed balanced on observable characterestics. Then, the difference in their liabilities is driven solely by the difference in their underlying circle rates. Since I restrict my sample to properties with a distance of 20 metres between them [6], I ensure that properties lie on the same side of the road and this granularity aids my empirical design. I also restrict my main sample to residential, owner-occupied properties that have a balanced panel from 2013–2024, to avoid any bias from new constructions.

I compare properties using a *within-property-pair-by-year* fixed effect specification with the logarithm of the 2011 circle rate as the main independent variable[7]. This allows me to estimate

[3] A circle typically comprises of a few parcels and could be as big as 1–2 blocks. The word 'circle' is a bit of a misnomer as they are not actually shaped as circles

[4] Circle rates are however set and updated by an external state-level commission every two years

[5] While it is possible that the baseline circle rate was driven by observable differences in properties between two neighbouring circles, these differences are likely to be small closer to the circle border.

[6] While this distance is very narrow, it effectively allows me to compare neighbours or near-neighbours. I discuss in Section 1 that in dense urban cities, this is not an implausible small distance.

[7] This design closely follows Dube et al. (2010)'s design comparing county-pairs with differences in minimum wages.

differences in the outcome within a pair of properties for the same year, due to a 1% increase in the circle rate. This also allows me to hold fixed unobservable characteristics within each property-pair that may drive compliance, such as access to public services, quality of roads, and other neighbourhood amenities.

In order for these estimates to be causal, the main identifying assumption is that these near-neighbour properties are observationally similar. Since liabilities are a hedonic function of the circle rate and observed property characteristics, I construct a "residual" term by residualizing liabilities with respect to the circle rate. This term then captures characteristics such as the size of the property, the height and age of the building and any amenities located within an apartment complex. I show that properties within a pair are balanced on this residual term. I also show that properties are balanced on observable characteristics such as the number of floors, the type of construction and the area in square metres of the property, variables I obtain from the Google Footprints database.

Next, I set up a simple conceptual framework that defines total revenues as being equal to average compliance times the average amount paid by compliers times the number of properties. I assume that the number of properties is fixed and that the compliance and taxes paid are linear and log-linear in circle rates, respectively. [8]. Then, the mean Revenue Maximizing Circle Rate (RMCR) can be obtained using the elasticity of compliance and taxes paid[9]. I then construct a counterfactual distribution of circle rates by scaling all existing circle rates by a common factor such that the mean of the resulting distribution equals the RMCR. This proportional scaling preserves the relative circle-rate schedule and is therefore equivalent to changing the statutory tax rate uniformly across properties while holding assessed values fixed.

I first estimate the effect on compliance due to increasing the circle rate, where I define compliance as an indicator equal to one if the property owner makes any payment in a given year. This is then an extensive-margin response to higher liabilities. I find that a 1% increase in the 2011 circle rate reduces compliance by 0.32 percentage points. Relative to a control mean of 0.69, this implies a compliance elasticity of -0.46 [10].

Is this fall in compliance offset by higher payments among those who do pay? On the intensive margin using only the sample of complier property-pairs, payments rise substantially. A 1% increase

[8] I test for non-linear responses and don't find any evidence of this. F-stat: 1

[9] I start with assuming that all properties in the city are charecterized by the same elasticities, essentially giving me 'pooled' measures that are weighted averages of property-level elasticities (Kumar and Liang, 2020)

[10] I also show in a robustness check that this estimate is very close to estimates I get from a classic Spatial Regression Discontinuity Design, following Best et al. (2025)

in the circle rate increases taxes paid by 1.29%. Combining this intensive-margin elasticity with the extensive-margin compliance elasticity implies that for the average property, the increase in payments among compliers is larger than the decline in compliance. Based on the model, the elasticity of compliance plus the elasticity of payments should exactly equal 0 at the revenue-maximizing point. Using my estimates, this sum is 0.83, suggesting that a 1% rise in the circle rate leads to a 0.83% increase in revenues.

Since I restrict the intensive margin to complier-property-pairs, a concern may be that this effect is driven by property-pairs differentially selecting into the sample. In a robustness check, I relax the fixed-effect structure so that pairs with only one payer also contribute to the estimate, and find a similar elasticity of 1.1. I also use a Poisson model[11] on the full sample of proeprties, and find a statistically similar but less precisely estimated, total revenue elasticity of 0.54. Together, these results suggest that selection into payment marginally affects the magnitude of the intensive-margin estimate, but does not overturn the conclusion that higher circle rates increase total revenues.

Extrapolating these elasticities, I find that the mean circle rate at which revenues are maximized is 834 USD. This amounts to all circle rates being scaled by 3.9, or a 300% increase. This is equivalent to a 290% increase in tax rates or in this case, the tax rate increasing to 0.975% from 0.25%, while keeping assessments unchanged[12].

Any enforcement activity could be underestimating the baseline compliance effect. I next make use of a policy rule, according to which the KMC sends reminder letters and threatens enforcement action once accrued liabilities cross a threshold of 117 USD. I restrict my baseline sample to properties below this threshold, who do not receive any letters. I reassuringly find that the compliance elasticity is not statistically different from the one estimated at baseline [13]. This enforcement by the KMC is very light touch and my elasticity estimates should be interpreted as those in settings with weak enforcement (Bachas and Soto, 2021; Best et al., 2015; Emran and Stiglitz, 2005).

A mechanism that would explain these results would be house price growth. If house prices in higher 2011 circle-rate areas, grew faster than their neighbouring house prices, then this wealth effect could affect both the fall in compliance and the amount paid by compliers. Using contemporaneous

[11] Since this sample now includes lots of 0s from non-compliant properties, I don't use a logged dependant variable.

[12] Although this is a fourfold increase, the new rate is not unusually high by international standards. The IMF reports that rates in market-value-based property tax systems typically range from 0.5–2%, the average effective tax rate on a median-valued owner-occupied home across the largest U.S. cities was 1.21% in 2025, São Paulo and Rio de Janeiro, levy residential property taxes at approximately 1% of assessed property value

[13] The KMC only does enforcement action via tax officers when total accrued liabilities cross a threshold of 10,000 USD, which is so high that very few residential properties are threatened with it. This is reassuring in my setting as it is not plausible that enforcement officers choosing to enforce in higher circle-rate areas biases my results

circle rates as proxies for house prices (Anagol et al., 2024), and adding it as an interacted term, I find no differential effects. It thus seems unlikely that my main results are caused by the growth rate of house prices [14].

It may then be expected that there are effects on where houses get constructed. Particularly, if more houses were being constructed in lower baseline rate areas this would be suggestive of tax avoidance behavior. Using the number of new properties that get added to the tax roll every year, I find that a 1% increase in the circle rate leads to 1.5–3.2 more houses being constructed on average within that circle. This suggests that the circle rate structure does affect city construction, but not in the direction avoidance behaviour would suggest. I also find in additional results that the COVID-19 pandemic and the resultant lockdown leads to a 1.5%-2% fall in compliance in 2020[15].

Next, I ease my assumption of a singular elasticity describing the whole city, and measure heterogenous responses. I follow the heterogeneous-treatment-effect specification as in Allcott et al. (2020) and allowing the effect of the circle rate to vary with the local within-pair mean circle rate level. I find that the elasticity of compliance becomes more negative and the elasticity of taxes paid becomes more positive, as the circle rate increases. [16].

I extrapolate these elasticities and using a model-based numerical estimation exercise, obtain a mean RMCR that takes these heterogenous elasticities into account. I find that this RMCR is now lower than the one suggested by the pooled exercise, all circle rates would now need to be scaled up by 2.25% or need a 125% increase, to maximise revenues. This is equivalent to a new tax rate of 0.5625%. I also measure local revenue-maximising rates using these elastcities, and in line with Gruber and Saez (2002), I find that the optimal tax rate schedule is declining.

Finally, I use both pooled and heterogenous elastcities to estimate changes in revenue, if the circle rate was to be updated. Aside from updating the circle rate to the two estimated revenue-maximising schedules, I also consider a scenario where the circle rate is simply updated to a more recent year. Accounting for heterogeity reduces all revenue estimates by half. Yet, revenue gains are large. Scaling all circle rates by 2.25, increases the average revenue per property by 34%, while using pooled

[14] I also find that places that had higher circle rates in 2011 did *not* grow substantially faster than places that had lower rates. In fact, they likely grew slower or almost at the same pace

[15] While there were several discount schemes allowing people to pay their liabilities late, these lasted until 2022. Compliance springing back to its pre-2020 levels in 2021 suggests that this steep fall could be attributed to a tightening of income contraints in 2020. This would be a hypothesis in line with Brockmeyer et al. (2021)

[16] These results are in line with Tourek et al. (2026), who find that increasing tax rates to those on top of the tax-liability schedule, reduces compliance but mechanically increases revenues. Their experiment is about progressivity, so they reduce rates for those at the bottom of the tax schedule and find that this increases compliance in the bottom. In my emperical exercise, rates rise for everybody.

elasticities and scaling all rates by 3.9% suggests a revenue increase of nearly a 100%. Considering heterogenous behavioural responses may thus be crucial to estimating revenue maximising rates.

This paper makes three contributions. First, it contributes to the literature on revenue-maximizing tax rates. A large literature shows that revenue-maximizing tax rates depend on taxpayers' behavioral responses to taxation (Saez, 2001; Badel and Huggett, 2017; Saez et al., 2012). However, aggregate elasticity estimates masks heterogeneity in underlying responses (Kumar and Liang, 2020). This distinction matters for revenue-maximizing taxation, Holter et al. (2019) and Guner et al. (2016) use structural models and find that it changes the Laffer rate, while Serrato and Zidar (2018) consider heterogeneity in estimating an optimal corporate tax rate and find that accounting for heterogeneity increases the rate. I contribute to this literature by considering a different type of tax and showing that accounting for heterogeneity substantially lowers the revenue-maximising rate.

Second, it contributes to the literature on property-tax assessment systems. I study an area-based assessment system, while most of the literature focuses on individual property-asessments. For instance, in a very similar setting of a city in India, and Duflo et al. (2026) find that updating individual property tax assessments leads to a large 30–60% increase in revenues. Avenancio-León and Howard (2022) and Hou et al. (2025) show taht imperfect and infrequent property assessments leads to regressivity. Knebelmann et al. (2026) show that algorithmic valuation improves assessment accuracy and equity in Senegal.

Third, it contributes to the literature on tax compliance and fiscal capacity in developing countries. Ajzenman et al. (2024) and Tourek et al. (2026) find that increasing the rate for richer properties and reducing the rate for poorer properties leads to increases in revenues. Existing work studies how enforcement, information, technology, social norms, tax rates, lottery-based rewards, collector incentives and progressivity affect compliance (Bergeron et al., 2024; Brockmeyer et al., 2021; Knebelmann et al., 2026; Best et al., 2025; Manwaring and Regan, 2024; Okunogbe and Tourek, 2024; Ferraz et al., 2024; Bergeron et al., 2024; Brockmeyer et al., 2021; Carrillo et al., 2021; Balán et al., 2022; Khan et al., 2016; Aparicio et al., 2011; Pomeranz, 2015; Ajzenman et al., 2024; Tourek et al., 2026). I contribute my showing that the compliance response can change heterogenously based on the assessment schedule.

I begin by describing the institutional setting in Section 1, describe my data in Section 2, and go over the conceptual framework, main results and heterogeneity in Sections 3, 4 and 5. In Section 8, I

conclude.

1 Urban Municipal Finances

In this section, I first describe the increasing importance of property taxes in India. Next, I describe the features of the property tax in Kolhapur, focusing on the ones I exploit to measure elasticities.

1.1 Recent developments

There are two broad sources of revenues for urban municipalities in India: own-source and government transfers. Historically, property taxes have formed about 30% of own-source revenue and 50% of own-tax-revenue, with some variation across cities (Awasthi et al., 2021). However, post-2016, after the passage of a Goods and Services Tax law that centralized sales tax collection, other sources of own-tax-revenue have significantly reduced. This has increased municipalities' reliance on property taxes and has brought local tax collection to the forefront as a policy issue [17].

This increased reliance has come along with reforms in property taxes including digitization of valuation exercises, digitizing village-level tax collection records, harmonizing tax bands, etc [18]. The paucity of easily-available data has however made studying the effects of these reforms difficult. In 2011, the Government of Maharashtra passed a Maharashtra Municipal Property Tax Board Act and *suggested* that all municipalities move away from the Rental Value Assessment[19] of properties and into the Capital Value Assessment method. The state also prescribed a partly hedonic formula with assigned weights to be used when assessing properties. Only two municipalities immediately took over this suggestion and made it law; Kolhapur was one of them. Thus, starting 2011, Kolhapur moved to assessing *owner-occupied properties* using the Capital Value Assessment Method [20].

1.2 Property Tax in Kolhapur: Valuation of Owner-Occupied Properties

On switching to the Capital Value Assessment method, each property 'a' in structure 'b' in year 't' was assessed based on the following formula:

$$\text{Valuation}_{a,b,t} = \text{Circle Rate}_{2011,c} \times \theta_{Category,b} \times \kappa_{Type,b} \times \mu_{Age,b} \times \lambda_{Height,b} \times \text{Square feet}_a \quad (1)$$

[17] See: https://fincomindia.nic.in/asset/doc/commission-reports/15th-FC/reports/studies/State%20of%20Municipal%20Finances%20in%20India.pdf.

[18] This includes policies such as SVAMITVA, for instance: https://svamitva.nic.in/

[19] The Rental Value Assessment method is resource-intensive and based on valuator discretion, as officials must estimate the actual or notional rent for all properties.

[20] Rental properties switched to using this method only in 2019

θ, κ, μ and λ are all weights assigned based on specific rules that I show in Table A1. Notice that all weights are common for all properties in structure 'b'. 'Square feet' refers to the area of the property 'a' and is the only feature that is property-specific, and lastly, 'Circle Rate' is a $ per square metre amount set by the state government and assigned to all properties in a 'circle' or a 'survey area'.

Circle Rates serve as a proxy for market property values and though they are likely much lower (Awasthi et al., 2021) they closely follow house price trends and have been trending closer to actual house prices over the past decade (Anagol et al., 2024). I present an excerpt of the document assigning circle rates to survey areas in Panel C and snapshots of a few survey areas in Panels A and B of Figure A1. Circle rate assignments are made by a state-level commission based on the demand and supply of properties in an area and are updated every other year [21].

However, circle rates in the property tax formula in Equation 1 are fixed at their 2011 value. Combined with the fact that properties were assessed for their characteristics last in 2007, the valuation of all owner-occupied properties has remained *unchanged* since 2011. The statutory property tax rate is then 0.25% of Valuation$_{a,b,t}$. This rate is constant across all properties in the municipality, though as valuation amounts differ, the liability or in other words, the effective tax rate for each property, differs.

1.3 Details on Circles or Survey Areas

Since I use the difference between two circle rates as my main identifying variation, it is important to note that circles or survey areas are not administrative divisions and not explicitly drawn for purposes of the property tax. They often criss-cross over roads and are fairly small, often the equivalent of one block in a standard city in the United States. In Figure A2, Panel A, I present a sample of circles superimposed on a zoomed-in Google Maps screenshot of the city. Outlines in red represent circles as drawn by the state commission. Notice that while circles adhere to the shape of large roads, they cross over smaller roads. This means that there is likely no difference in local government services or political institutions across neighbouring circles.

In Figure 1, Panel A, I present a shaded map of circles within the city. Darker colours present areas with a higher circle rate in 2011 and lighter colours are areas with a lower circle rate. I present a zoomed-in version of this map focusing on a portion of one ward in the city in Panel B, showing both

[21] Their purpose is to serve as a floor for the rates at which properties in the circle must be sold at and are reported for the purposes of a separate sales transaction tax. Anagol et al. (2024), for instance use these rates to check for bunching around transaction prices.

the compact nature of the circles and the jumps in rates. As rates are set based on the demand and supply of properties within a circle, it is likely that even within neighbouring circles, the one with the higher rate has more construction. As new properties get constructed, they get added to the roll and get freshly assessed. I constrain my main results to a balanced panel of properties to avoid any bias from newly constructed properties.

It is also possible that even between neighbouring circles, properties in the high-rate circle are different from properties in a lower-rate circle. However, at the border of two circles, this is less likely; houses at circle-borders that neighbour each other are likely to be observationally similar.

1.4 Building Codes and Property-Width Regulations

These distances may seem very small but under building codes in India and the state of Maharashtra, they are not implausible. Building code regulations allow for no gaps to exist between two adjacent plots up to an area of 450 metre^2 [22]. Average plot widths in mid-sized urban residential areas are about 6 metres, making it possible (and likely) that adjacent or near-adjacent plots have a distance of 8 metres between their centroids. Since the same building code also specifies that roads must be around 9–12 metres wide, it is also likely that the coordinates I compare, are on the same side of the road. I flesh out details on my distance thresholds, the main specification I use and the identifying assumptions that need to be satisfied for causality in Section 2.

1.5 Tax Bills and Enforcement

Tax Bills are generated on the 1st of April for the whole city, with a deadline to pay by the 30th of June, post which monthly penalties of 2% are levied. Before 2015, penalties were levied annually at 18% instead of accruing monthly. Tax bills are generated digitally and are physically delivered by post, while payments can be made online through a payment portal. Property tax debts need to be cleared in order to transfer property deeds, however enforcement action outside of this requirement primarily happens through letters.

Property owners with a cumulative accrued amount of over 10,000 rupees (around 100 USD), including penalties and unpaid bills are sent enforcement letters reminding them to make a payment. Once this accrued amount crosses 1,000,000 rupees (around 10,000 USD), enforcement officers make visits and threaten to lock up the premises of non-payers. This second threshold is high enough

[22] See page 221: https://www.mmrda.maharashtra.gov.in/sites/default/files/2023-10/UDCPR_compressed_2.pdf

that most properties that face this threat are large commercial properties. The first threshold allows me to separately check the effect of enforcement actions once the threshold of a 100 USD is crossed. However, outside of this, there does not seem to be any evidence of enforcement officers making visits and asking people to pay, as is seen in other low and middle income countries.

In the next section, I describe the various datasets used in this paper and the construction of all the primary variables used for analysis. I also describe the process of collecting and constructing the geographic variables necessary for my baseline spatial specification.

2 Data

I use three different sources of data to construct the main dataset with property-level variables (i) a property-level tax bills and payments history dataset, (ii) a property-level coordinates data for a subset of properties and (iii) a Google Footprints database with property-level characteristics. I further augment these with a dataset of circle rates and circle-level maps, digitized from scanned documents.

2.1 Property Tax Bills and Payment History

I start by collecting property-level bills and payment data from the Kolhapur Property Tax defaulters website, the purpose of which was to publicize names of defaulters. I then collect information from the Kolhapur Municipal Corporation's (KMC) property tax payment website once in December 2024 and use a different website to re-collect data on missing bills in April 2026 [23]. On all these sites, I sequentially search for property numbers starting from a 100 and collect all the information displayed. This gives me detailed demographic information such as the names of the property assessee and the occupier of the property, the full address, the circle number, ward number and division number; as well as information on bills and payments.

2.1.1 Tax Bills

Using the first source on defaulters, I obtain a unique property assessment number for each property, the ward it belongs to and information on tax bills such as the liability, the penalty accrued due to non-payment, the amount of the municipal tax owed and any discount offered on the bill by the municipality. I collect this information for all available years, though it starts getting well-populated only after 2011. This source for defaulters however, only contains tax bills for *never-payer* properties.

[23] The two sites I collected data from originally in December 2024 are not hosted by the KMC anymore. However, since tax year had not ended in 2024 when I first collected the data, payments are incomplete for Financial Year 2025, which allows me to retain my main sample only until FY 2024.

For those who always pay the tax (always-payers) or pay intermittently (ever-payers) the tax bill on the website disappears once the assessee pays the bill. This means that I don't observe any tax bills for always-payers and for ever-payers that paid their bill in FY 2025 [24].

I do however see the original bill for never-payers and for ever-payers who hadn't paid in FY 2025 [25]. I start with populating data for these two groups and out of 75,473 owner-occupied properties in the sample, this step gives me data for 23,069 properties[26].

Next, I use the tax bills data I re-collected in April 2026. New bills for properties are generated on the 1st of April, and re-collecting it immediately ensures that I have data on bills for properties *before* they are able to make a payment. This step gives me a large portion of the remaining data on liabilities and I am able to populate bills for 43,559 properties.

I am now left with 8845 owner-occupied properties with no data on bills, likely always- and ever-payers who paid very early in April 2026. For these properties, I simulate their bills using their payments and a few assumptions [27], however, I note that none of these properties with simulated bills are retained in my final geo-located sample.

2.1.2 Tax Payments

I collect information on the amount of the tax paid, the mode of payment, the exact date of payment and the receipt number.

2.1.3 Processing demographic information

Since all names and addresses are stored in Marathi using the Devanagari script, I first translate everything from Marathi to English using a Google Translate API. I then use the names of the owner or the assessee, and the name of the occupier to construct a variable marking properties as being owner-occupied. I also use assessee names along with regular expressions to classify properties as being 'commercial' or 'institutionally-owned' properties [28]. I make these distinctions as (i) rental properties underwent a valuation change in 2019 before which their valuations were not dependent

[24] Even if they didn't pay a bill in one intermittent year, but paid in the next year, the bill for the previous year disappears

[25] I see bills for never-payers for every single year, obtained from the defaulter list. For ever-payers, if I see a bill in FY2025, I am able to assume that the bill was the same for all years prior as liabilities have remained unchanged.

[26] Properties can ask to be re-evaluated, however, there are only about 1500 properties that have ever asked for re-evaluation. I am not able to see which properties these are, unless they are never-payers

[27] I use the rule that until 2015, penalties only accrued yearly on the entire amount defaulted. This means that the payment on the bill could be made at any point during the year and with no penalties. I now assign bills in two steps: (i) I tag all payments made per property before 2015 and (ii) assign them as a bill if payments were made for the same exact amount for 2 consecutive years. Using these assumptions for properties for which I have the actual tax bill, I find that I predict the bill accurately 77% of the time. This last step allows me to populate a tax bill for all properties in my sample.

[28] For instance, all properties ending with a 'Ltd', 'Limited', 'Corporation', 'Corpn' etc are marked as commercial. All schools, trusts, religious places, are marked as institutional

on the circle rate and are not constant over the past 15 years; (ii) commercial properties use a different circle rate, typically one meant for commercial spaces and since this rate is much higher, using the same rate as in the residential one may generate bias and (iii) institutional properties are those that are owned by the Kolhapur Municipal Corporation or other governments, these properties may have different incentives to comply or evade than purely residential properties.

2.2 Coordinate-level data

I obtain and construct two sets of coordinate-level data for a smaller sample of my data, (i) a Google Footprints database and (ii) a geolocated property-database. I then merge these two datasets so that each property in the final sample has a set of coordinates and a set of covariates attached to it.

2.2.1 Geolocating properties

I construct the geolocated property database using three sources. First, I obtain a sample of coordinates from an official KMC source. The KMC developed an app in 2026 which allowed the official property tax bill delivery officers to geotag each property they visited. I obtain coordinates for 4141 properties in my sample through this source. Since coordinates were located at the doorstep of each property, these are very accurate.

Second, I use the Maps Places API and find addresses for coordinates from the Google Footprints dataset. The Places API gives me an exact address of the coordinate, as well as a name of the building. I then manually match these addresses with the addresses from the property-tax dataset and am able to find coordinates for 670 properties this way. While the places API is useful, due to the poor quality of addresses in India, there are a lot of addresses with no matches and is often able to only find names of the closest apartment building.

I then move on to the third method of collecting coordinates where I manually find coordinates for addresses in my data using Google Maps [29]. In Figure A2 Panel B, I provide an example of a property on Google Maps that I manually geolocate. This gives me an additional 1534 properties and a total of 6345 properties with coordinates. I present all coordinates in my geolocated sample in Figure A4

All coordinates in my data have a minimum precision of six decimal points and a maximum precision of 11 decimal points. To construct pairs, I round up all coordinates to 6 decimal points.

[29] This method is very flexible and is particularly useful since several addresses in the data are structured in a non-standard way; some are specific and state the name of the building, the plot number or a colony number mixed with a plot number; some are very vague, with just the name of a street or a landmark. I am unable to geolocate properties with these very vague addresses but am able to use plot numbers and other details to get coordinates

2.2.2 Google Footprints data

The Google Footprints database for Kolhapur gives me useful information on the structures located on each coordinate. This includes the type of the structure - residential, commercial, religious, etc.; the number of floors; the area of the structure; the name of the road it is located on; the width of the road it is located on; the ward and the name of the area. I present a snapshot of the shapefile I obtain in Panel A, Figure A6 and a zoomed in snapshot in Panel B.

I then use a nearest-neighbour matching exercise between this data and the set of geolocated properties and retain characteristics only for those properties that find a match at or below 0.0005 miles.

2.3 Digitizing Circle-level shapefiles

Since there are no readily available shape files for the city at the circle-level, I first collect scanned maps of the city that mark circle areas using the city's 'Development Plan'. In Panels A and B of Figure A1, I present snapshots of one of the PDF maps I digitize.

I collect 18 maps from the KMC website, all of which join to make the whole city. Nine of these are for denser areas as dense areas tend to have closer circles. All circle areas are marked by dashed red lines. I first georeference all maps using control points on QGIS so as to be able to use their coordinate-level information. I then programmatically extract the red boundary linework from the PDFs and reconstruct circle polygons from these dashed outlines. Because the original boundaries are dashed rather than continuous, I use automated gap-closing steps to connect nearby dash segments before forming polygons. I separately extract the printed survey numbers from the PDF text layer, clean and standardize these labels, and spatially assign them to the corresponding polygons. While this step allows me to make maps as in Figure 1, there are still missing and mislabeled parcels. Due to this issue, I don't rely on these maps for my main analysis, but use it for exploratory purposes.

2.4 Digitizing Circle Rates

The other crucial piece of data I need are the circle-rates for 2011. I obtain hard copies of this data from the Stamps and Registration Department in Kolhapur and digitize it. I present a snapshot of the pdf I digitize in Panel C Appendix Figure A1. Each circle has a different rate for residential constructions, open land, commercial constructions and office spaces. I only extract the rate meant for residential purposes as that is the rate used in the property tax formula and thus the rate relevant to me. I

also collect this data from 2015 to 2024 as it is available on the Maharashtra Registration & Stamps Department's eASR portal. While I only use the 2011 rate for identification, the longer panel of the change in rates allows me to test for heterogeneity in outcomes between areas with higher and lower circle rate growth rates.

Circle rates are set by an external committee and the formula used to determine and update rates is not publicly available. There could then be a concern that boundaries are drawn and rates are set based on area characteristics. While my design of comparing houses located very close to each other should take care of this concern, in Section A, I conduct an exercise to check what predicts the circle rate. I find that no concerning factors emerge.

2.5 Final Sample

Out of the 75,473 owner-occupied properties in my initial sample, 38812 properties have a balanced panel and are not commercial or institutional properties. Out of these, I obtain coordinates for 6345 properties

The highest number of properties in the full balanced panel of properties lie in the middle, with an average circle rate of 183 USD. In Figure A5, Panel A, I present the distribution of the number of properties across circle rate bins. In Panel B, I present the corresponding amount of revenue collected for each circle rate bin and the potential revenue that could be collected if compliance was full. The largest amount of potential revenue to be collected also lies in the middle of the circle rate distribution, though there is a large amount of uncollected revenue even at the highest bin. In total, just from the balanced panel, there is 8 mn USD in potential uncollected revenue. In Section A.1, I plot descriptive relationships between key variables.

I also note that since the property tax formula uses rates from 2011, there is a large growth in circle rates over the past 15 years. In Figure A9, I present the distribution of circle rates in 2011 in the city in Panel A and the distributions for 2017, 2019, 2021 and 2023 in Panel B. The median circle rate, which is about 180 USD in 2011, is nearly 450 USD in 2023. This points at the likely large under-estimation of property taxes due to using rates from 16 years ago.

3 Conceptual Framework and Empirical Strategy

From the persepective of a policy maker, there are two things she could care about: compliance and the revenue collected [30]. The final revenue collected is then dependent on the compliance *and* the amount they pay while complying.

The amount by which compliance and taxes paid change with circle rates then determine the revenue function for this city. If it is necessary that taxpayers pay their whole liability every time, the amount of tax paid would be redundant as a singular measure for compliance would capture them both. However, assessees need not pay the entire amount while complying. In addition, since penalties increase with liabities, taxpayers who comply with a delay often pay an amount much higher than their liability.

I start by modifying the framework in Bergeron et al. (2024) [31] and add an additional term for the amount of the tax paid by compliers. The revenue collected at a circle rate 'cr', is then defined by

$$\mathrm{R(cr)} = \mathrm{N(cr)} \times \mathrm{C(cr)} \times \mathrm{A(cr)}^{+} \tag{2}$$

where $A(cr)^{+}$ is the average dollar amount paid by compliers and $C(cr)$ is the average compliance ranging between 1 and 0 at circle rate 'cr' and $N(cr)$ is the number of properties in a circle. Using the standard formula for elasticity, the elasticity of compliance is then defined as $\epsilon_{compliance} = \frac{\Delta C(cr)}{C(cr)} \times \frac{cr}{\Delta cr}$ or $\frac{dC(cr)}{dcr}\frac{cr}{C(cr)}$ and the elasticity of payments by compliers as $\epsilon_{payments} = \frac{\Delta A^{+}(cr)}{A^{+}(cr)} \times \frac{cr}{\Delta cr}$ or $\frac{dA^{+}(cr)}{dcr}\frac{cr}{A^{+}(cr)}$, where 'cr' is the dollar amount of the circle rate.

To obtain the circle rate at which revenue is the highest, I differentiate Equation 2 with 'cr',

$$\frac{dR(cr)}{dcr} = \mathrm{N(cr)}[\mathrm{C'(cr)} \times \mathrm{A(cr)}^{+} + \mathrm{C(cr)} \times \mathrm{A'(cr)}^{+}] \tag{3}$$

Holding N(cr) fixed, factoring out $\frac{C(cr)A^{+}(cr)}{cr}$ and substituting in the terms for elasticity, we get:

$$\frac{dR(cr)}{dcr} = [\epsilon_{compliance} + \epsilon_{payments}]\frac{C(cr)A(cr)^{+}}{cr} \tag{4}$$

Using the First Order Condition, revenue is maximized when $\frac{dR(cr)}{dcr} = 0$. Thus the circle rate at

[30] She could care about compliance as a standalone parameter, outside of total revenues collected, in order to address concerns about horizontal equity

[31] In their context, liabilities are fixed and compliers have to pay the full amount. Revenues are thus only a function of compliance and revenues are maximized when the elasticity of compliance equals -1

which revenue is the highest, it must be that:

$$\epsilon_{compliance} + \epsilon_{payments} = 0 \tag{5}$$

Then, revenue is increasing when $\frac{dR(cr)}{dcr} > 0$ or when $\epsilon_{payments} > \left|\epsilon_{compliance}\right|$ [32].

3.1 Obtaining Elasticities

In a standard setting, to measure the elasticity of compliance to liabilities, assuming a level-log model, we can estimate the following equation:

$$\text{Compliance}_{it} = \alpha_0 + \alpha_1 \text{Log(Tax Liability)}_i + \tau_t + \varepsilon_{it} \tag{6}$$

Where the elasticity at the mean for compliance is then given by:

$$\text{Elasticity} = \frac{\hat{\alpha_1}}{\overline{Compliance}} \tag{7}$$

And the elasticity of taxes paid can be estimated using:

$$\text{Log(Tax paid)}_{it} = \lambda_0 + \lambda_1 \text{Log(Tax Liability)}_i + \tau_t + \varepsilon_{it} \quad \text{for Compliance}_{it} = 1 \tag{8}$$

Where the elasticity is given by λ_1 since Eq 8 is a log-log regression.

However, there could be several omitted variables leading to bias in α_1 and λ_1. Properties with high tax liabilities could be located in areas with better public services, causing property-owners to comply. It could also be possible that property-owners with a high moral incentive or ability to pay the tax choose to locate in larger houses, leading to an upward bias in the estimate. In Figure A7, Panel D, I show that this hypothesis is likely true. It could also be possible that property-owners with means to evade the tax could be locating in larger houses, leading to a downward bias.

3.2 Causally identifying elasticities

To deal with this issue of endogeneity, I make use of the property tax valuation formula (Equation 1) and compare two properties that are located very close to each other but fall in different circles with differing circle rates in 2011. Properties that are located very close to each other have the same access

[32] This result is in principle similar to the one shown by Saez (2001), where teh revenue effect is captured by summing the average elasticities derived from the behavioral and mechanical effects.

to public services, and likely look similar to each other.

The identifying assumption is that *within* a pair of neighbouring properties, the assignment to either side of the circle border is random, thus randomly generating a difference in their compliance.

I restrict my baseline results to properties located very close to each other and that have a balanced panel. Formally, for a property 'i' in a circle with a circle rate 'cr' and a closely located property 'i' with a circle rate 'cr' :

$$\text{Tax Liability}_{i,i'} = \text{Circle rate}_{cr,cr'} \times v_{i,i'} \tag{9}$$

where $v_{i,i'}$ represents all the other weights on hedonic property characteristics used to calculate the tax liability as specified in Equation 1.

The identifying assumption here is then that for properties where $v_i = v_{i'}$, all differences in the liability are captured by the difference in circle rates. In my baseline results, I construct pairs of properties that lie within 0.01 miles or about 20 metres of each other in bordering 2011 circle rate-varying circles, and test if this is true.

To do this, I match all *coordinates* in my data using a nearest-neighbour matching algorithm. This results in distances ranging from 0.002 miles to 0.2 miles as it could be possible that the nearest-neighbour for a property is not a neighbouring one. However, I present my baseline results using 0.01 miles and show robustness to using 0.005 miles and 0.05 miles. I then construct pairs of properties between matched coordinates. Appendix Figure A10 illustrates this process in detail. I match coordinates A and B if they meet two conditions: (i) they are nearest neighbours and (ii) they have different circle rates in 2011. Then, if coordinate A has three properties and coordinate B has two properties, I form seperate 6 property-pairs. Since I restrict my baseline results to a balanced panel from 2013–2024, this results in 72 (6 x 12) observations for this sample.

Within each property-pair-year, I am now interested in the change in compliance due to the difference in baseline circle rates. This setup is similar in essence to Dube et al. (2010), where there are discontinuous changes in minimum wages across contiguous county-border-pairs in the United States. In their setting, the only thing changing across a pair was the minimum wage, in mine it is the circle rate in 2011 [33]. Then, using a pair-year fixed effect (δ_{pt}), I can measure compliance *within* a

[33] I also note that the institutional setting here is similar to that of Best et al. (2025), however, I do not use a regression discontinuity for my main specification as that would only allow me to capture a change in compliance due to having a higher circle rate. I show these results as robustness

property-pair-year. Formally, I estimate for all properties:

$$\text{Compliance}_{i(p)t} = \beta_0 + \beta_1 \text{Log(Circle Rate)}_{ip} + \delta_{pt} + \varepsilon_{i(p)t} \tag{10}$$

and

$$\text{Ln(Tax Paid)}_{i(p)t} = \gamma_0 + \gamma_1 \text{Log(Circle Rate)}_{ip} + \delta_{pt} + \nu_{i(p)t}, \quad \text{for Compliance}_{i(p)t} = 1 \tag{11}$$

Where 'p' refers to a property-pair, for property 'i' in year 't' . β_1 captures the marginal effect on compliance in percentage points and γ_1 captures the percentage change in taxes paid, due to a 1% change in the circle rate.

The key testable assumption for β_1 and γ_1 to be identified is now that within a pair, assignment to the higher circle rate must be unrelated to potential compliance outcomes, conditional on the pair-year fixed effect. In other words, properties in a pair must be balanced on observables that are used in calculating liabilities. I test for this assumption when discussing my results and show that my baseline results are robust to it. While this assumption specifies balance on observables, by ensuring that they are close to each other, I also hold fixed unobservables such as the access to and quality of public services that may affect compliance and be correlated to the circle rate. This specification thus estimates a pooled estimate from comparing properties across several different discontinuities.

The average compliance elasticity then, from Eq 10 is $\frac{\beta_1}{\overline{Compliance}}$ and the average elasticity of taxes paid from Eq 11 is γ_1. I can use these elasticities in Equation 5 to then estimate where the average property in this municipality lays along a revenue function.

3.3 At what circle rate is revenue maximised?

I next calculate the level of circle rates at which revenues are maximised under a uniform proportional scaling of the existing assessment schedule. Specifically, I consider a counterfactual in which every property's circle rate is multiplied by the same factor k, so that

$$\text{Circle Rate}_i' = k \times \text{Circle Rate}_{i,2011}.$$

Thus, the relative differences in circle rates across properties are preserved, while all assessments are increased by the same proportion. I refer to the mean of the resulting distribution as the 'revenue-

maximising circle rate'.

Using the condition derived in Eq. 5 and Eq. 10, with the full derivation provided in Section A.2, the revenue-maximising circle rate is:

$$\text{Circle Rate}^* = \exp\left[\overline{\text{Log(Circle Rate)}_{2011}} + \frac{-\frac{\beta_1}{\gamma_1} - \overline{\text{Compliance}}}{\beta_1}\right]. \tag{12}$$

Here, $\overline{\text{Log(Circle Rate)}_{2011}}$ is the mean log 2011 circle rate. Equivalently, the implied proportional scaling factor is

$$k^* = \frac{\text{Circle Rate}^*}{\exp\left(\overline{\text{Log(Circle Rate)}_{2011}}\right)},$$

so that a value of $k^* = 1.5$, for example, corresponds to increasing every property's circle rate by 50%.

3.4 What is the total change in revenue from changing circle rates?

I also derive the total change in revenue from changing circle rates in Section A.3, by differentiating Eq 2 by 'cr'. This results in

$$\Delta R \approx \Delta_{cr} A^+(cr)\left[\beta_1 + \gamma_1 C(cr)\right] \tag{13}$$

where β_1 and γ_1 are the elasticity of compliance and the elasticity of taxes paid by compliers, repectively. Δ_{cr} is the correponding percentage change in circle rates.

3.5 Heterogenous Effects

In the average effects I estimate above, there is an inbuilt assumption that all properties through the circle rate distribution respond similarly. The revenue-maximising circle rate is then built off this assumption.

I can go a step further and ease this assumption by allowing properties to respond differentially, based on where they lie in the rate distribution. For instance, there is evidence in the optimal taxation of incomes literature, that elasticities may be larger at both the lower- and the higher-end of the income distribution Saez (2001).

To do this, I exploit the fact that matched coordinate *groups* lie at different points of the circle-rate distribution and allow the effect of a circle rate increase to vary with the local circle-rate level. I primarily follow a standard heterogeneous-treatment-effect specification in which the treatment

variable is interacted with a pre-determined 'moderator', so that the marginal effect of treatment varies with the moderator. The moderator I use here is the position of a property on the rate-distribution, relative to the sample mean of the circle rate. This results in easily interpretable marginal effects.

To simplify notations, let $\ell \equiv$ Log(Circle Rate)

I start with defining a demeaned local circle rate for each property-pair 'p' within a matched coordinate pair group 'g':

$$Z^c_{g(p)} = \overline{\ell}_{g(p)} - \overline{\ell} \tag{14}$$

where $\overline{\ell}_{g(p)}$ is the average log circle rate among all properties in coordinate group $g(p)$, and $\overline{\ell}$ is the corresponding mean in the full estimation sample. $Z^c_{g(p)}$ estimates how different the circle rate within a property-pair is relative to the average circle rate in the sample [34]. $Z^c_{g(p)}$ thus ranges from negative to positive numbers, where properties with circle rates exactly at the mean have a $Z^c_{g(p)}$ value of 0.

Since the intensive-margin sample of payers is likely a smaller subset of the full sample, I define an analogous term for compliers as:

$$Z^a_{g(p)} = \overline{\ell}^a_{g(p)} - \overline{\ell}^a \tag{15}$$

where $\overline{\ell}^a_{g(p)}$ is the average log circle rate among paying properties in coordinate group $g(p)$, and $\overline{\ell}^a$ is the corresponding mean in the paid-tax estimation sample.

For the extensive margin, I now modify the baseline equation and estimate:

$$\text{Compliance}_{i(p)t} = \alpha_c + \beta_c \ell_{pt} + \theta_c \left(\ell_{ip} \times Z^c_{g(p)} \right) + \delta_{pt} + \varepsilon_{i(p)t}. \tag{16}$$

I continue to use property-pair-year fixed effects (δ_{pt}) and ℓ_{ip} is the log of the circle rate of property 'i' within a property-pair 'p', just as in Eq 10. Here θ_c is the additional coefficient, which estimates the heterogenous impact on compliance, relative to where the property lies in the circle-rate distribution since $Z^c_{g(p)}$ varies for each coordinate group.

For the corresponding intensive margin, I estimate the analogous specification among properties that pay positive taxes:

$$\text{Log(Tax Paid)}_{i(p)t} = \alpha_y + \beta_y \ell_{ip} + \theta_y \left(\ell_{ip} \times Z^a_{g(p)} \right) + \delta_{pt} + \nu_{i(p)t}, \quad \text{for Compliance}_{i(p)t} = 1. \tag{17}$$

[34] Note that within a property-pair, circle rates vary

Then, the marginal effect of an increase in log circle rates on compliance is given by $\beta_c + \theta_c Z^c_{g(p)}$, where effects can be measured at varying points of $Z^c_{g(p)}$, resulting in a distribution of effects. The corresponding marginal effect for taxes paid is given by $\beta_y + \theta_y Z^a_{g(p)}$. For both marginal effects, $Z^c_{g(p)}$ and $Z^a_{g(p)}$ can be changed by any small number to get the distribution, such that the effect of a 1% change in $\ell_p t$, is given at that point [35].

3.6 Heterogenous revenue maximizing circle rate

Just like with the pooled estimates, revenue is maximised when the sum of the two elasticities equals 0. I use this condition, alongwith the equation for compliance to solve for a circle rate where revenue is maximised, which I define similarly as I did in Section 3.3. I derive this formula in Section A.2. The revenue maximizing Log(Circle rate), defined as ℓ^*. ℓ^* should then solve:

$$\frac{\beta_c + \theta_c \left(\ell^* - \overline{\ell}_c\right)}{\bar{C} + \beta_c \left(\ell^* - \overline{\ell}_c\right) + \frac{1}{2}\theta_c \left(\ell^* - \overline{\ell}_c\right)^2} + \beta_y + \theta_y \left(\ell^* - \overline{\ell}_y\right) = 0. \tag{18}$$

Since this results in a cubic function, I solve this equation numerically over values of ℓ, and then convert the solution back into circle-rate units by exponentiating ℓ^*. This also gives rise to a uniform scaling factor, 'k' such that the distribution of all circle rates in the city increases by 'k', where ℓ^* is the geometric mean.

3.7 A distribution of revenue maximizing circle rates

I can also use these local elasticities to derive local revenue-maximising circle rates. Each circle rate can then be assigned its own '*CircleRate**', which would be a function of the elasticities derived at that circle rate. This will then result in a distribution of revenue maximising circle rates. I modifying Equation 12, such that

$$\text{Circle Rate}^*_{g(p)} = \exp\left[\overline{\text{Log(Circle Rate)}}_{g(p)} + \frac{-\frac{\beta_c+\theta_c}{\beta_y+\theta_y} - \overline{\text{Compliance}}_{g(p)}}{\beta_c + \theta_c}\right] \tag{19}$$

[35] In other words, these are the points at which marginal effects are evaluated at and is a 'cell' on the moderator grid. I use 0.01 for all baseline measures.

4 Main results

4.1 Defending the choice of the baseline distance

I start with estimating Equations 10 and 11 and present results using two distance thresholds, 8 metres and 16 metres, in Panels A and B of Table 1, respectively, though I consider 60 metres to be my preferred specification. I define compliance in two ways. One, is my baseline extensive margin measure, where compliance is a binary variable equalling 1 if the assessee pays *any* amount of the tax. The second, is a 'liability-clearing' measure, where compliance is a binary variable equalling 1 if the assessee pays *atleast* the liability clearing amount of the tax.

Before analysing my main results in Table 1, I examine the nearest-neighbour coordinate matched distances in my data as the thresholds I present results for are far narrower than usually seen in the literature. My econometric specification rests on comparing properties that are similar to each other, on account of being located very close to each other, such that they lie at the borders of their respective circle areas. In dense cities of India, properties often start looking very different from each other even if they are just one street away and this would introduce bias in my results. In Figure 2 Panel A, I present a sample of coordinate pairs, overlayed on circle areas where darker colors present higher circle rates. The two green circles and the two black 'Xes' that are numbered '658' and '670', respectively, represent two coordinate pairs. The pair labeled '658' has a distance less than 0.01 miles between them, corresponding to a distance less than 16.09 metres, which is also equivalent to taking about 20 steps between the centroid of two properties.

It is clear from this figure that the two coordinates in green are likely only separated by 1 property, they lie on the same side of the street and have varying circle rates. On the other hand, the property-pair labeled '670' has a distance of 0.011–0.05 miles between properties and is separated by 1 whole block. This pair is not in my main estimating sample. Despite being located in areas with varying rates and having a distance that is objectively not very large, in this context, they (i) are separated by several properties and (ii) do not lie at the border of two circles. I discuss and show that they are likely not similar looking properties in a later sub-section when I discuss the validity of my results. Once I constrain my sample to between-property distances below 16 metres, the number of observations that contribute to my results are a smaller percentage of the total observations in my data. In Panel B, I present the distribution of the share of observations at each distance, 16 metres corresponds to a share of 25%, and 8 metres to around 15%. They also correspond to the 30^{th} and 40^{th} percentile of the

distance measure itself.

4.2 What is the average effect on compliance?

How much does compliance on the extensive margin change with a change in circle rates? Using the baseline result in column (1), a 1% rise in the baseline circle rate leads to a 0.311 percentage point (pp) fall in compliance. The absolute value of the resultant elasticity using Equation 7 and the control mean of 0.701, is 0.44. Thus a 1% rise in the baseline circle rate leads to a 0.44% fall in compliance. This result holds at 0.01 miles as well. From Panel B, a 1% rise in the circle rate leads to a 0.322 pp fall in compliance, amounting to a nearly 0.46% fall in compliance.This point estimate of the fall in compliance is similar for all distances that lay between the two presented here. In Fig 4, I divide the distance between 0.005 miles and 0.01 miles into 10 percentiles and estimate Equation 10 within each percentile and present the estimated coefficient. In Section A I present and describe these estimates using a spatial Regression Discontinuity Design, with distance to the circle border as the running variable and a linear polynomial fit on both sides of the cut-off, similar to the design followed in Best et al. (2025). reassuringly, I find that RD estimates are very similar to the ones I obtain using this within-property-pair fixed effect. I find that a higher circle rate, reduces compliance by 0.21 – 0.48 percentage points at a distance of 0.018 miles and 0.024 miles. These results too attenuate toward 0 as the distance grows. In column (2), I present results on the liability-clearing value of compliance, where property-owners who pay less than their full liability are marked as being non-compliant.

While this extensive margin effect is important, even a partial payment of the liability allows the property to be marked as being compliant. While useful, this maybe overrestimating the fall in full compliance. A 1% rise in the baseline circle rate now leads to a 0.23 pp fall in liability-clearing compliance at both 0.005 miles and 0.01 miles. While these point estimates are lower than those in column (1), which is an unintuitive result, the control mean is much lower here than in the baseline. This amounts to slightly higher elasticities than in the baseline. A 1% rise in the circle rate now leads to a 0.58% and 0.59% fall in compliance, at 0.005 miles and 0.01 miles, respectively. This is a margin of compliance showing that aside from not paying entirely, another response to higher rates is people making incomplete payments.

4.3 What is the average effect on the amount of taxes paid?

The third margin I am interested in is now the elasticity of taxes paid itself, or the intensive margin effect. The dollar amount collected is very important for municipal tax policy and estimating revenue,

as discussed in Section 3. Here I ask, between a pair of *compliant* properties how does the amount paid vary as the circle rate increases by 1%?. Note, that this is different than a pure mechanical increase as proeprties pay penalties.

I present results on the intensive margin in column (3), Panels A and B, using Equation 11 and restricting to compliant properties [36]. These coefficients should be interpreted as elasticities as this is a log-log specification. A 1% increase in the baseline circle rate now leads to a 1.67% increase in the amount of taxes paid at 0.005 miles and 1.29% at 0.01 miles. Comparing these to the control mean of 87.94 and 73.64$, this amounts to an approximate 1.46$ and 1.02 $ increase in taxes paid.

This effect, however, combines the actual mechanical effect with properties selecting into payment. That is, the composition of property-pairs that pay could be affecting this estimate. While I decompose this later, in Table A3, column (1) Panel B, I ease the fixed effect and now allow property-pairs with one payer to remain in the regression. I now find an elasticity of 1.19.

4.4 Comparing the two effects

The reduction in compliance is completely offset by this increase in the amount of taxes paid due to a rate increase. Plugging the elasticity estimates of compliance and taxes paid on the intensive margin at 0.01 miles into Equation 5, with $\epsilon_{compliance}$ = -0.46 and $\epsilon_{payments}$ = 1.29, I get 0.83.

This is far higher than the optimal of 0, suggesting that the average property lies on a rising portion of the revenue function. I confirm that $\epsilon_{payments}$ is also greater than $\left|\epsilon_{compliance}\right|\Big|$, which is the other condition for a rising revenue curve. This remains true even if I use the more restrictive liability-clearing definition of compliance, plugging in $\epsilon_{compliance}$ = 0.59, I get 0.69.

Overall, a 1% increase in the circle rate leads to a 0.83% increase in revenues. In Table A3 I show additional results using the total amount paid, not restricting to compliers. In column 6, using the poisson, though the is significant at 11% (p value = 0.106), I find an elasticity of 0.54.

4.5 Are these estimates causal?

It is crucial for my identification to be causal that the pairs of properties I compare are observationally similar to each other, except for the difference in the circle rate that applies to them. In other words, the *only* reason two properties could have variation in their liabilities should be due to the difference in their circle rates, they should be similar on all other terms that determine their liabilities. I check if

[36] The smaller number of observations in column (3) now reflects both this restriction and the exclusion of property-pair-year cells with only one compliant property, which are singleton observations

this is true by checking if the residual term as defined in Equation 9 is similar across property pairs and present results in column (1), Panels A and B of Table 2. I also check if properties are balanced across the few characterestics I obtain from the Google Footprints database, and present results in columns (2), (3) and (4) of Table 2. I present the corresponding coefficient plot in Figure 5

The estimated residual for both distance thresholds is statistically not different from 0, supporting my estimation strategy. Further, in Figure A8, Panel A, I present this coefficient for distance thresholds through 0.05 miles. While the residual is not significantly different from 0 at the 95% level of signifcance through all distances, post 0.01 miles there is a clear fall in the point estimate, which is close to 0 before 0.01 miles. This confirms my earlier hypothesis that properties are likely not similar to each other as the distance between them grows. The residual turning negative is also indicative that as the circle rate increases, the liability increase for the higher circle rate property happens at a slower rate.

In columns (2), (3) and (4), I check for balance on the number of floors, the type of construction and the area of the property in square metres. 'Number of floors' are integers going from 1 to 12 and construction type is a categorical variable that is equal to 1 if the property is built with cement, 2 if it is partially built with cement, and 3 if it is built using other less-durable materials. I find that properties are balanced across all three characterestics, for both distance thresholds as the estimated coefficients are insignificant. These result confirm the validity of my estimating strategy.

4.6 Is there an impact of enforcement?

Due to effective enforcement, compliance responses to rate increases could be muted, causing the elastcities I find above to be under-estimates. I am able to check if this is true by making use of the policy rule of the KMC that stipulates that once total accrued liabilities (including any penalties) cross a threshold of a 100 USD, enforcement letters be sent to property-owners. These enforcement letters should be viewed as a light touch enforement action. Enforcement officers visit premises and do undertake stricter enforcement action but only when the liability crosses 10000 USD, this often applies only to commercial properties or very large residential properties, both of which do not exist in my sample.

I then estimate Equation 10 by restricting the sample to properties with total liabilities below 100 USD. This is a sample of properties that more or less regularly comply such that their cumulative liabilities never rise over the threshold. Since I am still comparing properties within pairs, both

properties in the pair would have to be below the threshold to remain in the sample, leading to a reduction in the sample size relative to the baseline. I present results in Panels A and B of Table 3 using all the three margins of compliance discussed before. I find that the response to a rate increase for non-enforced properties is not meaningfully different from the baseline. Compliance now drops by 0.32 pp due to a 1% increase in the circle rate, amounting to a 0.44% decrease from the mean. The estimate at 0.01 miles is slightly larger, a 0.373 pp decrease, however this decrease is not statistically different from the 0.32 pp decrease at baseline. On taxes paid, the estimated elasticities are larger than the baseline at 1.91% and 1.65%, though again, they are not statistically different from those at baseline.

4.7 Considering the dynamics

4.7.1 House Price Growth

A mechanism at play that would explain these results would be house price growth. If house prices in higher 2011 circle-rate areas, grew faster than their neighbouring house prices, then this wealth effect could affect both the fall in compliance and the amount paid by compliers. For instance, it is possible that compliers that see higher house price growths pay all their penatlies or always pay the full amount of the liability.

While I don't have direct house price data, circle rates act as good proxies for house prices (Anagol et al., 2024). To test, I add growth in circle rates and the interaction of this growth rate with the baseline circle rate as additional terms to the main specifification. I present these results in column (1) and (2) of Table 4, where Panel A has data for all years, while Panel B Is restricted to years before 2023 [37].

From column (2), Panel A, the coefficient on the baseline circle rate is the same as the one in Table 1, indicating that accounting for this growth rate does not have an impact on the overall magnitude of the fall in compliance. The coeffeicient on the interaction term then indicates that homeowners in areas with higher baseline rates and higher growth rates, relatively have a small 0.01 pp fall in compliance. So a property in a high growth area has an overall 0.352 pp (-0.339 - 0.0133) fall in compliance. This magnitude is similar in panel B, column 2. Relatively high growth properties have an overall 0.35 pp (-0.368 + 0.0181) fall in compliance.

The result on the amount of taxes paid, however, is more confounding. From column (1) Panel A, compliers in high growth rate and higher baseline circle rate areas pay 0.049% *less*. Since this amount

[37] I make this distinction as I show in Figure A15, that the circle rate in 2011 was positively correlated with circle rates from 2015–2022, but turns negative in 2023 and 2024. Since this could be an outlier change, I separate the sample.

is purely mechanical, this must imply that these properties have lower liabilities than properties in lower growth rate and lower baseline circle rate areas. This in turn is only possible if the growth rate of circle areas and the baseline circle rate is inversely correlated. I find suggestive evidence of this in column (3), a 1% increase in the baseline circle rate leads to only a 0.52% increase in the average circle rate from 2015–2024. This result is not as stark in Panel B column (3), which suggests that a 1% increase in the circle rate in 2011, leads to a 0.97% increase in the average circle rate from 2015–2022.

This discussion above helps highlight two main things, (i) it seems unlikely that my main results are caused by the growth rate of house prices and (ii) places that had higher circle rates in 2011 did *not* grow substantially faster than places that had lower rates. In fact, they likely grew slower or almost at the same pace

4.7.2 Construction

As a result of this fact I present regarding the differential rates at which places grew, it may be expected that there are effects on where houses get constructed. Particularly, if more houses were being constructed in lower baseline rate areas this would be suggestive of tax avoidance behavior. I check for this in column (4) of Table 4, where the dependant variable is 'Construction'. 'Construction' refers to the number of new properties that get added to the tax roll every year[38], within the same circle area.

I find that a 1% increase in the circle rate leads to 1.5–3.2 more houses being constructed on average within that circle. This is a large effect based on the control means, and amounts to a 750% – 533 % increase. This suggests that the circle rate structure does affect city construction, but not in the direction avoidance behaviour would suggest.

4.8 The effect of COVID on compliance

In my main specification I estimate the outcome variable within a pair-year. This means that it is possible to re-estimate these effects separately for each year and specifically isolate the effect of the COVID-19 pandemic and the resultant lockdown. To check, I club all years pre- and post-2020, and test 3 separate regressions: pre, 2020 and post. I present results in Figure 6 with Panels A and B restricting results to 0.005 miles and 0.01 miles, respectively. I also present results for a version where I test each year separately in Figure A13.

[38] I note that 'construction' is a bit of a misnomer, as it is possible that properties get constructed and then don't enter teh tax roll. However, this is not as likely because new properties have to be registered with the official housing authority and get automatically added to the tax roll, per teh municipality.

Through both figures, it is clear that COVID had a large negative impact on compliance. The fall in compliance nearly triples, a 1% rise in the circle rate now has a 1 pp fall in compliance. This pattern is also visible in the year-by-year regressions. The lockdown in India lasted intermittently through the 2020 and would have potentially led to tightening of income constraints and a lowering of compliance. This would be a hypothesis in line with Brockmeyer et al. (2021) and Bergeron et al. (2024).

4.9 Robustness

Lastly, I check if the results in Panel A Table1 are robust to modifications in the sample and specifications. My main results are restricted to a balanced panel of owner-occupied properties, with no weights and ward-year fixed effects. In Table 5 I present results by weighting the regression using inverse weights for the number of pairs by proeprty in Panel B, using an unbalanced panel in Panel C and adding fixed effects for property characteristics in Panel D, for both 0.005 miles and 0.01 miles.

In the baseline regression, within a pair could be a property that has only one pair and a property that got matched to multiple other properties. Properties that get matched to multiple properties may then be over-weighted in the sample. I address this concern by adding inverse-weights for the number of pairs by property, such that properties that appear fewer times in the whole sample get smaller weights. From Panel B, I find that at 0.005 miles, the compliance is the compliance response is the as the baseline and at 0.01 miles, the point estimate is nearly the same but is noisier.

In Panel C I include properties that start appearing in my sample later than 2013. I restrict my main results to a balanced panel of properties to make sure that no properties that get constructed as a response to circle rates are in my sample. I do this as this construction response could be partially an avoidance response if more properties get constructed at higher circle-rates than at lower circle-rates and then property owners in these newly constructed houses are non-compliant. I do find that my point estimates are now slightly higher in columns(1) and (2), but not statistically different that the baseline. In Panel D, I add fixed effects for property characteristics such as the number of floor and the type of construction of the property and find that the coefficient is exactly the same as the baseline.

5 Heterogeneous responses across rates

The main results discussed above are a result of pooling all the various circle-rate within-pair discontinuities in my data. The elasticities I derive are then a result of assuming a linear-log functional form and additionally suggest that all properties in the city repond similarly to a rate increase.

Based on Section 3.5, I go one step further and test for heterogeneous responses to a 1% increase in circle rates, at different points of the circle rate distribution. I use the same sample used to estimate column (1), Panel B of Table 1 and start with estimating Equation 16. I present results in Column (1), Table A5. However, the marginal effect of an increase in the baseline circle rate, captured at various points of the distribution of $Ln(DemeanedLocalCircleRate)$, would give us the heterogeneity of interest. I present the plot of marginal elasticities in Panel A, Figure 7. The x-axis here is $Ln(DemeanedLocalCircleRate) + \overline{Ln}(BaselineCircleRate)$ and the estimated elasticities are the marginal effects divided by mean compliance. Next, using the sample as in column (3), Panel B of Table 1, I estimate 17. Results are presented in Column (1), Table A5, but again, the estimates of interest are the marginal effects at the range of points represented by I estimate marginal effects using $Ln(DemeanedLocalCircleRate : PaidSample)$. I plot the resultant elasticities or marginal effects in Panel B, Figure 7.

Results on marginal effects suggest two main things: (i) Compliance becomes more negative at higher baseline circle rates, that is, compliance becomes more elastic. (ii) Taxes paid also gets more elastic as the circle rate increases. I re-estimate these elasticities using a non-linear moderator in Figure A17 and completely non-parametrically in Figure A14.

I also estimate Equation 16 using the share of taxes paid and the amount of taxes paid (including zeros) in Fig A18. While the marginal effects on the amount of taxes paid is noisy, this represents total revenues and becomes positive after a circle rate of 210. The share of taxes paid rises with the circle rate, suggesting that intensive margin effects on compliance are partly driven by compliers paying a higher share of the tax they owe.

6 What is the revenue maximising circle rate?

I now use the elasticities calculated in Section 4.4 and 5, extrapolate them and find-model based points at which revenue is maximised.

6.1 Using pooled elasticities

I start with using Eq 12 to calculate the Revenue Maximising Circle Rate (RMCR). Since this exercise is a result of a small 1% change to *all* circle rates, the whole circle rate schedule of the city shifts up by a uniform proportion or scaling factor('k'), such that the mean of the new distribution is the RMCR [39].

[39] I use Eq 19 to do a version of this exercise that uses local elasticities to measure local RMCRs, such that each baseline circle rate gets a different RMCR and is not just a proportional rise in the schedule. I plot this figure in Appendix Figure

From Table 1, $\beta_1 = 0.32$ $\gamma_1 = 1.29$ and $\overline{\text{Compliance}} = 0.69$. In this sample, $\overline{\text{Log(Circle Rate)}}$ is 5.358. Substituting all these numbers gives me a circle rate of 843 USD. The corresponding scaling factor is 3.9. The RMCR schedule is thus 300% more than the current circle rate schedule.

I plot this schedule in Figure 8, Panel A. The blue line shows the RMCR as a result of using pooled elasticities. The dashed grey line is a 45 degree line representing teh baseline circle rate.

6.2 Using heterogenous elasticities

Using Eq 18, I numerically estimate a similar scaling factor that would lead to a circle-rate schedule that maximises revenues, except this schedule now accounts for heteogenous elasticities at the bottom and top of the circle rate distribution.

I find that the scaling factor 'k' is now much lower at 2.25, or a 125%. This leads to a heterogenous RMCR of 480 USD, which means that the schedule shifts up by a 125% such that the mean of the new schedule is 480$. I plot this distribution in Figure 8, Panel A, using the red line.

6.3 Using heterogenous elasticities by ward

I use the RMCR condition in Eq 18 and the elasticities measured in Section 5 to find scaling factors and RMCRs seperated by ward. Ward 15, ends up with the largest change, amounting to 350%, while ward 48 actually suggests a schedule 2.6% lower than the current schedule. I present these figures in Figure A20. Within each ward, not all circle rates are identified, leading to differences in the range of the predicted schedule.

7 Impact on revenues due to changing area-based assessments

Next, I use Eq 13 and calculate the potential revenue to be gained under 8 different scenarios. Five of these are circle rate updates to alternate years, 2016, 2018, 2020, 2022 and 2024. Then, I consider a yearsly 2% increase, scaling to the pooled RMCR of the schedule scaling up by more than 200% and the heterogenous RMCR where the schedule scales up by around a 150%. For each of these, I use both pooled and heterogenous elasticities to measure the total change in revenues. I plot these changes in Figure 9

In Panel A, using pooled elasticities, I find that scaling to the Revenue Maximising Circle Rate has a large 100% increase in revenues. The yearly 2% increase, which also reprensets a much smaller change to circle rates than suggested by teh RMCR, results in the smallest revenue gain of 11%.

A19. Due to a low compliance response at smaller circle rates, it assigns a very high RMCR value, making this plot appear regressive.

Updating the base year results in changes ranging from 56–63%, but don't linearly rise due to the way circle rates in the city grew. Since circle rates dropped in 2024, the change in 2024 is also the lowest.

These patterns are somewhat consistent in panel B, which uses heterogenous elasticities as well. However, scaling to the heterogenous RMCR now has a much smaller 34% increase in revenues. This increase is quantitatively very similar to the increase suggested by updating the base year. Thus, allowing properties to respond heterogenously seems to have a substantial impact on total revenues collected. In Figure A21 I plot revenues from ward-level RMCRs. Interestingly, ward no. 48 which had a decrease in the RMCR schedule, continues to experience a 1.04% increase in total revenues.

8 Conclusion

In this paper I examine behavioral responses to a higher tax liability, using both the intensive and extensive margin. Doing this in a setting with near-stagnant liabilities allow me to estimate the effects of updating area-based assessments to its revenue-maximized-rate. These revenues are a key source of local government fiscal capacity as they are necessary for providing public services and to reduce dependance on the centre. Increasing revenues is thus a major policy concern and has in recent years, led to several reforms such as digitising assessments. I find that higher rates reduce property-tax compliance, but not enough to offset the increase in payments among those who comply. Average estimates thus suggest that properties in Kolhapur remain on the upward-sloping portion of the revenue curve. Using a model-based extrapolation, I find that the circle rate schedule needs to be scaled up by 200% for revenues to be maximised, assuming no non-linear responses to an increase in liabilities. I then allow responses to vary heterogenously based on where in the circle rate distribution a property lies, and numerically estimate a RMCR schedule that maximises revenues, taking tehse varying elasticities into account. This results in a lower scaling of 150%. I also find ward-level differences in the RMCR. The potential gains from updating the CR schedule are large, and though updating to the RMCR has the largest revenue gains, even updating the base year leads to substantial gains.

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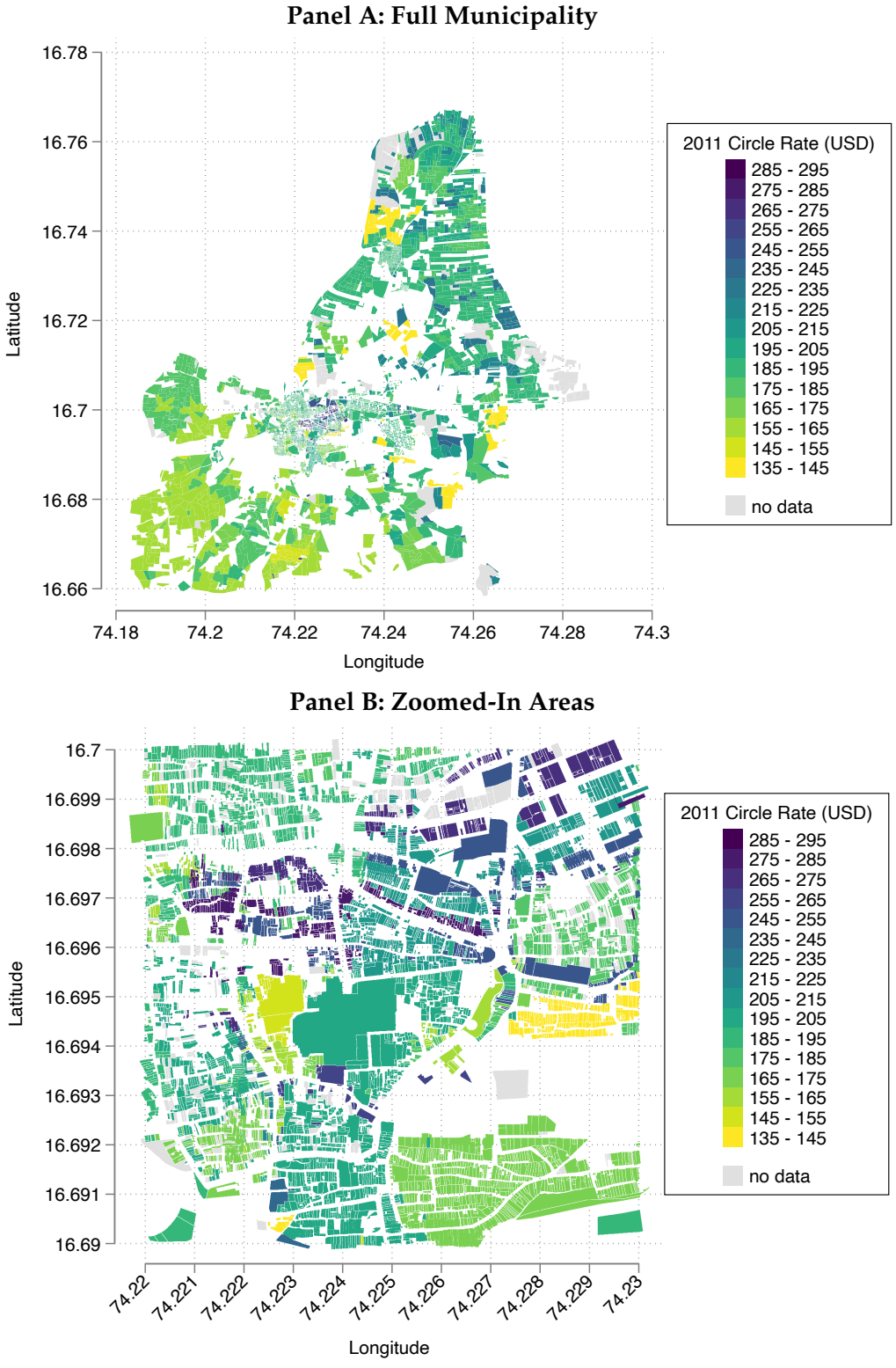
9 Figures

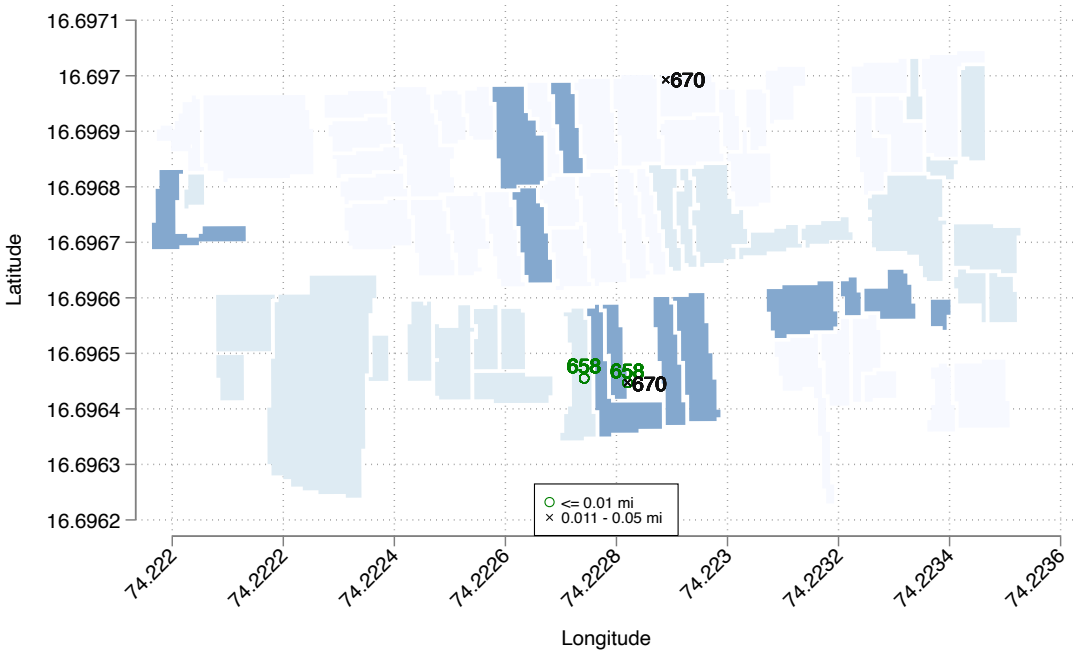
Figure 1
Discontinuities in Baseline Circle Rates



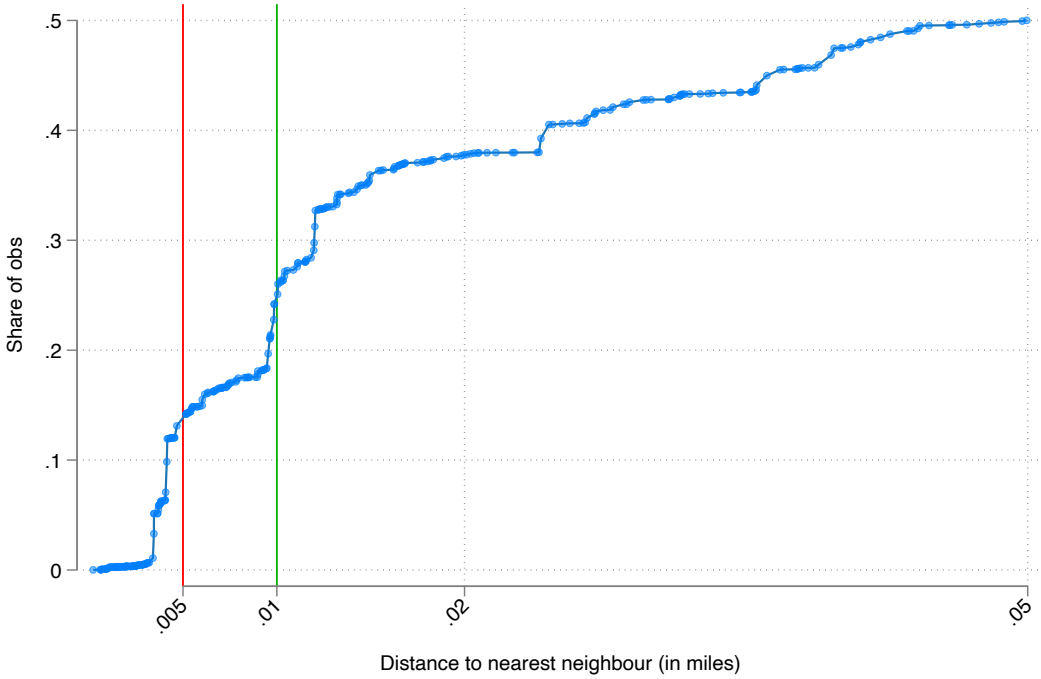
Notes: This figure presents survey areas shaded according to their 2011 circle rates. Panel A shows the full municipality of Kolhapur, while Panel B presents a zoomed-in view. Darker areas have higher circle rates than lighter areas. Circle rates range from USD 135 to USD 295. Areas shown in grey could not be matched to the circle-rate data.

Figure 2
Picking the Threshold Nearest-Neighbour Matched Distance

Panel A. An Example of property pairs at different distances



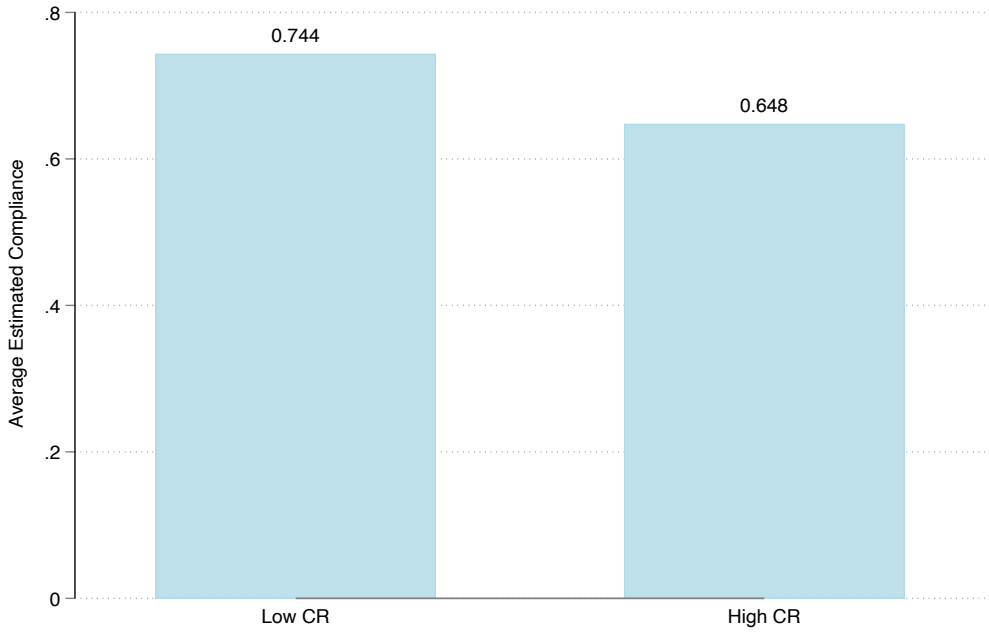
Panel B. Cumulative share of observations at each distance



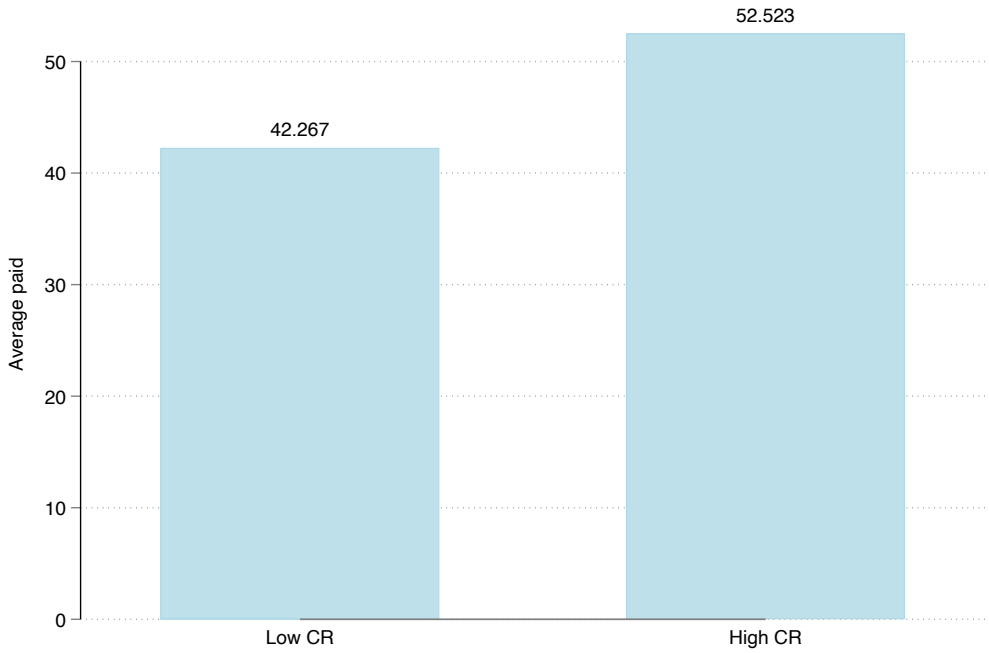
Notes: This figure presents a sample of property pairs at varying distances in Panel A and the cumulative share of observations at each distance in Panel B. In Panel A, the green labels present a pair of coordinates with a distance of 0.01 miles between them and the black labels present a pair of coordinates with a distance of 0.011 to 0.05 miles between them. Coordinates are overlaid on circle areas. with darker shades representing higher circle rates in 2011. In Panel B, each point is the cumulative share of observations at that distance threshold. All distances between coordinates are a result of a nearest neighbour matching algorithm. In Panel B, the red line and the green line represent distances used in the main set of results: 0.005 miles and 0.01 miles.

Figure 3
Average Treatment Effects on Compliance and Amount Paid

Panel A. Compliance



Panel B. Amount Paid



Notes: This figure presents average predicted binary compliance and the average predicted amount of taxes paid in Panels A and B, respectively. Compliance is measured using the following equation:

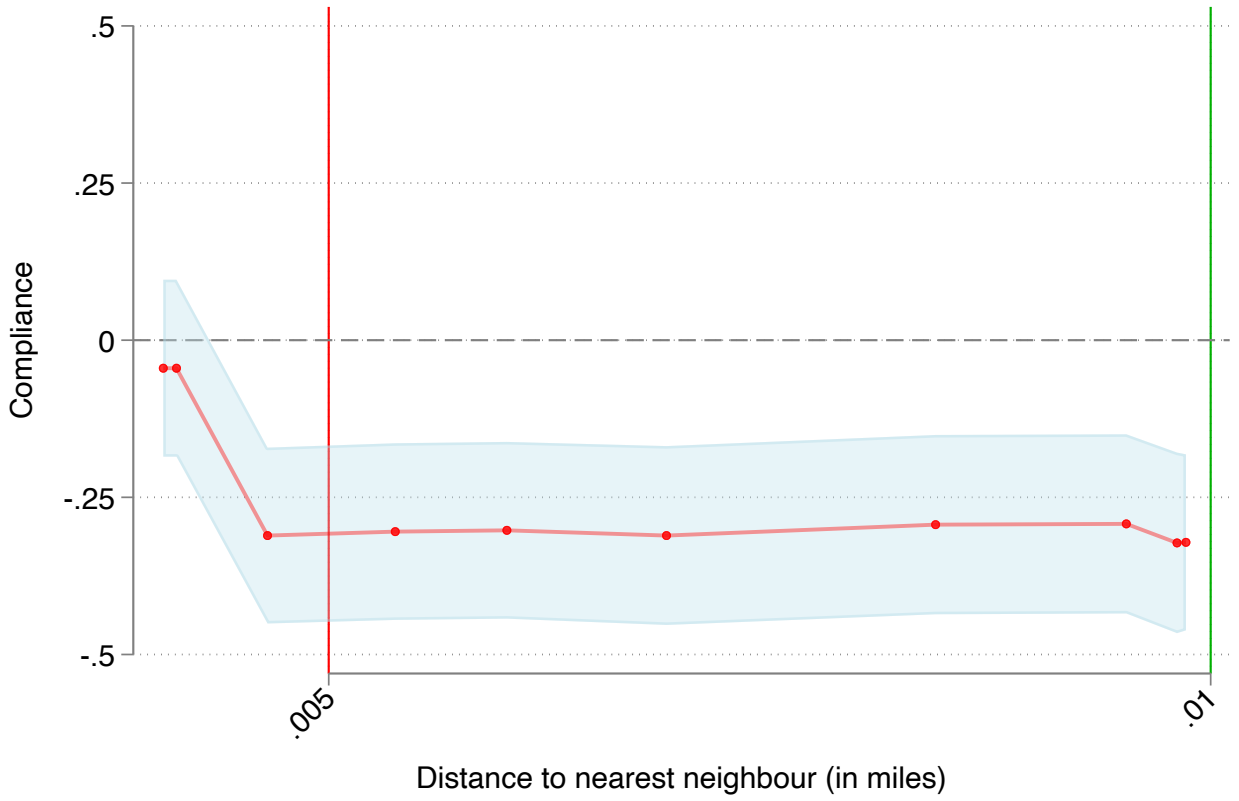
$$\text{Compliance}_{i(p)t} = \beta_0 + \beta_1 \text{Log(Circle Rate)}_{ip} + \delta_{pt} + \varepsilon_{i(p)} \tag{20}$$

and taxes paid for compliers using:

$$\text{Log(Tax Paid)}_{i(p)t} = \beta_0 + \beta_1 \text{Log(Circle Rate)}_{ip} + \delta_{pt} + \varepsilon_{i(p)} \tag{21}$$

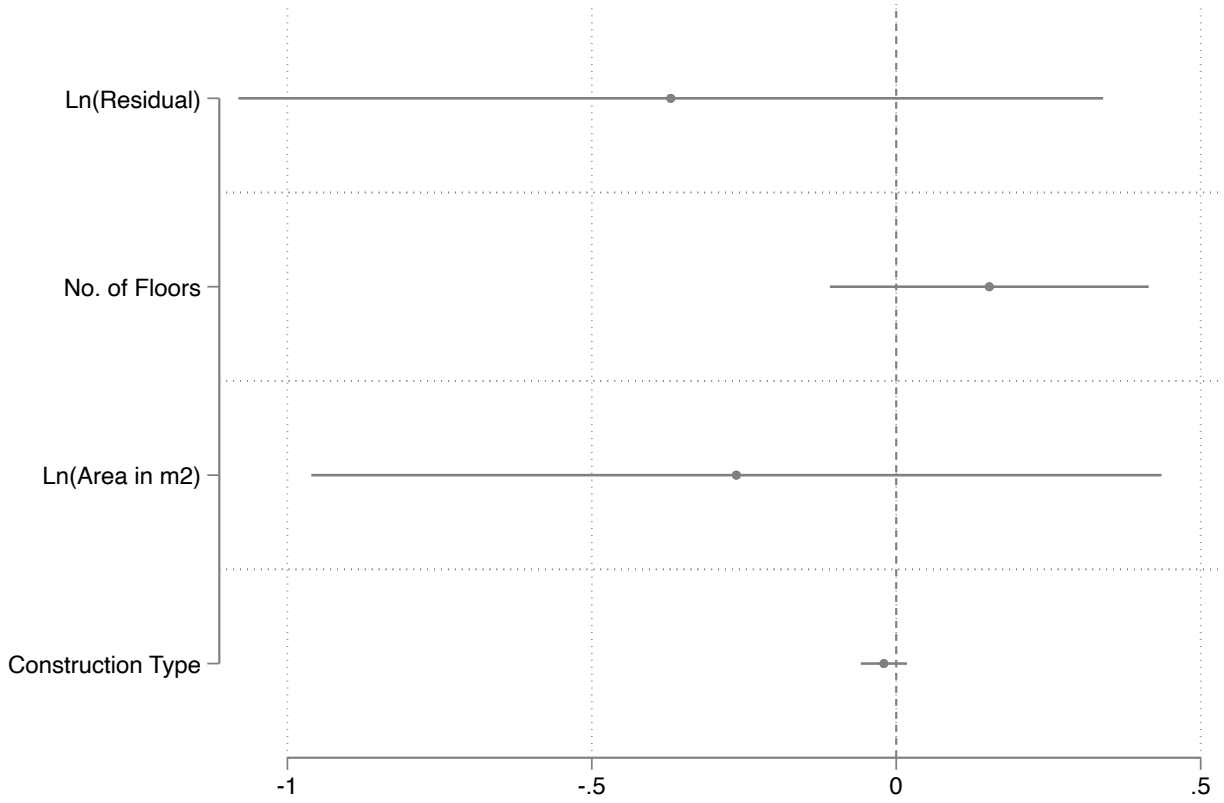
Within each pair 'p', I mark the property with the higher circle rate as being a 'High CR' property and the one with the lower circle rate as being a 'Low CR' property. I estimate $\widehat{Compliance}$ and $\widehat{Log(TaxPaid)}$, exponsentiate Log(Tax Paid) and present means seperately for High and Low CR properties.

Figure 4
Estimated Compliance through 0.01 miles



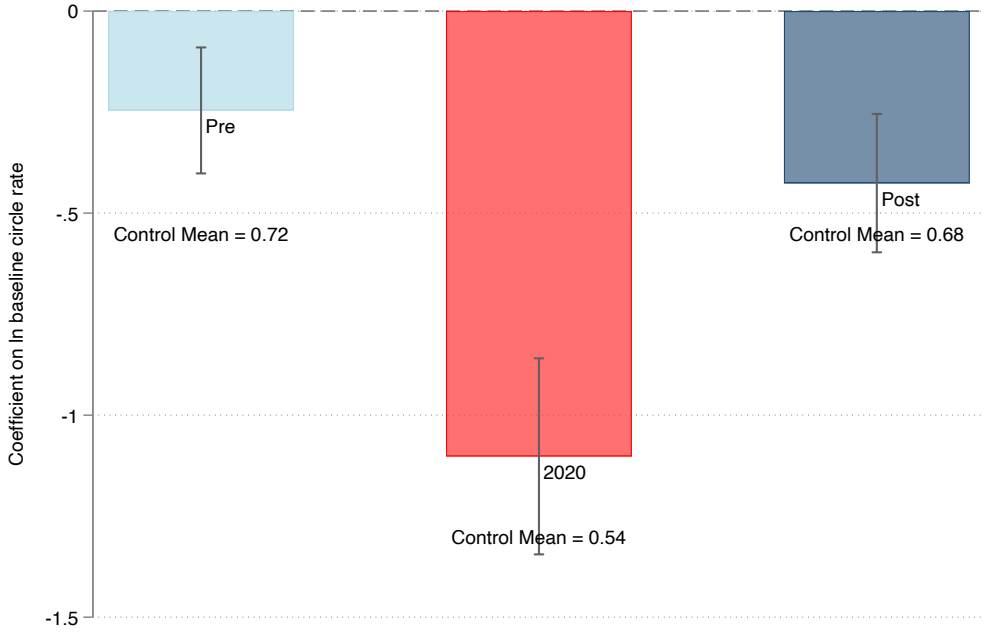
Notes: This figure presents the β_1 from estimating the Equation 20 for varying distance thresholds. $Compliance_{i(p)t}$ refers to the compliance status of a property 'i' in year 't' within property-pair 'p'. Property pairs are formed using a nearest-neighbour distance matching algorithm. Each point in this graph represents the estimated β_1 from changing the distance threshold that defines the property-pairs. Compliance is defined as an indicator for whether a property paid *any* tax in a given year. β_1 should be interpreted as the change in compliance due to a 1% change in the 2011 Circle Rate. All specifications include property-pair-by-year and ward fixed effects. he red and green vertical lines indicate the two distance thresholds used for the main results, 0.005 and 0.01 miles, respectively. The shaded light-blue region denotes 95% confidence intervals.

Figure 5
Balance on Observables

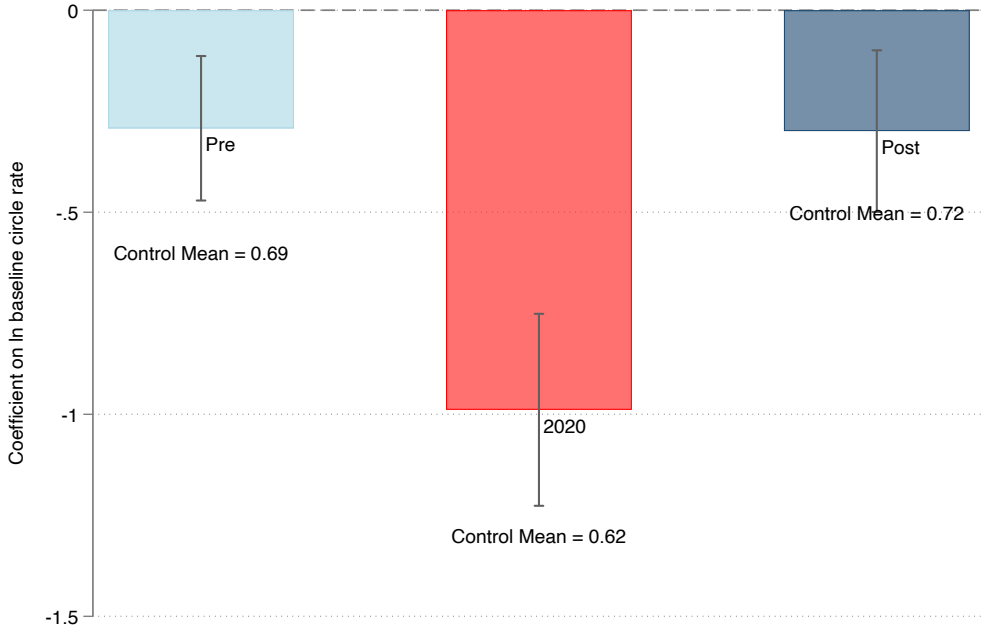


Notes: This figure reports balance tests for observable property characteristics within matched property pairs. Each point is the coefficient on log baseline 2011 circle rate from the main property-pair-by-year and ward fixed-effects specification, estimated separately with the listed characteristic as the dependent variable. Ln(Residual) is the residual component of the valuation formula after accounting for the circle rate, as defined in Equation 9. Number of floors, construction type, and built-up area are from Google Footprints where available. Whiskers show 95 percent confidence intervals.

Figure 6
A Large Fall in Compliance during COVID-19



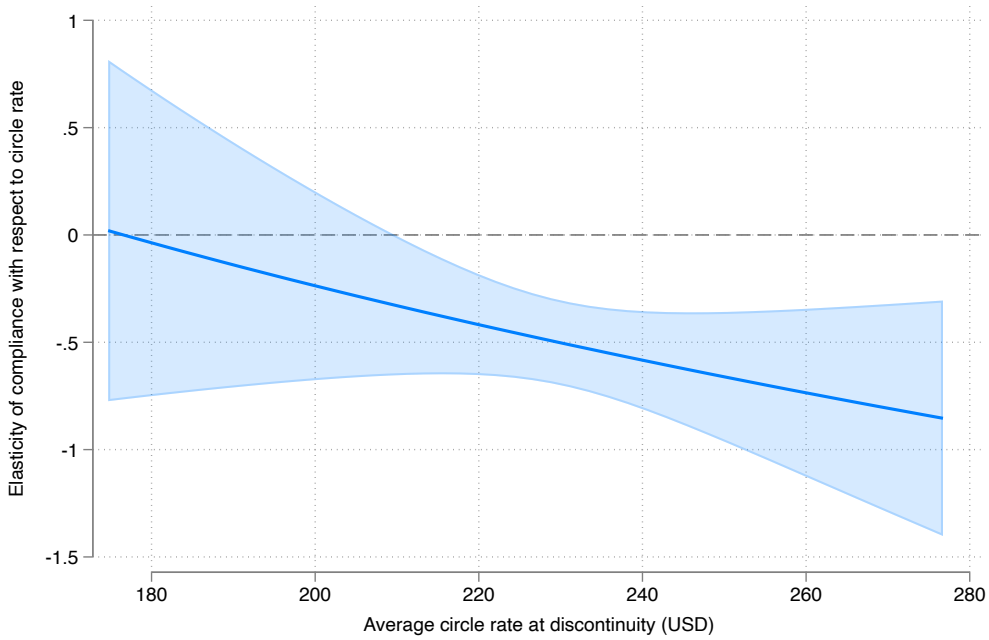
Panel A. 0.005 miles



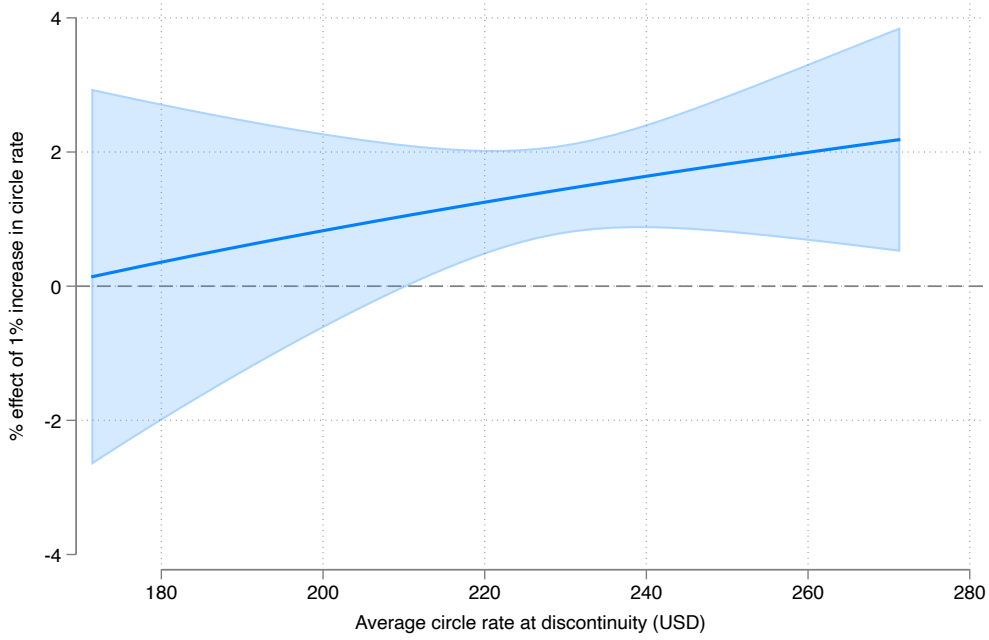
Panel B. 0.01 miles

Notes: This figure presents the coefficient from estimating Equation 20 seperately for years pre-2020, post-2020 and 2020. For 2020, the specification uses only property-pair and ward fixed effects. Panel A presents estimates using a threshold distance of 0.005 miles and Panel B presents estimates using a distance of 0.01 miles. Confidence intervals are presented at the 95% level. Fig A13 presents a version of this figure where Equation 20 is estimated seperately for each year.

Figure 7
Elasticity of Compliance and Taxes Paid: Heterogeneous Responses across the CR distribution



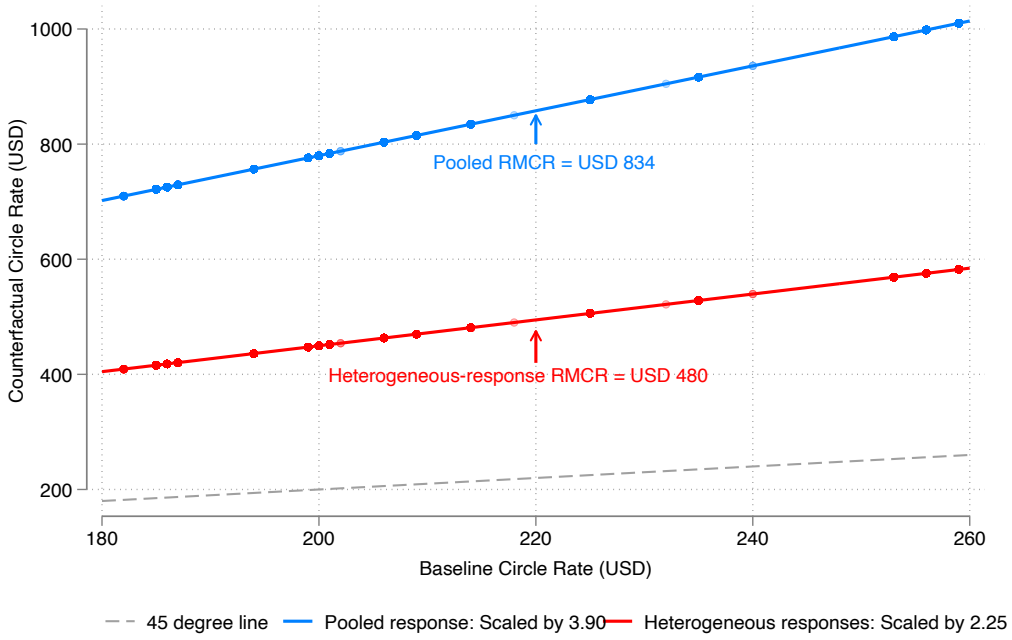
Panel A. Elasticity of Compliance



Panel B. Elasticity of Taxes Paid

Notes: This figure presents the results from estimating Equation 16 in Panel A and converting the estimated semi-elasticities into elasticities and 17 in Panel B. In Panel A, compliance is defined as a binary variable equal to 1 if a property makes any payment in a given year. In Panel B taxes paid is the dollar amount of taxes paid for compliant properties. The dark blue line presents estimated elasticities at each point. The shaded blue area represents 95% confidence intervals.

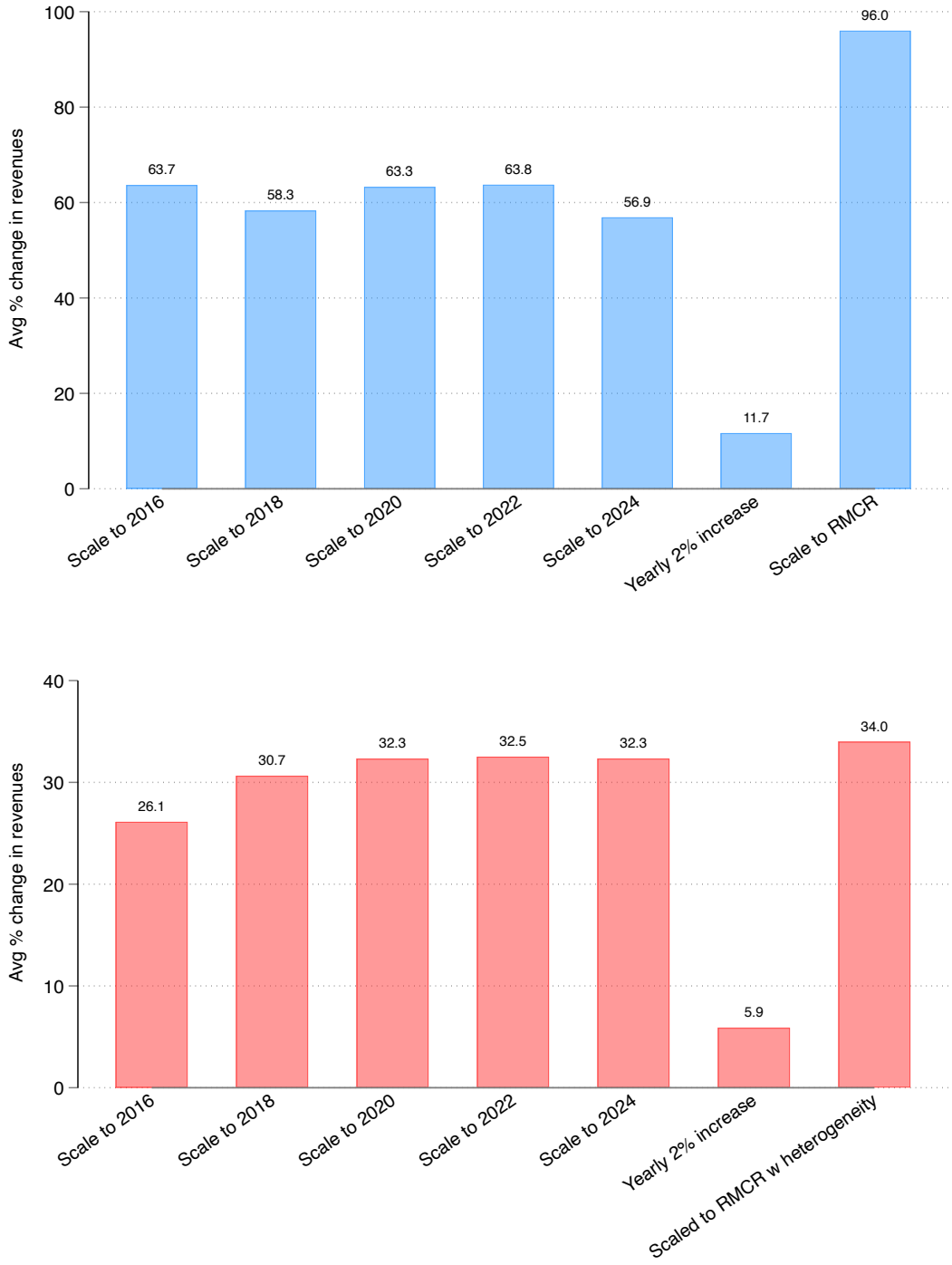
Figure 8
How should the circle rate schedule be scaled such that revenue is maximised?



Panel A. City-wide schedule change using average and heterogenous responses

Notes: Notes: Notes: This figure presents model-implied revenue-maximizing circle-rate schedules. It compares a uniform citywide scaling based on pooled elasticities (blue line) with a uniform scaling that allows compliance and payment elasticities to vary over the baseline circle-rate distribution (red line). The dashed grey line is the 45-degree line, where the counterfactual circle rate equals the 2011 baseline circle rate. The pooled schedule scales baseline circle rates by 3.9, implying an average RMCR of about USD 834; the heterogeneous-response schedule scales baseline circle rates by 2.25, implying an average RMCR of about USD 480.

Figure 9
Percentage Change in Revenue



Notes: This figure reports predicted percentage changes in revenue under alternative assessment-update scenarios. Panel A uses pooled compliance and payment elasticities from the main specification. Panel B uses heterogeneous elasticities estimated over the 2011 circle-rate distribution. The base-year update scenarios replace 2011 circle rates with observed rates in 2016, 2018, 2020, 2022, or 2024. The remaining scenarios apply a mechanical 2 percent annual update, scale the schedule to the pooled RMCR in Panel A, or scale the schedule to the heterogeneous-response RMCR in Panel B. Predicted revenue changes are calculated using Equation 13 and combine the extensive-margin compliance response with the intensive-margin payment response. Positive values indicate higher predicted revenue relative to the 2011 baseline schedule.

10 Tables

Table 1
How does Compliance change due to an Increase in Circle Rates?

Panel A: Distance of 0.005 miles between properties

	(1)	(2)	(3)
	Compliance		Ln(Tax Paid)
	Extensive Margin	Liability Clearing	Intensive Margin
Ln(Baseline Circle Rate)	-0.311***	-0.231**	1.677***
	(0.0714)	(0.0875)	(0.377)
Obs	10390	10390	5068
N clusters	63	63	60
Control Mean	0.701	0.395	87.72
Elasticity	-0.443	-0.585	
Pair x Year, Ward FE	Y	Y	Y

Panel B: Distance of 0.01 miles between properties

	(1)	(2)	(3)
	Compliance		Ln(Tax Paid)
	Extensive Margin	Liability Clearing	Intensive Margin
Ln(Baseline Circle Rate)	-0.322***	-0.232***	1.291***
	(0.0715)	(0.0761)	(0.400)
Obs	18938	18938	9068
N clusters	145	145	131
Control Mean	0.696	0.387	79.08
Elasticity	-0.462	-0.599	
Pair x Year, Ward FE	Y	Y	Y

Notes: This table reports estimates of the effect of increasing the baseline 2011 circle rates on compliance outcomes. In columns 1 and 2, the dependent variable is an indicator for compliance. Column 1 defines *Compliance* as an indicator equal to one if a property makes any tax payment in a given year. Column 2 defines *Compliance* as an indicator equal to one if a property pays at least its recorded liability in a given year. Column 3 reports the effect on *Ln(Tax Paid)* or the logarithm of tax paid, measured in dollars. Recorded tax payments may include penalties for late payment. Since *Ln(Tax Paid)* is undefined for properties with zero payment, column 3 is restricted to properties that make a positive payment. The smaller number of observations in column 3 reflects both this restriction and the exclusion of property-pair-year cells with only one compliant property, which are singleton observations. All estimates are based on Equation 20, with Panel A and B restricting to properties that have a nearest neighbour within 0.005 miles and 0.01 miles, repectively, corresponding to 8.04 and 16.09 meters. The sample is restricted to property pairs with different baseline circle rates and to owner-occupied, non-commercial, and non-institutional properties. All specifications include property-pair-by-year and ward fixed effects. Standard errors are clustered two-way at the circle-rate and coordinate-circle-rate levels. Significance levels: * p<0.10, ** p<0.05, *** p<0.01.

Table 2
Testing the Main Assumption: Balance between Pairs of Properties

Panel A: Distance of 0.005 miles between properties

	(1) Ln(Residual)	(2) Number of Floors	(3) Construction Type	(4) Ln(Area in m2)
Ln(Baseline Circle Rate)	-0.0201	-0.0125	-0.000139	-1.17e-29
	(0.371)	(0.0164)	(0.000172)	(.)
Obs	10390	10232	10232	1080
Control Mean	0.762	1.273	1.005	3.904

Panel B: Distance of 0.01 miles between properties

	(1) Ln(Residual)	(2) Number of Floors	(3) Ln(Area in m2)	(4)
Ln(Baseline Circle Rate)	-0.370	0.153	-0.0203	-0.263
	(0.359)	(0.132)	(0.0192)	(0.349)
Obs	18938	18704	18704	4746
Control Mean	0.934	1.730	1.015	3.371

Notes: This table presents balance on observables across properties, within a property-pair. The dependent variable is a residual term, the number of floors and the type of property construction in column 1, 2 and 3, respectively. The residual term is the residual of the tax liability once the circle rate is accounted for within the property valuation formula, as defined in Equation 9. The number of floors and property-construction type are variables collected from the Google Footprints database, matched to the sample coordinate database. Number of floors are integers going from 1 to 12 and construction type is a categorical variable that is equal to 1 if the property is built with cement, 2 if it is partially built with cement, and 3 if it is built using other less-durable materials. All estimates are based on Equation 20, with Panel A and B restricting to properties that have a nearest neighbour within 0.005 miles and 0.01 miles, repectively, corresponding to 8.04 and 16.09 meters. The sample is restricted to property pairs with different baseline circle rates and to owner-occupied, non-commercial, and non-institutional properties. All specifications include property-pair-by-year and ward fixed effects. Standard errors are clustered two-way at the circle-rate and coordinate-circle-rate levels.

Table 3
Are these Elasticities driven by Enforcement Action?

Panel A: Distance of 0.005 miles between properties

	(1)	(2)	(3)
	Compliance		Ln(Tax Paid)
	Extensive Margin	Liability Clearing	Intensive Margin
Ln(Baseline Circle Rate)	-0.326***	-0.298***	1.917***
	(0.0676)	(0.0893)	(0.366)
Obs	8864	8864	4682
N clusters	58	58	58
Control Mean	0.73	0.423	83.89
Elasticity	-0.447	-0.704	
Pair x Year, Ward FE	Y	Y	Y

Panel B: Distance of 0.01 miles between properties

	(1)	(2)	(3)
	Compliance		Ln(Tax Paid)
	Extensive Margin	Liability Clearing	Intensive Margin
Ln(Baseline Circle Rate)	-0.373***	-0.297***	1.655***
	(0.0670)	(0.0834)	(0.356)
Obs	14928	14928	8118
N clusters	135	135	123
Control Mean	0.74	0.424	66.35
Elasticity	-0.504	-0.700	
Pair x Year, Ward FE	Y	Y	Y

Notes: This table repeats the main specification after restricting the sample to properties below the KMC reminder-letter threshold. Properties with cumulative unpaid liabilities below INR 10,000, approximately USD 117 using an exchange rate of INR 85 per USD, do not receive reminder letters. Columns 1 and 2 use compliance outcomes: any payment and liability-clearing payment, respectively. Column 3 uses log tax paid among properties with positive payments. Panels A and B restrict the matched-pair distance to 0.005 miles and 0.01 miles, respectively. The sample is restricted to property pairs with different baseline circle rates and to owner-occupied, non-commercial, and non-institutional properties. All specifications include property-pair-by-year and ward fixed effects. Standard errors are clustered two-way at the circle-rate and coordinate-circle-rate levels. Significance levels: * p<0.10, ** p<0.05, *** p<0.01.

Table 4
Circle Rate Growth and Construction

Panel A: Sample: 0.01 mi

	(1) Ln(Tax Paid)	(2) Compliance	(3) Ln(Circle Rate)	(4) Construction
Ln(Baseline Circle Rate)	0.970**	-0.339***	0.525***	3.210***
	(0.477)	(0.0883)	(0.0531)	(0.416)
Circle Rate Growth	0.267***	0.0740***		
	(0.0963)	(0.0221)		
Ln(Baseline Circle Rate) × Circle Rate Growth	-0.0492***	-0.0133***		
	(0.0167)	(0.00391)		
Obs	5166	10196	12596	13916
Control Mean	3.352	0.715	6.010	0.609
Pair x Year, Ward FE	Y	Y	Y	Y

Panel B: Sample: 0.01 mi and Year less than 2023

	(1) Ln(Tax Paid)	(2) Compliance	(3) Ln(Circle Rate)	(4) Construction
Ln(Baseline Circle Rate)	1.243**	-0.368***	0.977***	1.502***
	(0.510)	(0.0958)	(0.0702)	(0.207)
Circle Rate Growth	0.00867	-0.0893**		
	(0.270)	(0.0429)		
Ln(Baseline Circle Rate) × Circle Rate Growth	-0.00344	0.0181**		
	(0.0519)	(0.00798)		
Obs	3224	7154	9554	15872
Control Mean	3.205	0.677	6.074	0.288
Pair x Year, Ward FE	Y	Y	Y	Y

Notes: This table examines whether the main effects are explained by subsequent growth in circle rates or by new construction. Columns 1 and 2 use log tax paid and compliance as the main dependant variables and include circle-rate growth and its interaction with the baseline 2011 circle rate. Column 3 tests the relationship between baseline 2011 circle rates and contemporaneous circle rates. Column 4 uses new construction, measured as the number of new properties added to the tax roll in a circle area as the outcome. Panel A uses the 0.01-mile matched sample with all available years. Panel B restricts to years before 2023. Circle rate data starts from 2015, so all tables are restricted to 2016–2024. All specifications include property-pair-by-year and ward fixed effects. Standard errors are clustered two-way at the circle-rate and coordinate-circle-rate levels. Significance levels: * p<0.10, ** p<0.05, *** p<0.01.

Table 5
Robustness

	(1) 0.005 mi	(2) 0.01 mi
Panel A: Baseline		
Ln(Baseline Circle Rate)	-0.311***	-0.322***
	(0.071)	(0.071)
Observations	10390	18938
Control Mean	0.701	0.696
Elasticity	-0.443	-0.462
Panel B: With weights		
Ln(Baseline Circle Rate)	-0.301**	-0.347
	(0.117)	(0.214)
Observations	10390	18938
Control Mean	0.701	0.696
Elasticity	-0.429	-0.499
Panel C: Unbalanced panel		
Ln(Baseline Circle Rate)	-0.339***	-0.324***
	(0.070)	(0.065)
Observations	13366	25182
Control Mean	0.681	0.688
Elasticity	-0.498	-0.471
Panel D: Property fixed effects		
Ln(Baseline Circle Rate)	-0.308***	-0.295***
	(0.072)	(0.074)
Observations	10234	18706
Control Mean	0.706	0.698
Elasticity	-0.436	-0.423

Notes: This table reports robustness checks for the extensive-margin compliance outcome. Columns 1 and 2 use matched-pair distance thresholds of 0.005 miles and 0.01 miles. Panel A reports the baseline specification. Panel B weights observations inversely by the number of properties at the coordinate, giving greater weight to coordinates with fewer properties. Panel C uses an unbalanced panel. Panel D adds property fixed effects. Compliance is an indicator for whether a property makes any tax payment in a given year. Standard errors are clustered two-way at the circle-rate and coordinate-circle-rate levels. Significance levels: * p<0.10, ** p<0.05, *** p<0.01.

A Appendix

A.1 Descriptive Plots for the municipality

Since the relationship between circle rates and liabilities are key to my analysis, I plot this relationship in Figure A7, Panel C. Properties at higher circle rates have higher liabilities. I also show in Panels A and B that properties located in areas with higher circle rates are also more compliant and pay more tax. In Panel D, I show that compliance also increases as liabilities increase. These plots are purely descriptive as they are binned scatterplots, with each dot representing the mean of the respective bin.

A.2 Deriving the circle rate at which revenue is maximized

A.2.1 Using Average Effects

At the optimal circle rate, substituting the two elasticities derived from Equations 10 and 11 into Equation 5, we get:

$$\frac{\beta_1}{C^*} + \gamma_1 = 0,$$

where C^* denotes compliance at the revenue-maximizing circle rate.

Using the estimating equation for compliance, this can be written as:

$$-\frac{\beta_1}{\gamma_1} = \beta_0 + \beta_1 \log(CircleRate^*) + \bar{\delta},$$

where $\bar{\delta}$ denotes the average pair-year fixed effect.

Since

$$\bar{C} = \beta_0 + \beta_1 \overline{\log(CircleRate)} + \bar{\delta},$$

we can recover the effective intercept as

$$\beta_0 + \bar{\delta} = \bar{C} - \beta_1 \overline{\log(CircleRate)}.$$

Then,

$$-\frac{\beta_1}{\gamma_1} = \bar{C} - \beta_1 \overline{\log(CircleRate)} + \beta_1 \log(CircleRate^*).$$

Rearranging this, we get

$$\log(CircleRate^*) = \overline{\log(CircleRate)} + \frac{-\frac{\beta_1}{\gamma_1} - \bar{C}}{\beta_1}.$$

The implied revenue-maximizing circle rate is therefore

$$CircleRate^* = \exp\left[\overline{\log(CircleRate)} + \frac{-\frac{\beta_1}{\gamma_1} - \bar{C}}{\beta_1}\right].$$

A.3 Deriving the total change in revenue due to a change in rates

To calculate the change in revenue from a 1 percent increase in circle rates, start from

$$R(cr) = C(cr) \times A^+(cr), \tag{22}$$

where $C(cr)$ is compliance and $A^+(cr)$ is the amount paid by compliers.

Differentiating revenue with respect to the circle rate gives

$$\frac{dR(cr)}{dcr} = C'(cr)A^+(cr) + C(cr)A^{+\prime}(cr). \tag{23}$$

Since the compliance equation is estimated with $\log(cr)$, the compliance semi-elasticity is

$$\frac{dC(cr)}{d\log(cr)} = \beta_1. \tag{24}$$

Using the fact that

$$\frac{d\log(cr)}{dcr} = \frac{1}{cr},$$

we get

$$C'(cr) = \frac{dC(cr)}{dcr} = \frac{\beta_1}{cr}. \tag{25}$$

Similarly, since the payment equation among compliers is log-log,

$$\frac{d\log A^+(cr)}{d\log(cr)} = \gamma_1. \tag{26}$$

Therefore,

$$A^{+\prime}(cr) = \frac{dA^+(cr)}{dcr} = \frac{\gamma_1 A^+(cr)}{cr}. \tag{27}$$

Substituting these terms into the revenue derivative gives

$$\frac{dR(cr)}{dcr} = \frac{\beta_1}{cr}A^+(cr) + C(cr)\frac{\gamma_1 A^+(cr)}{cr}. \tag{28}$$

Factoring out $\frac{A^+(cr)}{cr}$, this becomes

$$\frac{dR(cr)}{dcr} = \frac{A^+(cr)}{cr}\left[\beta_1 + \gamma_1 C(cr)\right]. \tag{29}$$

For a 1 percent increase in circle rates,

$$\Delta cr = 0.01cr. \tag{30}$$

Using a first-order approximation,

$$\Delta R \approx \frac{dR(cr)}{dcr} \times 0.01cr. \tag{31}$$

Substituting the expression for $\frac{dR(cr)}{dcr}$,

$$\Delta R \approx 0.01A^+(cr)\left[\beta_1 + \gamma_1 C(cr)\right]. \tag{32}$$

Evaluated at the sample mean, this becomes

$$\Delta R \approx 0.01\bar{A}^+\left[\beta_1 + \gamma_1 \bar{C}\right], \tag{33}$$

where $\bar{A}^+$ is mean tax paid among compliers and $\bar{C}$ is mean compliance.

A.3.1 Using Heterogenous Effects

I next allow the behavioral response to changes in circle rates to vary with the baseline circle-rate environment. Let M_i denote the centered baseline circle-rate for property i, which is held fixed throughout the counterfactual exercise. The estimated marginal effect of log circle rates on compliance for a property of type M_i is

$$\beta_c(M_i) = \beta_c + \theta_c M_i, \tag{34}$$

while the corresponding elasticity of payments among taxpayers is

$$\beta_y(M_i) = \beta_y + \theta_y M_i. \tag{35}$$

Thus, the estimated behavioral response differs across properties according to their baseline circle-rate environment, but a property's moderator M_i does not change when its circle rate changes.

To calculate a citywide revenue-maximizing circle-rate adjustment, I consider a common proportional scaling of the existing circle-rate schedule. Let k denote the common scaling factor, such that

$$cr_i(k) = k, cr_{i0}, \tag{36}$$

where cr_{i0} is property i's baseline circle rate. Since

$$\log cr_i(k) - \log cr_{i0} = \log k, \tag{37}$$

all properties experience the same proportional increase in circle rates.

Given the estimated heterogeneous compliance response, expected compliance under scaling factor k is

$$C_i(k) = C_{i0} + \beta_c(M_i) \log k, \tag{38}$$

where C_{i0} denotes expected compliance at the baseline circle-rate schedule. This calculation assumes that the type-specific compliance semi-elasticity $\beta_c(M_i)$ remains constant as circle rates are scaled.

For payments conditional on paying,

$$\log A_i^+(k) = \log A_{i0}^+ + \beta_y(M_i) \log k, \tag{39}$$

or equivalently,

$$A_i^+(k) = A_{i0}^+ k^{\beta_y(M_i)}. \tag{40}$$

Expected revenue from property i is therefore

$$R_i(k) = C_i(k) A_i^+(k), \tag{41}$$

and aggregate expected revenue is

$$R(k) = \sum_i \left[C_{i0} + \beta_c(M_i) \log k\right] A_{i0}^{+} k^{\beta_y(M_i)}. \tag{42}$$

The revenue-maximizing common scaling factor is then

$$k^* = \arg\max_k R(k). \tag{43}$$

Equivalently, the first-order condition is

$$\sum_i A_i^{+}(k^*) \left[\beta_c(M_i) + C_i(k^*)\beta_y(M_i)\right] = 0. \tag{44}$$

I solve equation (43) numerically. The resulting k^* gives the revenue-maximizing uniform proportional adjustment to the existing circle-rate schedule while allowing the behavioral responses to differ according to baseline circle-rate conditions. The corresponding citywide revenue-maximizing circle rate can be summarized by scaling the baseline geometric mean circle rate:

$$CR_{\text{het}}^* = k^* \exp\left(\overline{\log cr_0}\right). \tag{45}$$

A.4 Was the circle rate in 2011 driven by political economy effects?

There could be a concern that the circle rate in 2011 is driven by

in Table A2, I present results from a mutivariate regression with electoral-ward level characteristics as independent variables and the circle-level circle rate in 2011 as the dependant variable. There are 77 electoral wards in the municipality, so each ward could contain multiple circle-rates, however, due to an absence of covariates at the parcel-level, I use aggregate electoral-ward estimates. These estimates should then be interpreted as ward-level averages. No conerning pattern emerges from this estimation.

One concern could be that circles with a politically powerful elected representative helps set higher rates for the circle. I use ward-level elections from 2009, to construct a 'party alignment' variable, which is a dummy equal to 1 if the elected representative in 2009 belonged to the same party as the elected part at the state-level (the Indian National Congress). Party Alignment does not seem to drive the circle rate as the coefficient is indistinguishable from 0.

A.5 Regression Discontinuity

An alternate specification that could be used to measure the effect on compliance due to a higher circle rate could be given by:

$$\mathrm{Y}_i = \alpha + \tau\,\mathrm{High}_i + \beta_1 D_i + \beta_2\left(\mathrm{High}_i \times D_i\right) + \varepsilon_i, \tag{46}$$

where D_i is the running variable, which is the distance of property 'i' to a *shared border.* $High_i$ is a binary variable that equals 1 when property 'i' lies in the higher circle rate side of the border. τ is then the coefficient of interest, and can be interpreted as the effect on the dependant variable, due to being in a higher circle-rate circle. I estimate this specification by clustering standard errors at the circle-level and using triangular kernels.

In order to estimate this, I have to now structure my data differently. Instead of matching each property with a

neighbouring property, I now form coordinate pairs. I restrict pairs to those that lie in neighbouring circles with varying circle rates. Within a coordinate pair, I mark all properties that are in the higher circle-rate circle as being $High_i$, and consequently use the distance of the coordinate to the shared border as the running variable. There are thus a few notable differences in this data structure relative to my baseline. One, the distance measure is now not a distance to the paired property but rather a distance to the shared border and two, since each property-year only appears in my data once, this reduces the power of my estimating sample.

In Figure A11, I present RD estimates on compliance, plotted using linear polynomials on both sides of the running variable. I restrict the distance bandwidth to 0.018 miles, 0.024 miles, 0.031 miles and 0.049 miles in Panels (a)–(d), and a similar pattern as the baseline emerges. Compliance reduces in the high circle-rate circles, and the effect disappears as the sample bandwidth increases to 0.05 miles. Since I use triangular weights, properties that lie closer to the border have higher weights in the estimation than properties that lie farther. From the plots, it seems likely the effect is driven by these very closely located properties. In Figure A12, I present both the conventional and bias-corrected point estimates at different bandwidths ranging from 0.010 miles to 0.050 miles.

In Table A4, I present the resultant point estimates for compliance in Panel A, taxes paid in Panel B and the residual term in Panel C. The large drop in the number of observations and the resultant lack of power lead to most estimates being not statistically different from 0 at the 95% level of confidence. However, the signs and point estimates of all coefficients are in line with my baseline estimates. Properties at the higher circle-rate side of the discontinuity have 0.18 pp and 0.48 pp lower compliance using 0.012 miles and 0.018 miles as bandwidths. They simultaneously pay 1.30% and 2.311% more tax.

B Appendix Tables and Figures

B.1 Tables

Table A1
Multipliers by Property Characteristics

Item	Weight
Category	
Bungalow; penthouse; row house; club house / other facilities used by members of a co-operative housing society	1.50
Two-storeyed apartments	1.25
Room or apartment and similar; swimming tank	1.00
Attached garage	0.50
Society office	0.30
Car parking in stilt / basement / podium	0.25
Type	
Superior RCC building	1.50
Other RCC building	1.00
Pucca construction other than RCC	0.70
Semi-permanent building	0.40
Kuchha building	0.30
Age	
Residential buildings after 01.04.2008	1.00
Non-residential buildings after 01.04.2008	1.50
Residential buildings after 1992 and before 01.04.2008	0.75
Non-residential buildings after 1992 and before 01.04.2008	1.25
Residential buildings before 1992	0.50
Non-residential buildings before 1992	1.10
Height	
Basement excluding car parking; lower first floor; upper first floor; ground floor; 1st to 4th floor	1.00
5th to 10th floor	1.05
11th to 20th floor	1.10

Notes: This table reports the multipliers used in Kolhapur's formula-based property valuation system. Multipliers vary by property category, construction type, building age, and floor height. These weights enter the assessed valuation formula together with the circle rate and property area. Source: Kolhapur Municipal Corporation property tax rules, available at https://web.kolhapurcorporation.gov.in/department?deptid=2

Table A2
Predictors of the circle rate

	(1) Baseline circle rate
Numebr of Households	4.106
	(2.964)
Owner occupied properties	3.181
	(2.818)
Commercial properties	-15.19
	(20.23)
Institutional properties	1.730
	(31.35)
Total population	-1.089
	(1.064)
Population age 0–6	-18.16***
	(4.902)
Illiterate population	6.851***
	(1.765)
Main cultivators	-18.40**
	(7.248)
Main agricultural labourers	7.089
	(9.272)
Total workers	-1.715
	(3.583)
Main workers	2.249
	(2.937)
Main household industry workers	8.122**
	(3.668)
Scheduled Tribe population	3.871
	(8.410)
Total population	0.0389
	(0.274)
Scheduled Caste population	-0.983**
	(0.465)
Party Alignment	-173.4
	(339.1)
Constant	20275.2***
	(1919.5)
Observations	702
R-squared	0.159

Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table presents estimates from a multivariate regression using the baseline 2011 circle rate at the circle-level as the dependant variable and electoral ward characterestics as independant variables. Each electoral ward contains several circles. This specification includes controls for administrative wards, which are bigger and contain several electoral wards. *Households* is the number of households, *Owner occupied*, *Commercial*, and *Institutional* are the number of properties that are owner-occupied, commercial and institutional, respectively. Other covariates include total population, population ages 0–6, illiterate population, total workers, main workers, main cultivators, main agricultural labourers, main household industry workers, Scheduled Tribe population and Scheduled Caste population. *Party Alignment* is a dummy variable equal to one if the ward-level representative in 2009 belonged to the same party as the majority party in the state.

Table A3
Additional Outcomes

Panel A: 0.005 miles

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Ln(Tax Paid)	Share Tax Paid		Tax Paid			
Ln(Baseline Circle Rate)	1.587***	5.328***	1.659***	129.2***	57.01***	1.047***	1.107***
	(0.256)	(0.489)	(0.148)	(20.91)	(14.85)	(0.320)	(0.321)
Obs	10784	5068	9508	5068	10390	9508	15278
Compliers only		Y		Y			
Control Mean	81.78	2.62	1.67	87.72	58.11	56.48	56.48
Elasticity		2.034		1.473	0.981		
Pair x Year, Ward FE		Y	Y	Y	Y	Y	
Pair, Year, Ward FE	Y						Y
Poisson			Y			Y	Y

Panel B: 0.01 miles

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Ln(Tax Paid)	Share Tax Paid		Tax Paid			
Ln(Baseline Circle Rate)	1.195***	4.714***	1.495***	69.49	26.08	0.545	0.607*
	(0.321)	(0.587)	(0.145)	(46.90)	(27.75)	(0.337)	(0.343)
Obs	18495	9068	17288	9068	18938	17288	26419
Compliers only		Y		Y			
Control Mean	81.27	2.10	1.40	79.08	53.76	56.11	56.11
Elasticity		2.245		0.879	0.485		
Pair x Year, Ward FE		Y	Y	Y	Y	Y	
Pair, Year, Ward FE	Y						Y
Poisson			Y			Y	Y

Notes: This table reports the effect of baseline 2011 circle rates on alternative payment outcomes or alternate specifications. Column 1 uses log of taxes paid. Columns 2 and 3 use the share of tax paid, defined as tax paid divided by recorded liability. Columns 4–7 use tax paid in levels. Columns marked Compliers only restrict the sample to propert-pairs with positive payments. Columns marked Poisson are estimated using Poisson pseudo-maximum likelihood to accommodate zero payments.Columns marked as Pair, Year, Ward FE use pair, year and ward fixed effects as opposed to twoway pair-year fixed effects. Panel A restricts matched property pairs to 0.005 miles; Panel B uses 0.01 miles. Standard errors are clustered two-way at the circle-rate and coordinate-circle-rate levels.

Table A4
Compliance, amount paid and balance on residual: Regression Discontinuity

Panel A: Compliance

	(1)	(2)	(3)	(4)
	0.012	0.018	0.031	0.049
		Compliance - Extensive Margin		
High	-0.18	-0.48	-0.11	0.01
	(0.46)	(0.47)	(0.33)	(0.23)
Obs	178	499	1645	2746
Control Mean	0.612	0.752	0.693	0.652
Shared Boundary FE	N	N	N	N

Panel B: Amount of Tax Paid

	(1)	(2)	(3)	(4)
	0.012	0.018	0.031	0.049
		Ln(Tax Paid) - Intensive Margin		
High	1.306	2.311***	1.263	0.580
	(1.241)	(0.487)	(1.195)	(1.046)
Obs	109	375	1140	1790
Control Mean	148.560	64.192	56.005	70.493
Shared Boundary FE	N	N	N	N

Panel C: Residual

	(1)	(2)	(3)	(4)
	0.012	0.018	0.031	0.049
		Ln(Residual)		
High	0.63	1.37	1.20	0.73
	(0.74)	(0.80)	(0.91)	(0.92)
Obs	178	499	1645	2746
Control Mean	138.468	72.513	59.238	71.284
Shared Boundary FE	N	N	N	N

Notes: *Notes:* This table presents regression discontinuity results for the extensive margin measure of compliance, the intensive margin measure of the amount of taxes paid and the residual term, across different bandwidths from columns (1) – (4). Each coefficient comes from estimating the following equation

$$\text{Compliance}_i = \alpha + \tau\,\text{High}_i + \beta_1 D_i + \beta_2\,(\text{High}_i \times D_i) + \varepsilon_i,$$

where High is equal to 1 if a coordinate has the higher circle rate relative to its matched coordinate. D_i is the running variable, which is distance to the circle rate border. I present only τ and use triangular distance weights. These results are then equivalent to using Stata's 'rdrobust'.

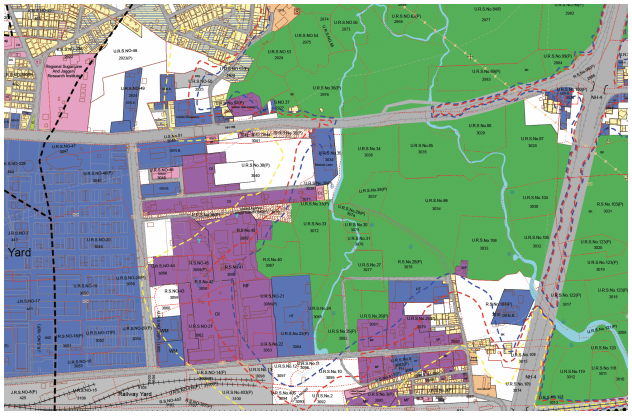
Table A5
How do outcomes vary by position on the circle rate distribution?

	(1) Compliance	(2) Ln(Tax Paid)
Ln(Baseline Circle Rate)	-0.254**	1.082**
	(0.101)	(0.515)
Ln(Baseline Circle Rate) × Ln(Demeaned within-Pair Circle Rate)	-1.323	4.453
	(0.989)	(4.728)
Obs	18938	9068
Control Mean		
Pair x Year, Ward FE		

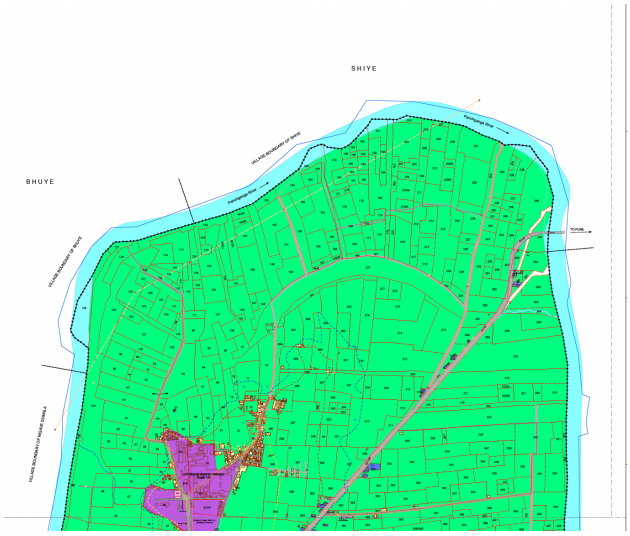
This table reports heterogeneous-response specifications that allow the effect of baseline circle rates to vary with the local position in the circle-rate distribution. I test Equation 16 and 17 in columns 1 and 2 respectively. Column 1 uses compliance as the dependent variable. Column 2 uses log tax paid among compliers. The interaction terms combine log baseline circle rates with demeaned local log circle rates, separately for the compliance and paid samples. The coefficients are used to construct the marginal-effect and elasticity plots in the heterogeneity figures. All specifications include property-pair-by-year and ward fixed effects. Standard errors are clustered two-way at the circle-rate and coordinate-circle-rate levels.

B.2 Figures

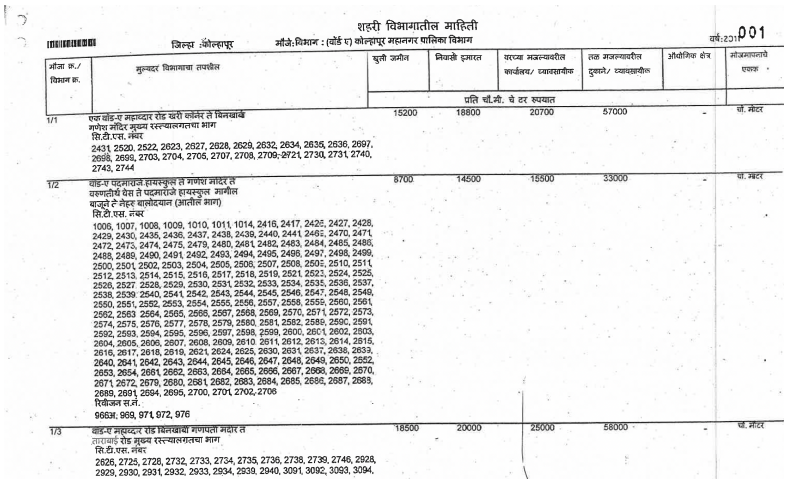
Figure A1
Maps of Survey Areas and Circle Rates



Panel A: Survey area map



Panel B: Survey area map-2



शहरी विभागातील माहिती

वर्ष:2011

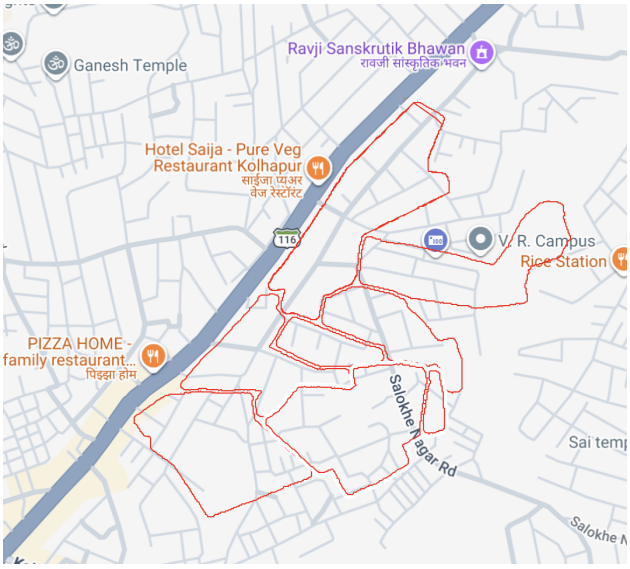
जिल्हा :सोलापूर मौजे:भिमान : (वर्ग ए) सोलापूर महानगर पालिका विभाग

मौजा क्र./ विभाग क्र.	मूल्यदर विभागाचा तपशील	खुली जमीन	निवासी इमारत	वरच्या मजल्यावरील कार्यालय/ व्यावसायीक	तळ मजल्यावरील दुकाने/ व्यावसायीक	औद्योगिक क्षेत्र	मोजमापनाचे एकक
		प्रति चौ.मी. चे दर रुपयात					
1/1	एच वॉड-ए महाव्दार रोड खरी कॉर्नर ते भिमवार्ड गणेश मंदिर मुख्य रस्त्यालगतचा भाग सि.टी.एस. नंबर 2431, 2520, 2522, 2623, 2627, 2628, 2629, 2632, 2634, 2635, 2636, 2697, 2698, 2699, 2703, 2704, 2705, 2707, 2708, 2709, 2721, 2730, 2731, 2740, 2743, 2744	15200	18800	20700	57000	-	चौ. मीटर
1/2	वॉड-ए पद्मावती सयस्कुल ते गणेश नादर ते वरणटोपी पेस ते पद्मावीजे हायस्कुल मागील बाजूने ते सेहर वालीदयान (आतील भाग) सि.टी.एस. नंबर 1006, 1007, 1008, 1009, 1010, 1011, 1014, 2416, 2417, 2425, 2427, 2428, 2429, 2430, 2435, 2436, 2437, 2438, 2439, 2440, 2441, 2465, 2470, 2471, 2472, 2473, 2474, 2475, 2479, 2480, 2481, 2482, 2483, 2484, 2485, 2486, 2488, 2489, 2490, 2491, 2492, 2493, 2494, 2495, 2496, 2497, 2498, 2499, 2500, 2501, 2502, 2503, 2504, 2505, 2506, 2507, 2508, 2509, 2510, 2511, 2512, 2513, 2514, 2515, 2516, 2517, 2518, 2519, 2521, 2523, 2524, 2525, 2526, 2527, 2528, 2529, 2530, 2531, 2532, 2533, 2534, 2535, 2536, 2537, 2538, 2539, 2540, 2541, 2542, 2543, 2544, 2545, 2546, 2547, 2548, 2549, 2550, 2551, 2552, 2553, 2554, 2555, 2556, 2557, 2558, 2559, 2560, 2561, 2562, 2563, 2564, 2565, 2566, 2567, 2568, 2569, 2570, 2571, 2572, 2573, 2574, 2575, 2576, 2577, 2578, 2579, 2580, 2581, 2582, 2585, 2590, 2591, 2592, 2593, 2594, 2595, 2596, 2597, 2598, 2599, 2600, 2601, 2602, 2603, 2604, 2605, 2606, 2607, 2608, 2609, 2610, 2611, 2612, 2613, 2614, 2615, 2616, 2617, 2618, 2619, 2621, 2624, 2625, 2630, 2631, 2637, 2638, 2639, 2640, 2641, 2642, 2643, 2644, 2645, 2646, 2647, 2648, 2649, 2650, 2652, 2653, 2654, 2661, 2662, 2663, 2664, 2665, 2666, 2667, 2668, 2669, 2670, 2671, 2672, 2679, 2680, 2681, 2682, 2683, 2684, 2685, 2686, 2687, 2688, 2689, 2691, 2694, 2695, 2700, 2701, 2702, 2706 रिवीजन स.नं. 966अ, 969, 971, 972, 976	8700	14500	15500	33000	-	चौ. मीटर
1/3	वॉड-ए महाव्दार रोड भिमवाशी गणपती नादर ते सि.टी.एस. नंबर 2626, 2725, 2728, 2732, 2733, 2734, 2735, 2736, 2738, 2739, 2746, 2928, 2929, 2930, 2931, 2932, 2933, 2934, 2939, 2940, 3091, 3092, 3093, 3094,	18500	20000	25000	58000	-	चौ. मीटर

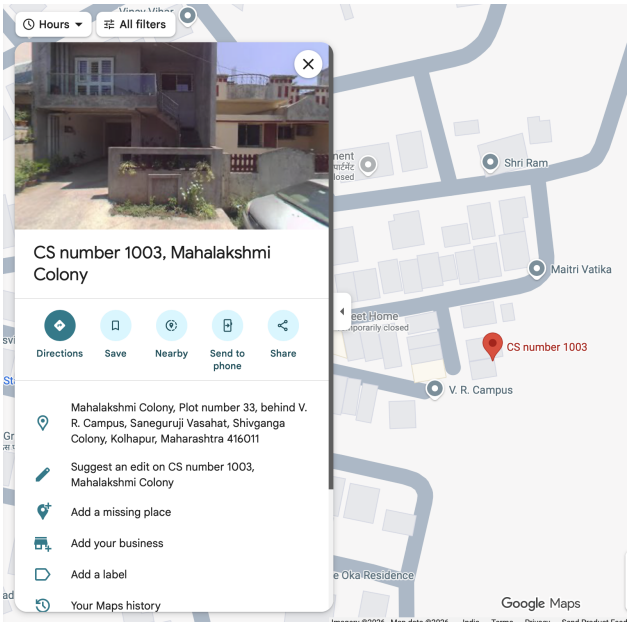
Panel C: Circle rates: 2011

Notes: This figure presents screenshots of survey area pdfs in Panel A and B and a snapshot of the pdf displaying circle rates per survey area in panel C. In Panels A and B, outlines in light red mark survey areas and the text in black marks the survey number. Other dashed red, blue, black, yellow, lines present other administrative divisions within the city

Figure A2
Parcels and Address Example



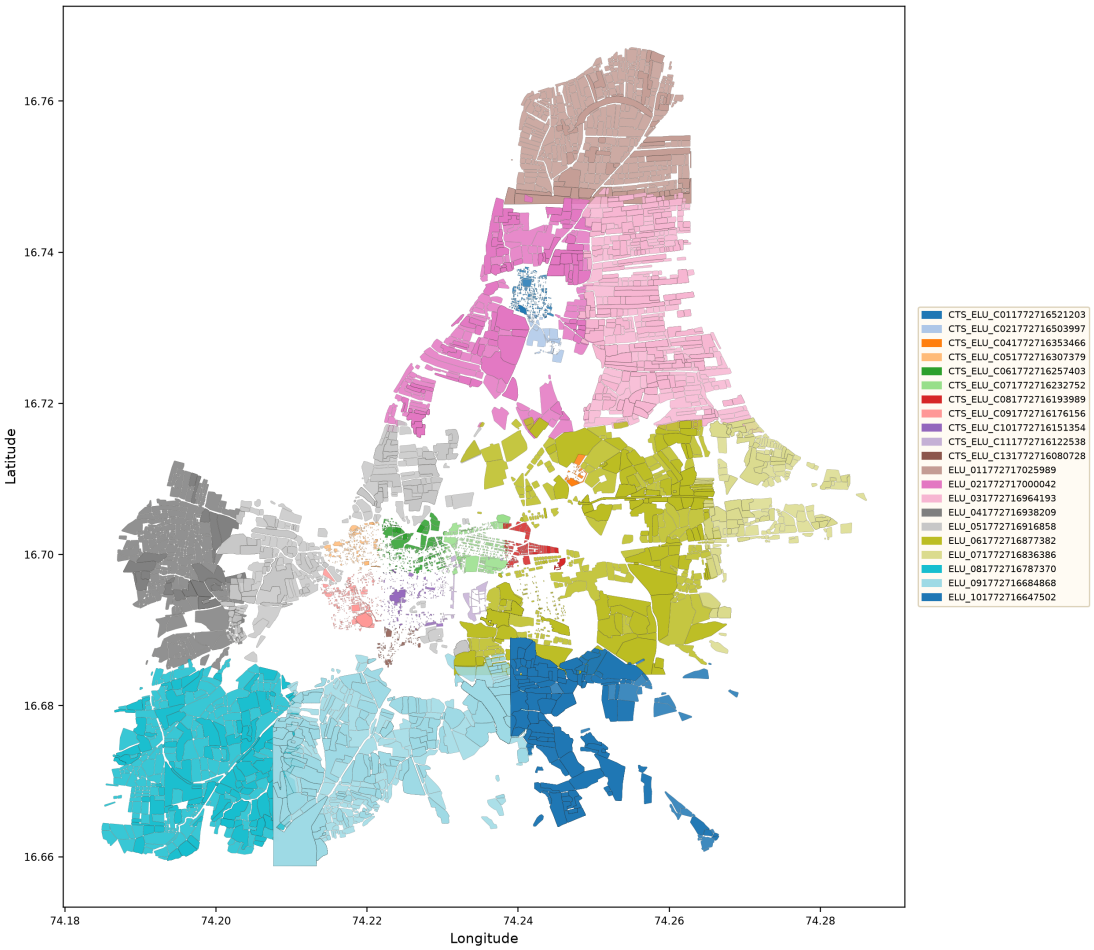
Panel A: Sample of parcels on Google Maps



Panel B: An example address

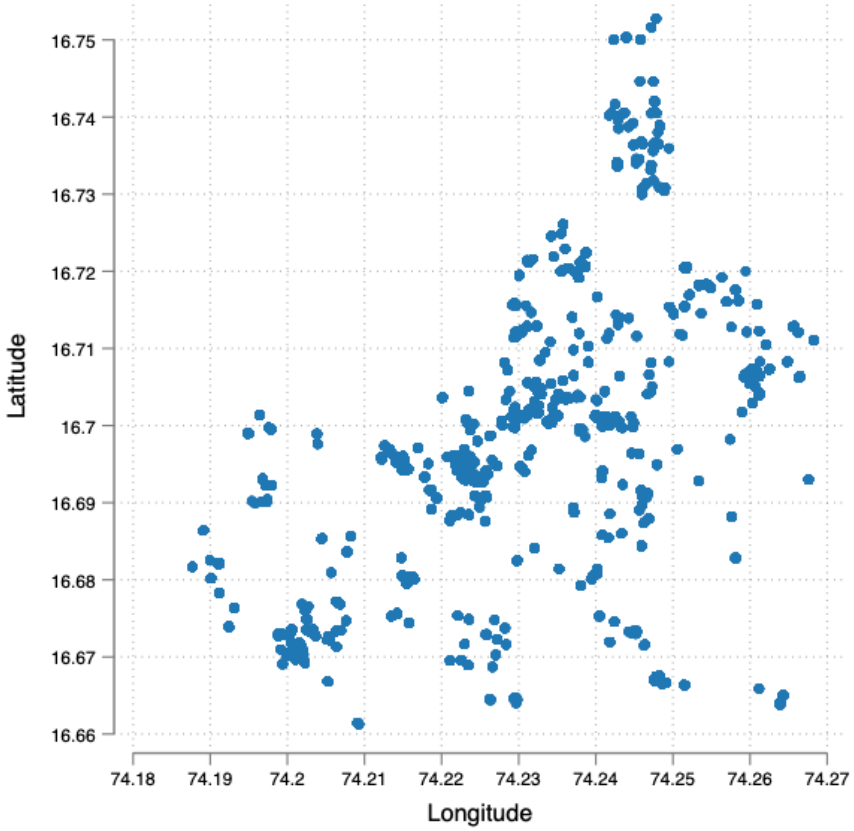
Notes: This figure presents a sample of parcels imposed on a Google Maps screenshot of the city in panel A and an example of a coordinate collected manually, from an address using Google Maps.

Figure A3
Digitising Maps



Notes: This figure shows the 18 maps that were combined and digitized to form the full map of circle boundaries. Each map is color-coded to be a different color

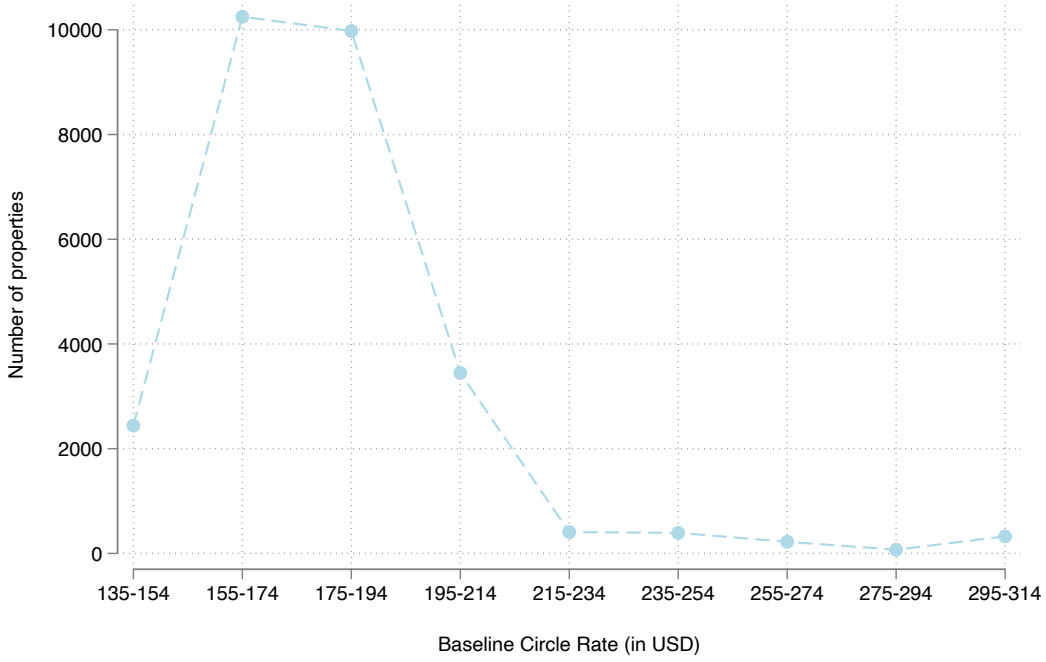
Figure A4
Coordinates in the sample



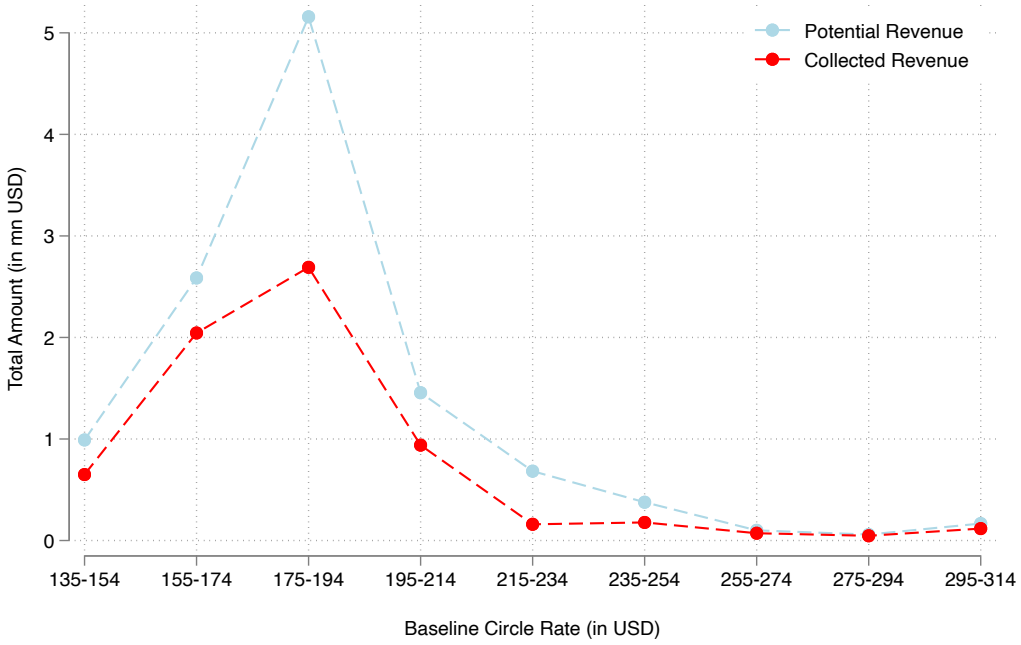
Notes: Each blue dot in this figure represents a coordinate in the final sample. Each coordinate could have mulltiple properties.

Figure A5
Distribution of Properties and Potential versus Actual Revenue

Panel A. Number of properties by circle rate bins

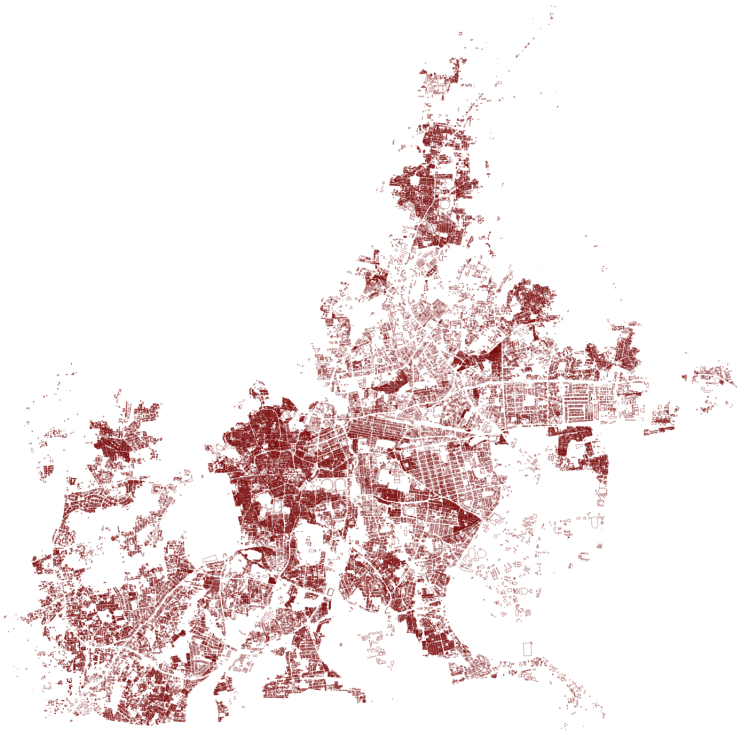


Panel B. Potential and actual revenue by circle rate bins



Notes: Panel A shows the distribution of owner-occupied properties across 2011 circle-rate bins. Panel B shows actual revenue, defined as the sum of tax payments and penalties collected, and potential revenue, defined as the sum of assessed tax liabilities, within each bin. The sample is restricted to properties observed in the balanced panel from 2013 to 2024. Circle rates are expressed in U.S. dollars using an exchange rate of INR 85 per U.S. dollar. Within this sample, there is about 8 million USD in uncollected, potential revenue.

Figure A6
Google Footprints data

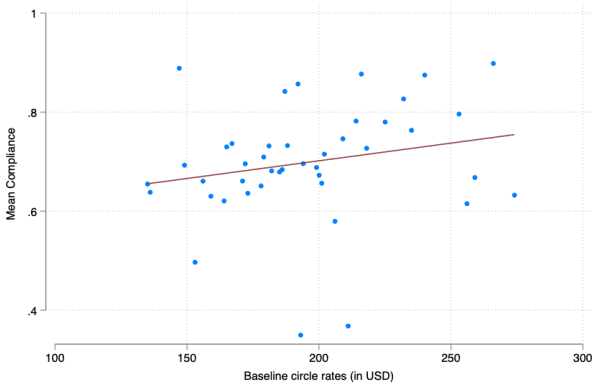


Panel A: Snapshot of the whole city

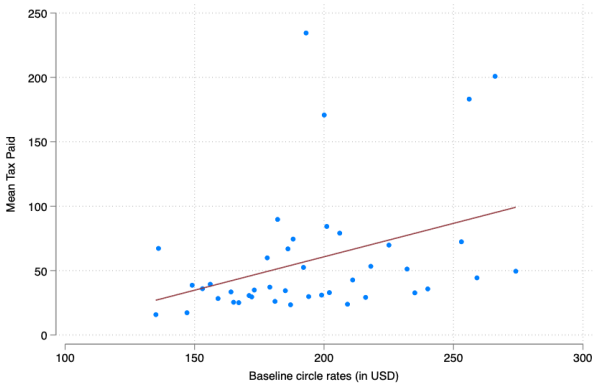
Panel B: Zoomed in

Notes: This figure presents Google Footprints shapefile for the whole city of Kolhapur in panel A, and a zoomed in snapshot in Panel B. In panel B, each rectangle or square is a property parcel.

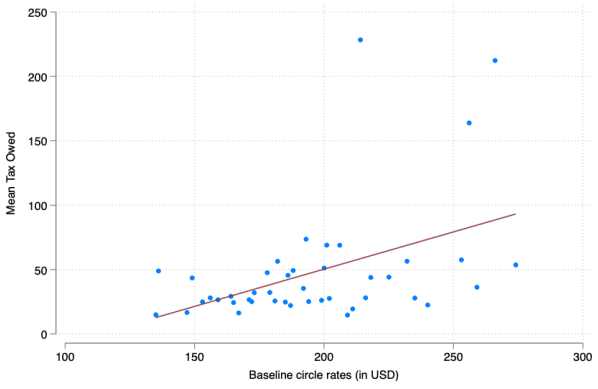
Figure A7
Descriptive Plots



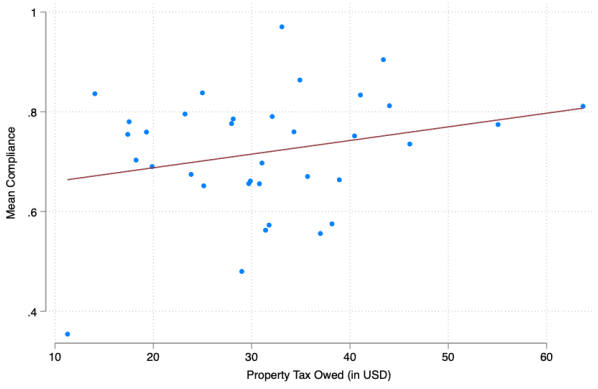
Panel A. Compliance



Panel B. Tax Paid



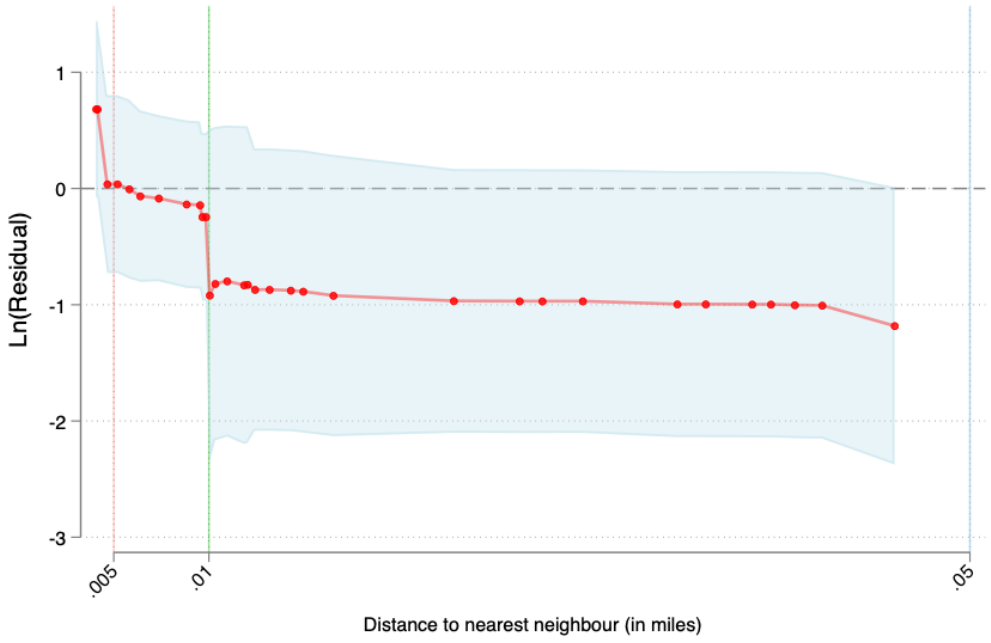
Panel C. Tax Owed



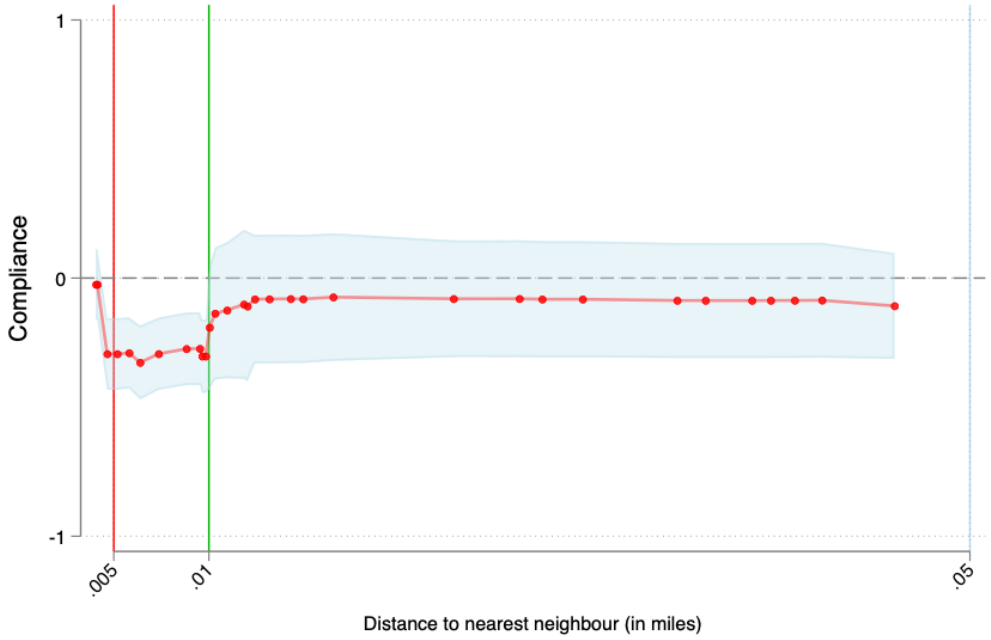
Panel D. Compliance and Tax Owed

Notes: This figure presents descriptive binned scatter plots plots of the properties in the final sample. Panels A, B and C plot the mean circle rate in 2011 against mean compliance, taxes paid and taxes owed, respectively. Panel D plots mean compliance against the property tax owed.

Figure A8
Distribution of the Residual and complaince after 0.01 miles



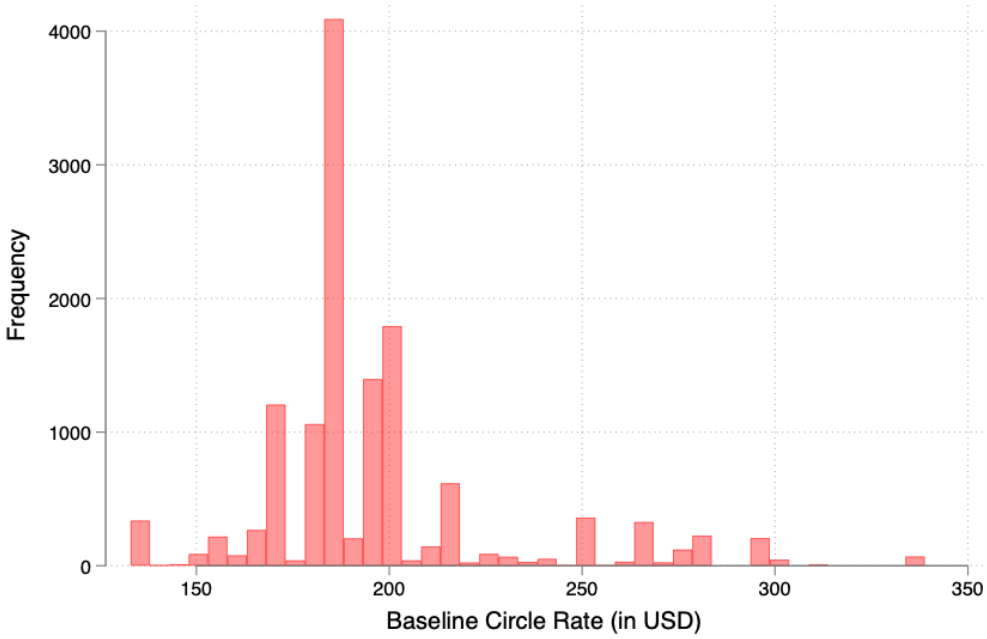
Panel A. Residual



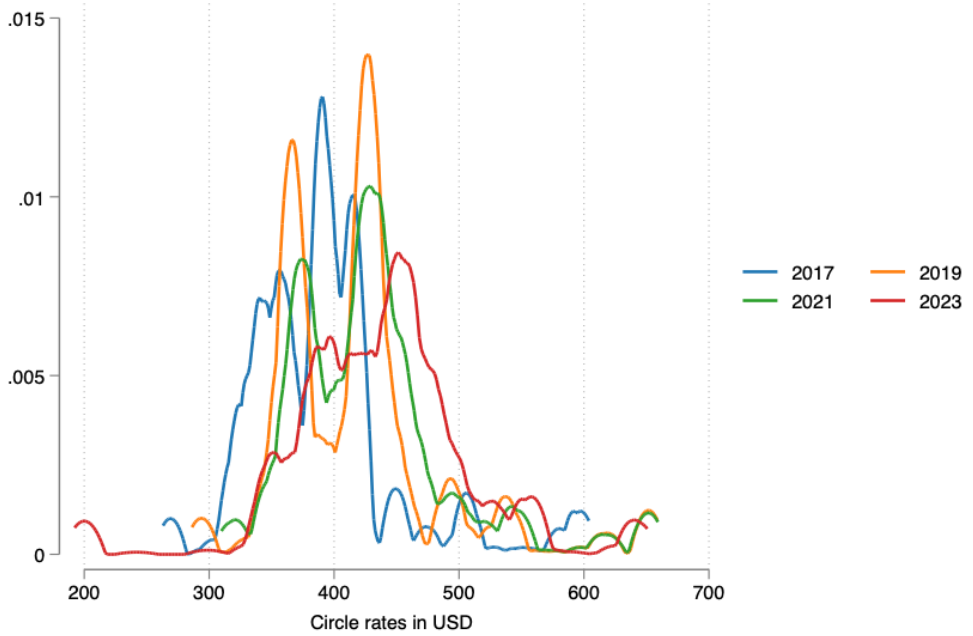
Panel B. Compliance

Notes: This figure shows how the balance and compliance estimates vary as the matched-pair distance threshold expands beyond the main 0.01-mile cutoff. Panel A plots coefficients for the residual from the valuation formula, defined in Equation 9. Panel B plots coefficients for compliance. Confidence intervals are shown at the 95 percent level. The figure illustrates that balance is strongest at short distances and that the compliance effect attenuates as more distant pairs are included.

Figure A9
Circle Rate Distributions



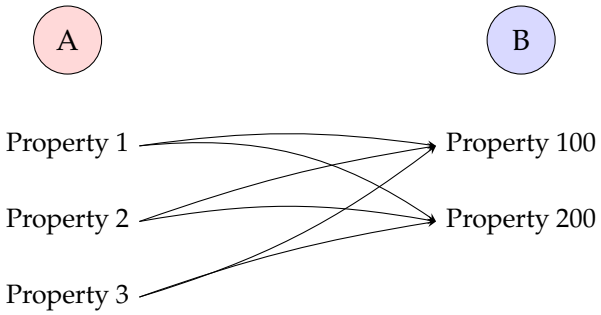
(a) Circle rate distribution: 2011



(b) Circle rate distributions: 2015–2024

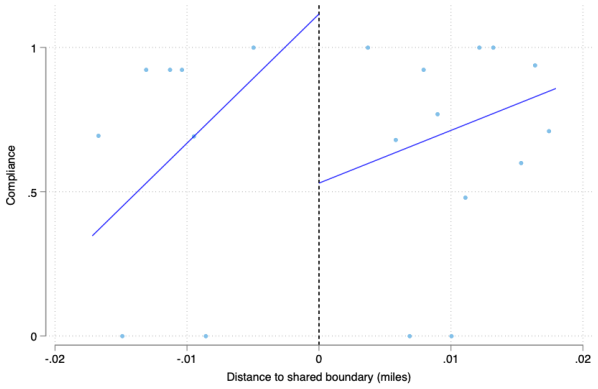
x *Notes:* This figure presents the distribution of circle rates in Kolhapur. Panel A plots the distribution of baseline 2011 circle rates. Panel B plots density distributions for later observed circle-rate schedules in 2017, 2019, 2021, and 2023. Circle rates are expressed in U.S. dollars using an exchange rate of INR 85 per U.S. dollar.

Figure A10
Illustration of pair formation

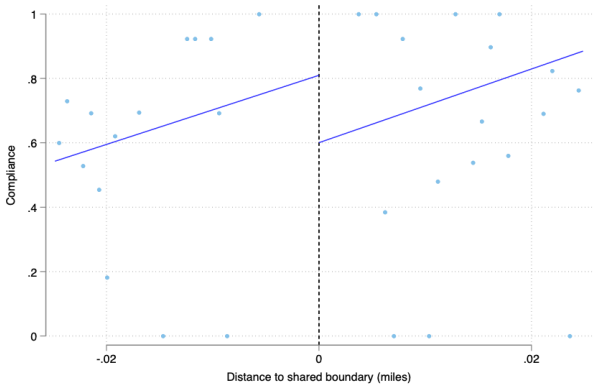


Notes: This figure provides a simplified illustration of the nearest neighbour matching between coordinates and the formation of property-pairs. The two circles A and B represent two different coordinates that got matched using a nearest-neighbor matching algorithm. properties 1, 2 and 3 are 3 properties in coordinate A and properties 100 and 200 are 2 properties located in coordinate B. The arrows between the 5 properties represent the number of pairs being formed, which is 6 in this illustration.

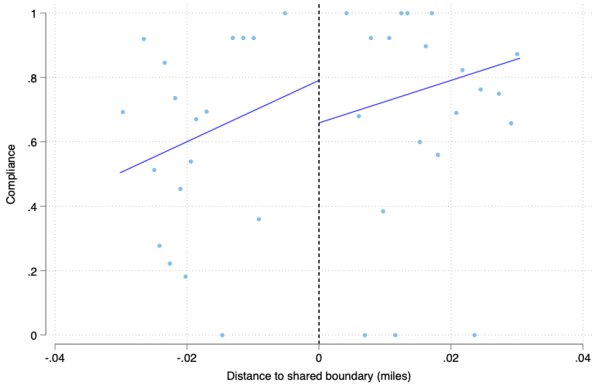
Figure A11
Regression Discontinuity estimates across bandwidths



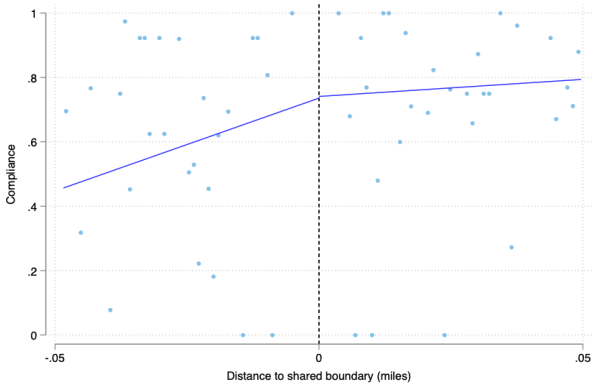
(a) Bandwidth = 0.018 miles



(b) Bandwidth = 0.024 miles



(c) Bandwidth = 0.031 miles



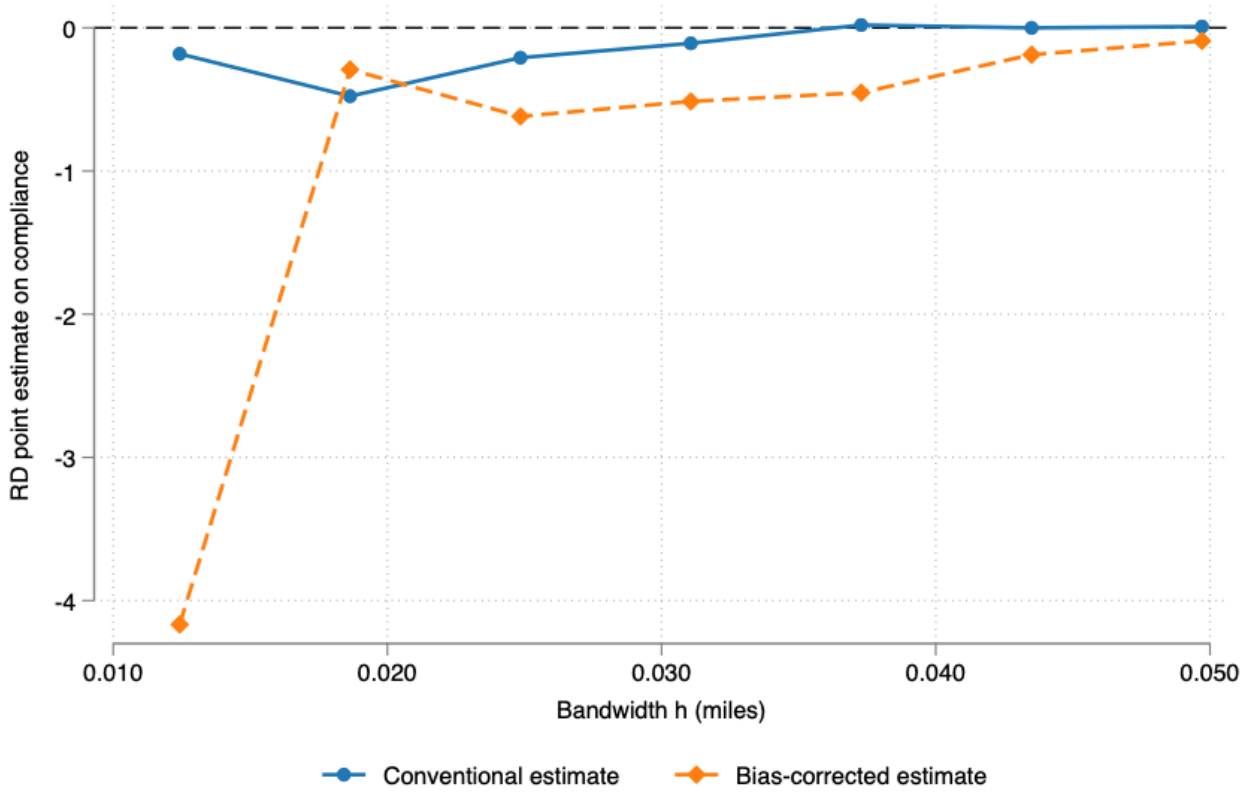
(d) Bandwidth = 0.049 miles

Notes: This figure presents regression discontinuity plots for different bandwidths across Panels (a) – (d). In each figure I estimate

$$\text{Compliance}_i = \alpha + \tau\,\text{High}_i + \beta_1 D_i + \beta_2\,(\text{High}_i \times D_i) + \varepsilon_i,$$

where High is equal to 1 if a coordinate has the higher circle rate relative to its matched coordinate. D_i is the running variable, which is distance to the circle rate border. All specifications use linear polynomials to fit points of both sides of the discontinuity and triangular kernel weights, thus assigning higher weights to observations that lie closer to the border. Observations to the right of the vertical line at 0 are 'treated' observations or those where High is equal to 1. Each point is then a binned mean of the outcome variable, compliance. These plots are formed using Stata's 'rdplot' command.

Figure A12
Regression Discontinuity estimates across bandwidths

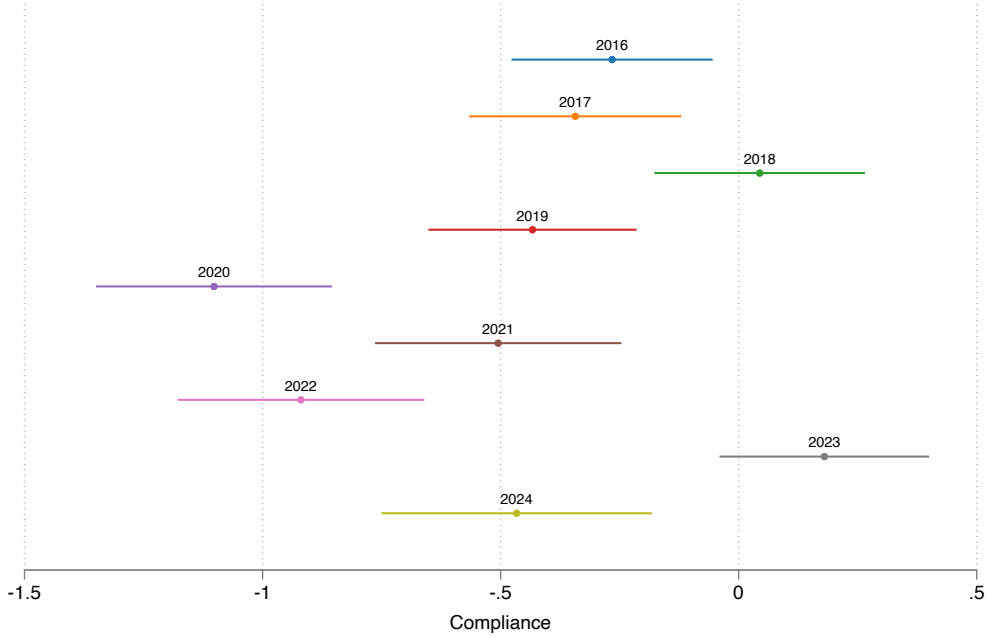


Notes: This figure presents estimates from testing:

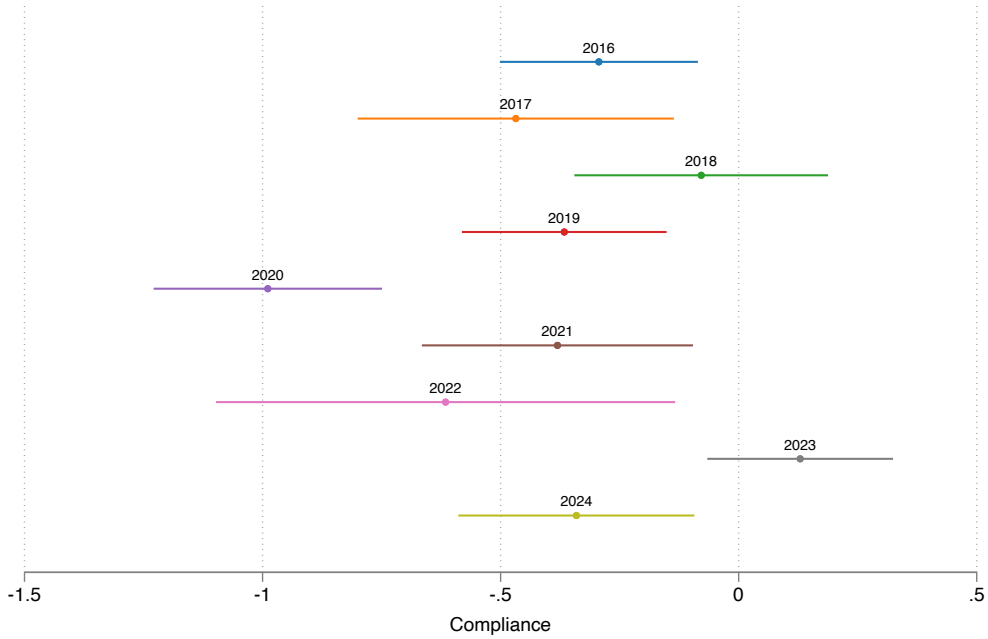
$$\text{Compliance}_i = \alpha + \tau\,\text{High}_i + \beta_1 D_i + \beta_2\,(\text{High}_i \times D_i) + \varepsilon_i$$

for different distance bandwidths. Distances in miles are presented on the x-axis. Each point plots τ, where blue dots are conventional estimates and orange dots are bias-corrected estimates from 'rdrobust' on stata. All specifications use linear polynomials to fit points of both sides of the discontinuity and triangular kernel weights.

Figure A13
Compliance through time



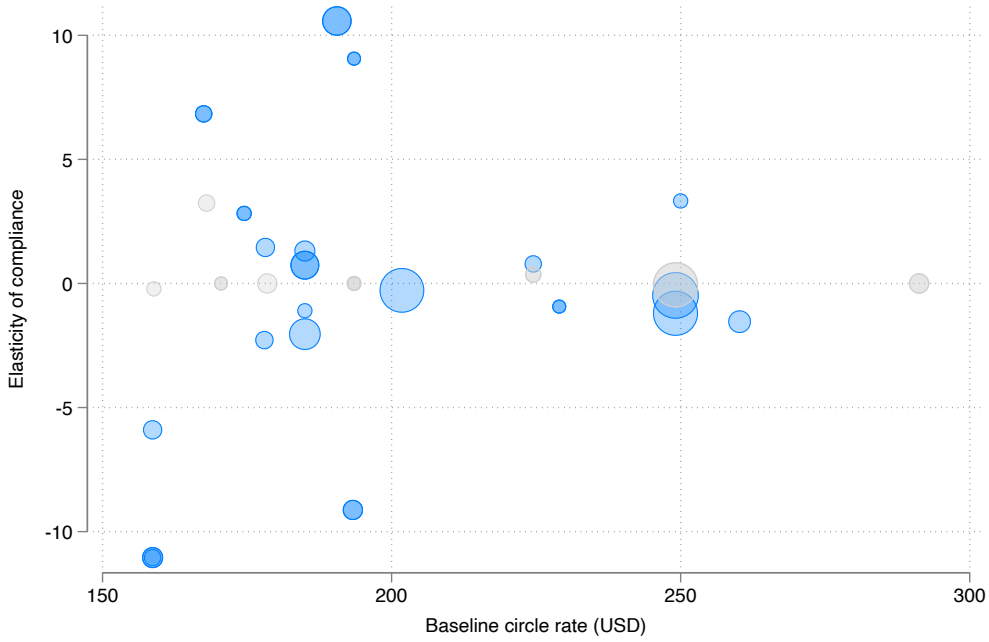
Panel A. 0.005 miles



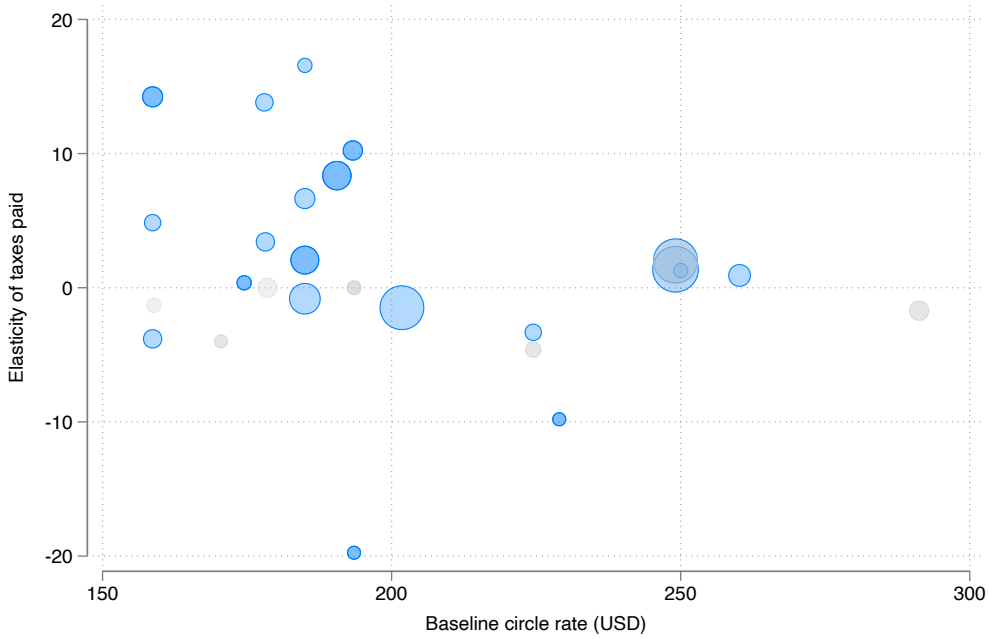
Panel B. 0.01 miles

Notes: This figure presents the coefficient on compliance from estimating a modified version of Equation 10. Each specification uses only property-pair fixed effects and estimates a seperate estimate for every year. Panel A presents estimates using a threshold distance of 0.005 miles and Panel B presents estimates using a distance of 0.01 miles. Confidence intervals are presented at the 95% level.

Figure A14
Elasticity of compliance and taxes paid: non-parametric



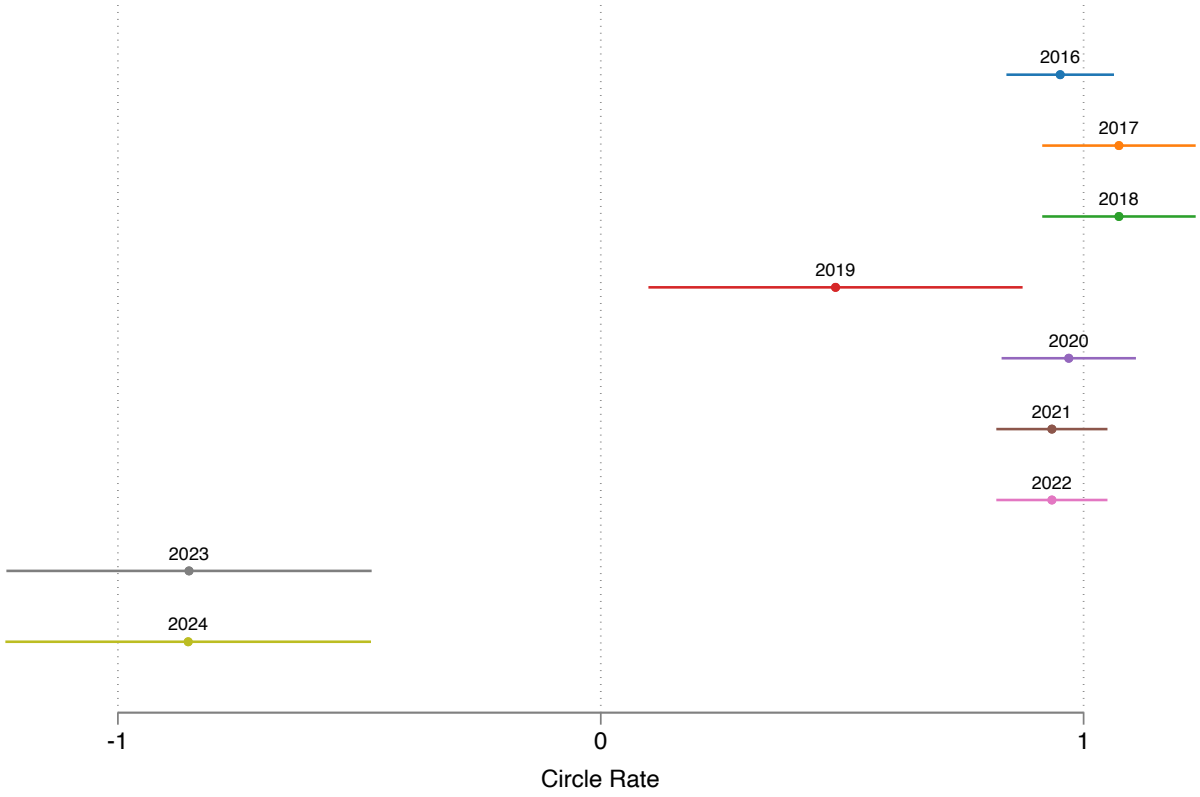
Panel A. Elasticity of Compliance



Panel B. Elasticity of Taxes Paid

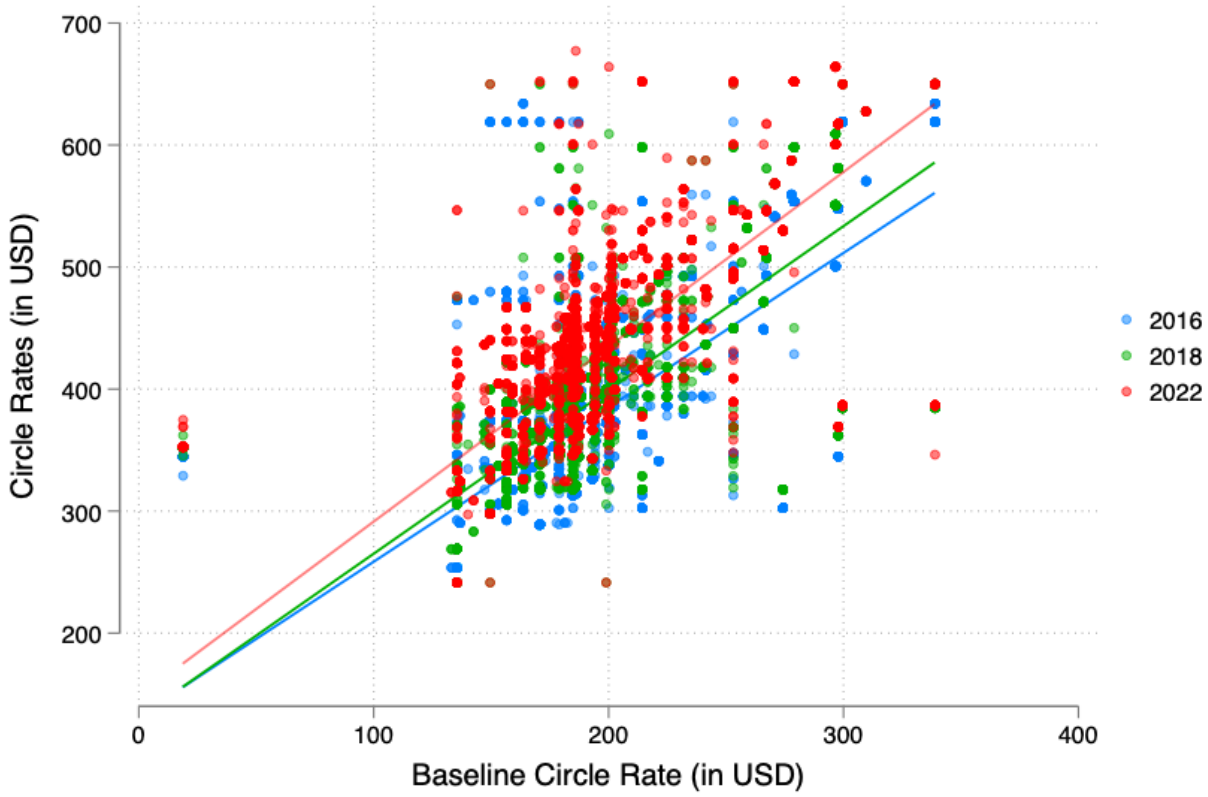
Notes: This figure presents the coeffcients and elasticities from estimating specifications analogous to Equation 10. In Panel A, each point presents the corresponding β divided by the mean of compliance within that sample. In Panel B, each point presents the estimated β by using taxes paid as the dependant variable and a sample of compliers. Estimates are presented at the mean of the circle-rate within the pair. Each point is weighted by the number of properties in a coordinate-pair and larger circles represent a larger sample. The dashed blue line is the line of best fit, weighted by the number of properties. In Panel B, one point with an elasticity greater than 200 is not shown here for ease of representation, but is used to calculate the line of best fit.

Figure A15
Dynamics: How are contemporaneous circle rates related to baseline 2011 circle rates?



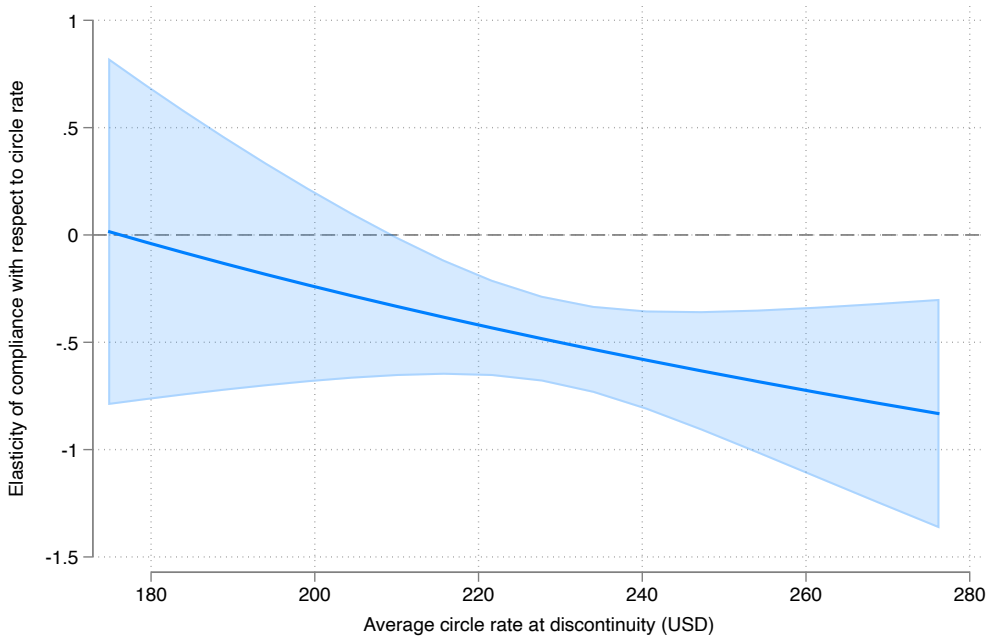
Notes: This figure reports dynamic estimates of the relationship between baseline circle rates and contemporaneous circle rates using the 0.01-mile matched-pair sample. Points show year-specific or event-time coefficients from modified versions of the main matched-pair specification. Confidence intervals are shown at the 95% level. The figure is used to assess the evolution of teh relationship between circle rates in 2011 and those from 2016–2024.

Figure A16
Baseline circle rate versus circle rates in 2016. 2018, 2022

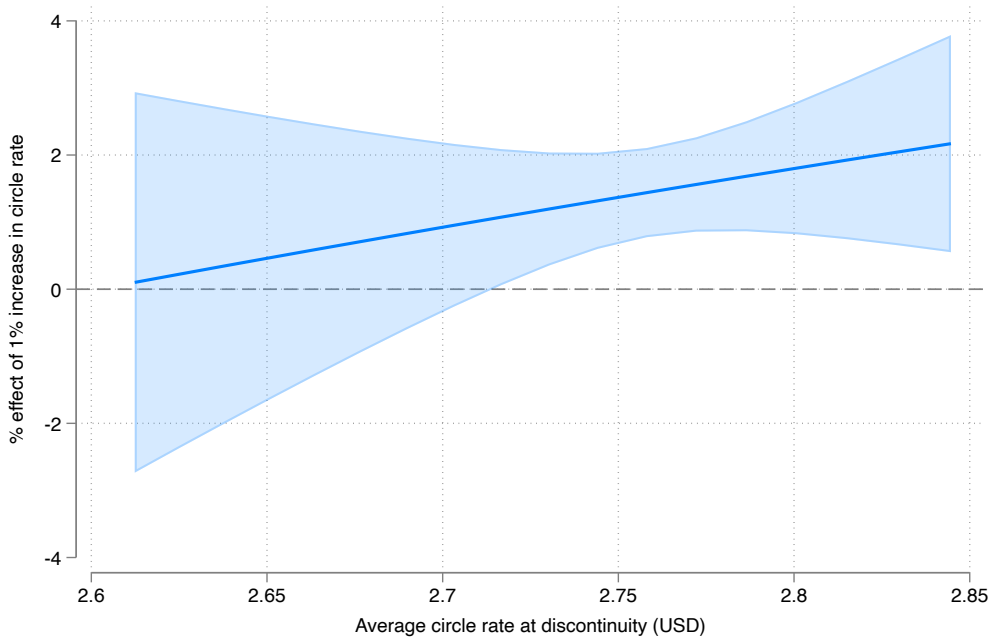


Notes: This figure plots baseline 2011 circle rates against later circle rates in selected years. Each point represents a circle-rate area. The figure is used to assess whether areas with high baseline circle rates also experienced faster subsequent growth in administratively assigned circle rates.

Figure A17
Elasticity of Compliance and Taxes Paid: heterogeneity, nonlinear moderator



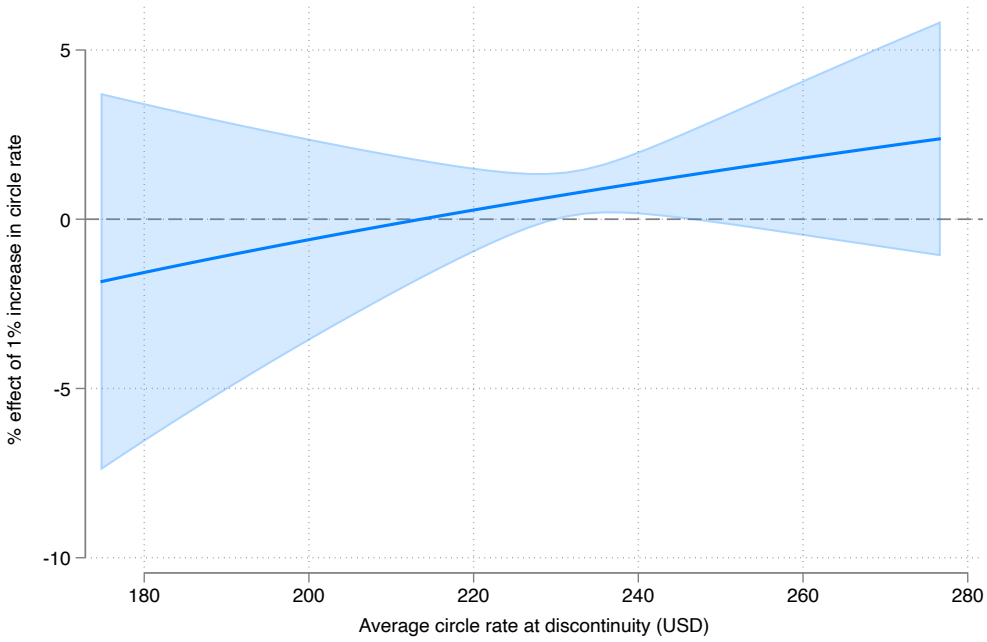
Panel A. Compliance



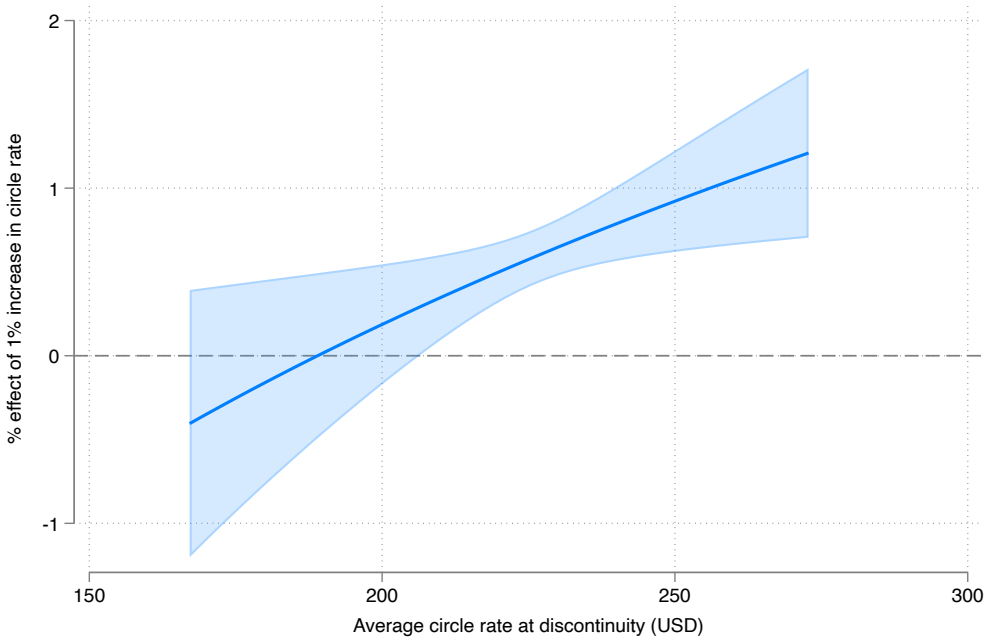
Panel B. Elasticity of Taxes Paid

Notes: This figure presents the results from estimating a modified version of Equation 16 in Panel A and 17 in Panel B, with a non-linear moderator. In Panel A, compliance is defined as a binary variable equal to 1 if a property makes any payment in a given year. In Panel B taxes paid is the dollar amount of taxes paid for compliant properties. The dark blue line presents estimated elasticities at each point. The shaded blue area represents 95% confidence intervals.

Figure A18
Elasticity of taxes paid and share of taxes paid: heterogeneity



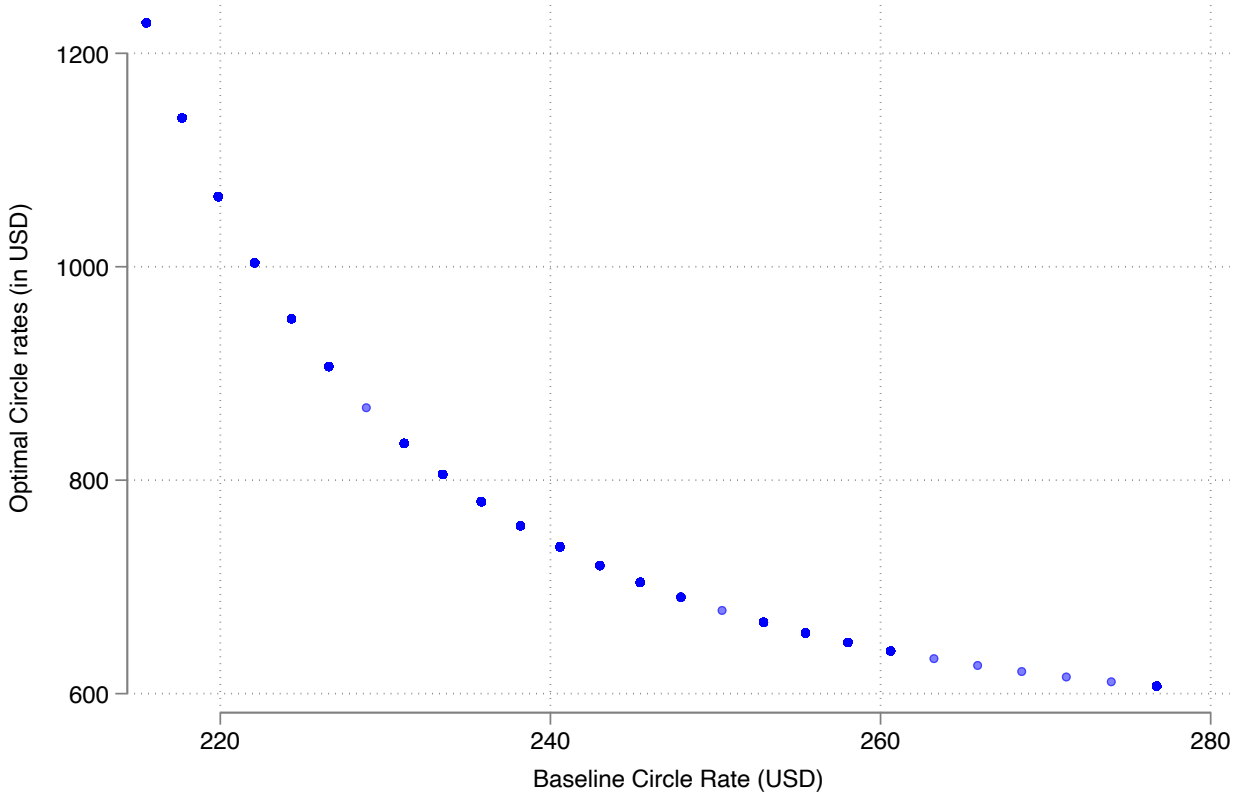
Panel A.Taxes Paid: Including non-compliers



Panel B. Share Tax Paid

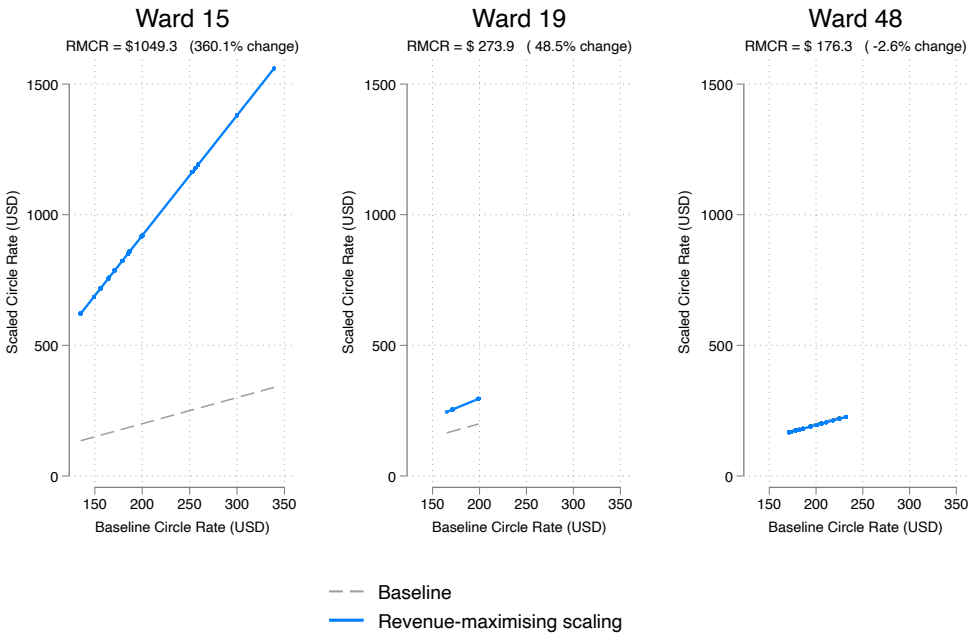
Notes: This figure presents the results from estimating Equation 16 in Panel A with dependant variable being the level of taxes paid (including zeros)in Panel A and the share of taxes paid in Panel B. The dark blue line presents estimated elasticities at each point. The shaded blue area represents 95% confidence intervals.

Figure A19
Local RMCRs measured using heterogenous elasticities



Notes: This figure plots local revenue-maximizing circle rates implied by heterogeneous elasticities across the baseline circle-rate distribution. Each point maps a baseline 2011 circle rate to the RMCR implied by the estimated local compliance and payment elasticities. The figure is an appendix exercise because local estimates at the lower end of the circle-rate distribution imply very high RMCRs, reflecting relatively weak estimated compliance responses in that part of the distribution.

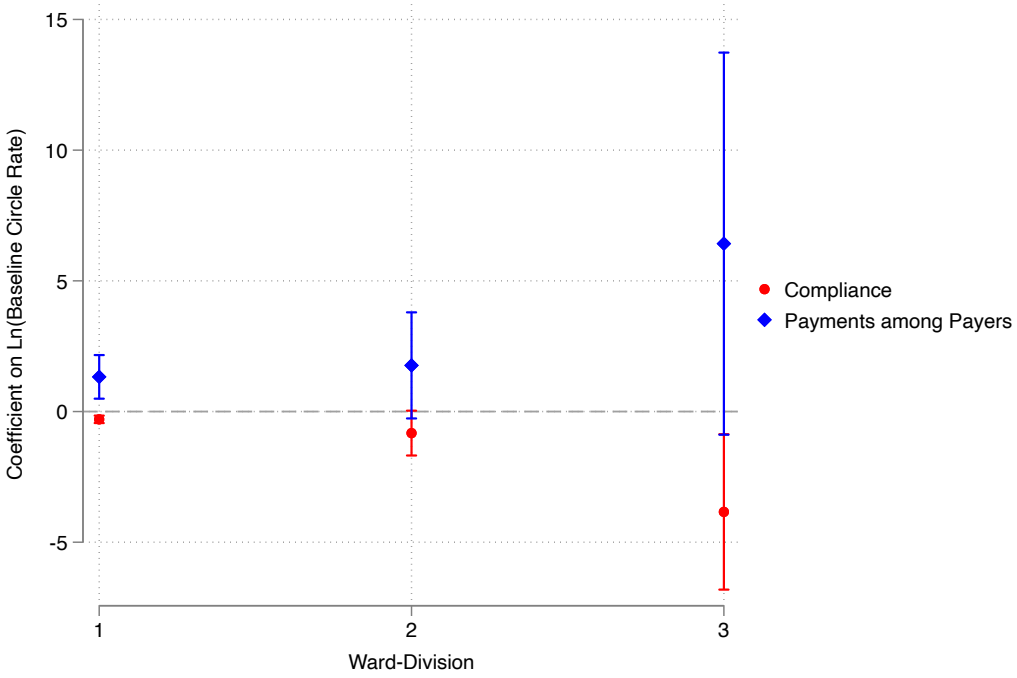
Figure A20
Ward-level circle rate schedule for revenue maximisation



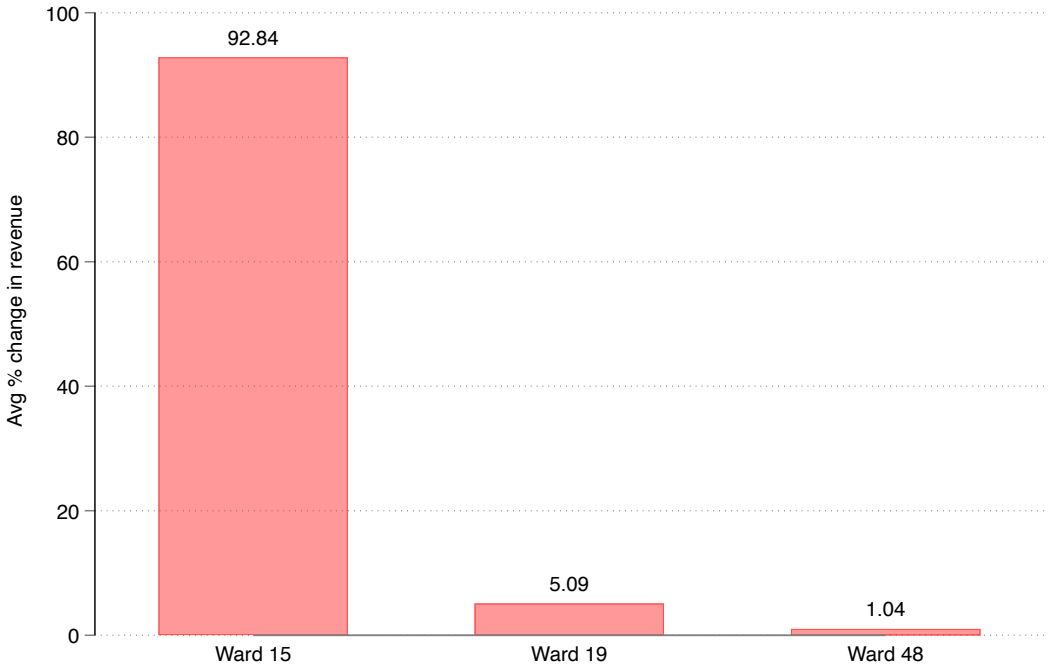
Ward-level updates using ward-specific average responses

Notes: Notes: Notes: This figure s ward-specific RMCR schedules for selected wards using ward-specific average responses. Not all baseline circle rates are observed within every ward. I plot the corresponding revenue changes from updating to the ward-level elasticities in Figure A21

Figure A21
Ward-specific RMCR



Panel A. Ward-level elasticities



Panel B. Ward-level changes in revenue

Notes: This figure reports ward-level heterogeneity in behavioral responses and implied revenue gains. Panel A plots ward-division-specific coefficients on log baseline circle rates, estimated separately for compliance and log tax paid among payers; whiskers show 95 percent confidence intervals. Panel B reports predicted average percentage changes in revenue from moving selected wards to their ward-specific RMCR schedules. Calculations combine the estimated extensive-margin compliance response with the intensive-margin payment response. Positive values indicate higher predicted revenue relative to the baseline schedule.